

Asymmetric Information in Wage Contracts: Evidence from an Online Experiment

Daniel Herbst
University of Arizona

Link to paper



Link to slides



Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?

Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?
- What would be the premium for a contract that...

Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?
- What would be the premium for a contract that...
 - Pays restaurant servers when they're stiffed on the tip?

Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?
- What would be the premium for a contract that...
 - Pays restaurant servers when they're stiffed on the tip?
 - Pays lawyers when they lose a case?

Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?
- What would be the premium for a contract that...
 - Pays restaurant servers when they're stiffed on the tip?
 - Pays lawyers when they lose a case?
 - Pays academics when they're rejected from journals?

Insuring Labor Product

- Most workers' labor product depends on uncertain factors
- What kinds of contracts might reduce that uncertainty?
- What would be the premium for a contract that...
 - Pays restaurant servers when they're stiffed on the tip?
 - Pays lawyers when they lose a case?
 - Pays academics when they're rejected from journals?
- Employers provide *implicit insurance* through **fixed wages**

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
 - Rideshare driver paid per mile
 - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
 - Limo driver paid per hour
 - Staff photographer with annual salary

← Earnings Risk

Insurance →

Hourly wage + tips

Annual salary + bonus

Base pay + commission

Are these contracts socially optimal?



Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
 - Rideshare driver paid per mile
 - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
 - Limo driver paid per hour
 - Staff photographer with annual salary

Earnings Risk

Insurance

Asymmetric information $\Rightarrow$ market distortions:

- **Moral hazard:** fixed wage induces less effort
- **Adverse selection:** less productive workers sort into fixed wages

$\Rightarrow$ Suboptimal wage contracts

Public policies can mitigate these distortions:

- Taxes tips, commissions, or bonuses
- Hourly wage subsidies
- Employment classification rules
- Portable benefits programs
- Minimum wage

$\Rightarrow$ Promote implicit insurance in wage contracts

How do moral hazard and adverse selection influence wage contracts?

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
 - Two-stage RCT separates moral hazard, adverse selection, and wage effects
 1. Offer workers a *choice* between randomized hourly wage or piece rate
 2. Randomly increase hourly workers' wages to match highest wage offer
 - Find evidence of both moral hazard and adverse selection into hourly wages
 - MH $\equiv$ Treatment effect: take-up of hourly contract $\Rightarrow$ output $\downarrow$ by 6.3% relative to the mean
 - AS $\equiv$ Selection into treatment: wage offer $\uparrow$ by 10% $\Rightarrow$ marginal worker's productivity $\uparrow$ by 1.4%
(2nd stage top-ups $\Rightarrow$ estimates are independent of potential wage effects)

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
2. Develop model of fixed-wage labor markets under asymmetric info
 - Clarifies roles of AS and MH in shaping equilibrium, allowing for monitoring costs
 - Equilibrium and efficient shares of hourly work are determined by three curves:
 1. $\bar{w}(\theta)$: **reservation wage** for an hourly position at quantile θ
 2. $MV(\theta)$: **marginal value** of output among workers at a given reservation wage
 3. $AV(\theta)$: **average value** of output among workers with lower reservation wage

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
2. Develop model of fixed-wage labor markets under asymmetric info
3. Use MTE framework to estimate equilibrium and efficient hourly wages
 - Let experimental wage offers serve as **instrument** for hourly contract take-up $\Rightarrow$
 - Hourly labor supply at given wage, $S(w)$ = propensity score
 - Marginal value under hourly pay, $MV_1(\theta)$ = potential outcome in treated state
 - Marginal value under piece rate, $MV_0(\theta)$ = potential outcome in untreated state
 - $\Rightarrow$ MTE of hourly contracts map directly into model
 - Flexible framework can be applied to other markets (e.g., restaurant servers, barbers, etc.)
 - Application: adverse selection in rideshare driving (Angrist et al., 2021)

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
2. Develop model of fixed-wage labor markets under asymmetric info
3. Use MTE framework to estimate equilibrium and efficient hourly wages
4. Calculate DWL of inefficient wages and welfare impact of tax/subsidy
 - Underprovision of fixed-wage jobs $\Rightarrow$ Welfare loss of \$0.04 per hour of labor
 - Piece-rate tax $\leq 14\%$ $\Rightarrow$ raises \$1 of govt revenue at social cost of \$0.87 or less
 $\Rightarrow$ *Eliminating* taxes on output-based pay (e.g., tips) likely has low social returns!

Existing Literature

- Asymmetric Information in Labor Markets
 - Spence (1973), Weiss (1980), Gibbons and Katz (1991), Farber and Gibbons (1996), Shimer (2005), Ales and Maziero (2016), Pallais (2016)
- Selection and Incentive Effects of Wage Contracts
 - Lazear (2000), Nagin et al. (2002), Gaynor et al. (2004), Shearer (2006), DellaVigna and Pope (2018), Kantarevic and Kralj (2016), Angrist et al. (2021), Brown and Andrabi (2021), Emanuel and Harrington (2024)
- Measuring Adverse Selection & Moral Hazard
 - Chiappori and Salanie (2000), Karlan and Zinman (2009), Bryan et al. (2015), Einav et al. (2010), Kowalski (2023), Herbst and Hendren (2024)
- Selection and Marginal Treatment Effects
 - Black et al. (2022), Mogstad et al. (2018), Huber (2013), Carneiro et al. (2011), Heckman and Vyctlacil (2005, 2007), Björklund and Moffitt (1987)

Outline

- ① **Experimental Design and Results**
- ② **Model of Asymmetric Information**
- ③ **Estimates of Marginal Value and Welfare Loss**
- ④ **Optimal Fixed-Wage Subsidy and Piece-Rate Tax**
- ⑤ **Extensions**

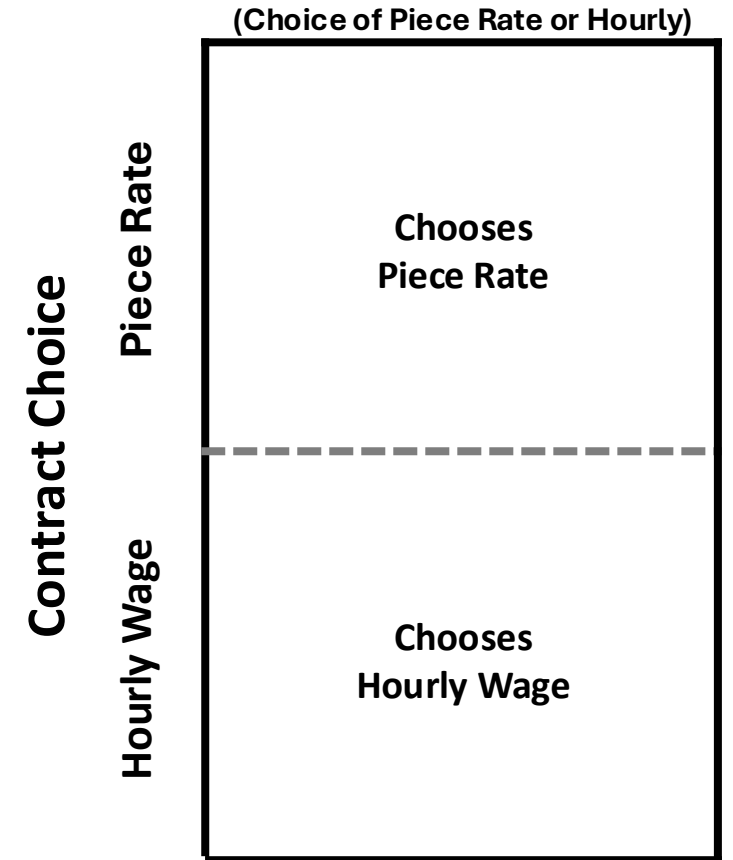
Outline

- 1 Experimental Design and Results**
- 2 Model of Asymmetric Information**
- 3 Estimates of Marginal Value and Welfare Loss**
- 4 Optimal Fixed-Wage Subsidy and Piece-Rate Tax**
- 5 Extensions**

Detecting Moral Hazard and Adverse Selection

- Suppose worker i is offered a choice:
 - Remain on piece rate ($H_i = 0$)
 - Switch to an hourly wage ($H_i = 1$)
- Potential outcomes:
 - Y_{1i} : Worker i 's potential output under hourly wage
 - Y_{0i} : Worker i 's potential output under piece rate

$$Y_i = H_i Y_{1i} + (1 - H_i) Y_{0i}$$

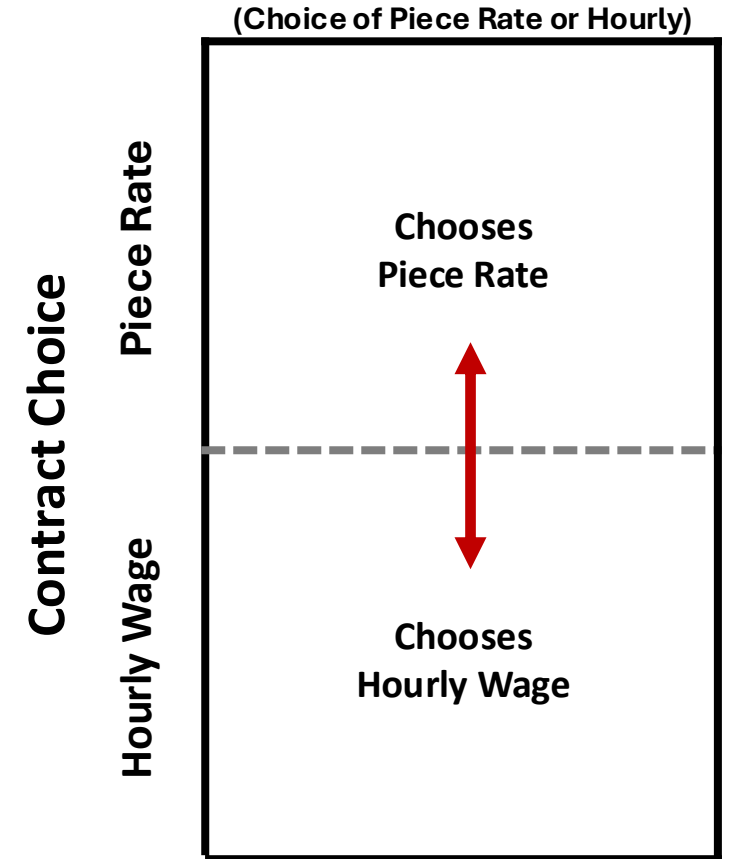


Detecting Moral Hazard and Adverse Selection

- Suppose worker i is offered a choice:
 - Remain on piece rate ($H_i = 0$)
 - Switch to an hourly wage ($H_i = 1$)
 - Potential outcomes:
 - Y_{1i} : Worker i 's potential output under hourly wage
 - Y_{0i} : Worker i 's potential output under piece rate
- $Y_i = H_i Y_{1i} + (1 - H_i) Y_{0i}$
- Comparing across self-selected groups combines treatment and selection:

$$E[Y_i | H_i = 1] - E[Y_i | H_i = 0] =$$

$$\underbrace{E[Y_{1i} - Y_{0i} | H_i = 1]}_{\text{Treatment on the Treated}} + \underbrace{E[Y_{0i} | H_i = 1] - E[Y_{0i} | H_i = 0]}_{\text{Selection on } Y_0}$$

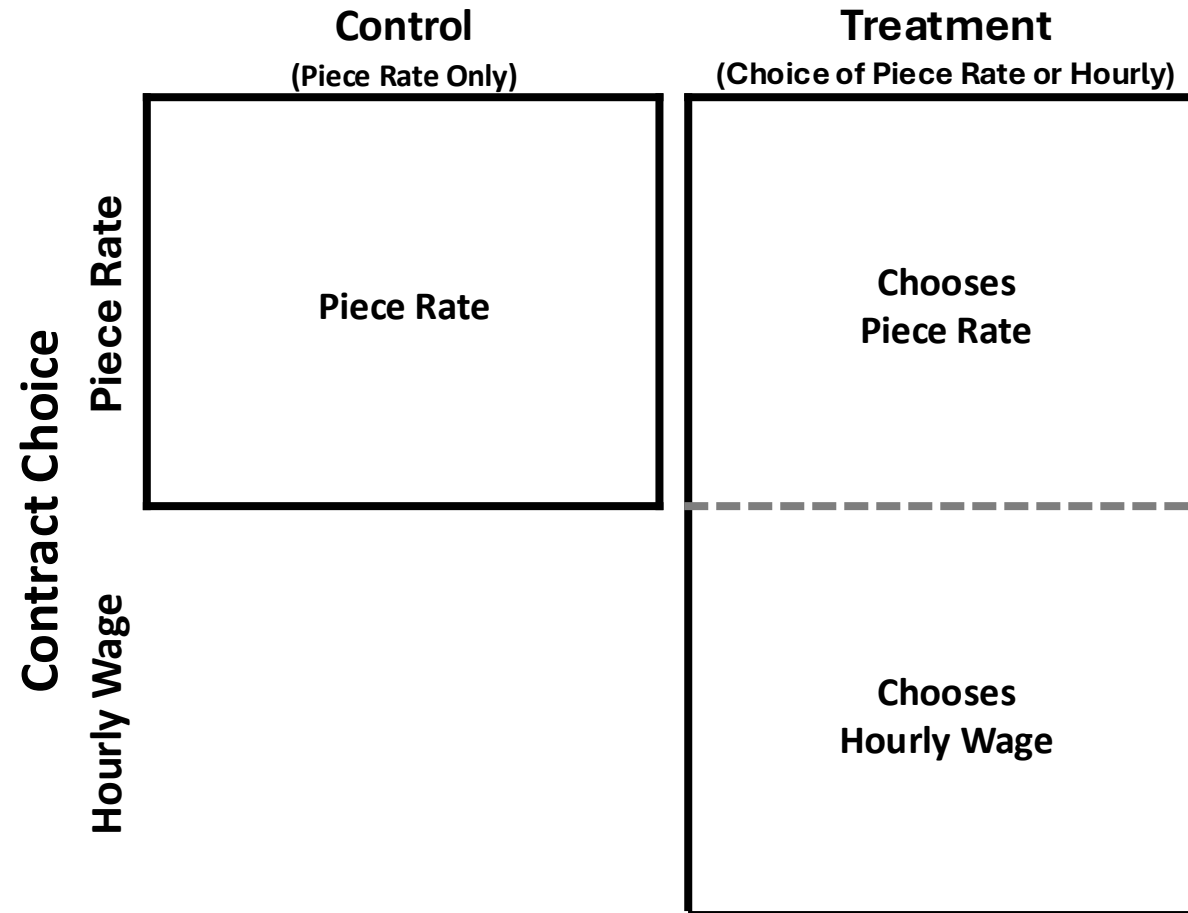


Experimental Design: Single Wage Offer

Simplified Experimental Design

Randomize workers into two offer sets:

- **Treated workers ($W_i = 1$):** offered choice
 - Remain on piece rate ($H_i = 0$)
 - Switch to an hourly wage ($H_i = 1$)
- **Control workers ($W_i = 0$):** no hourly offer
 - Remain on piece rate ($H_i = 0$)

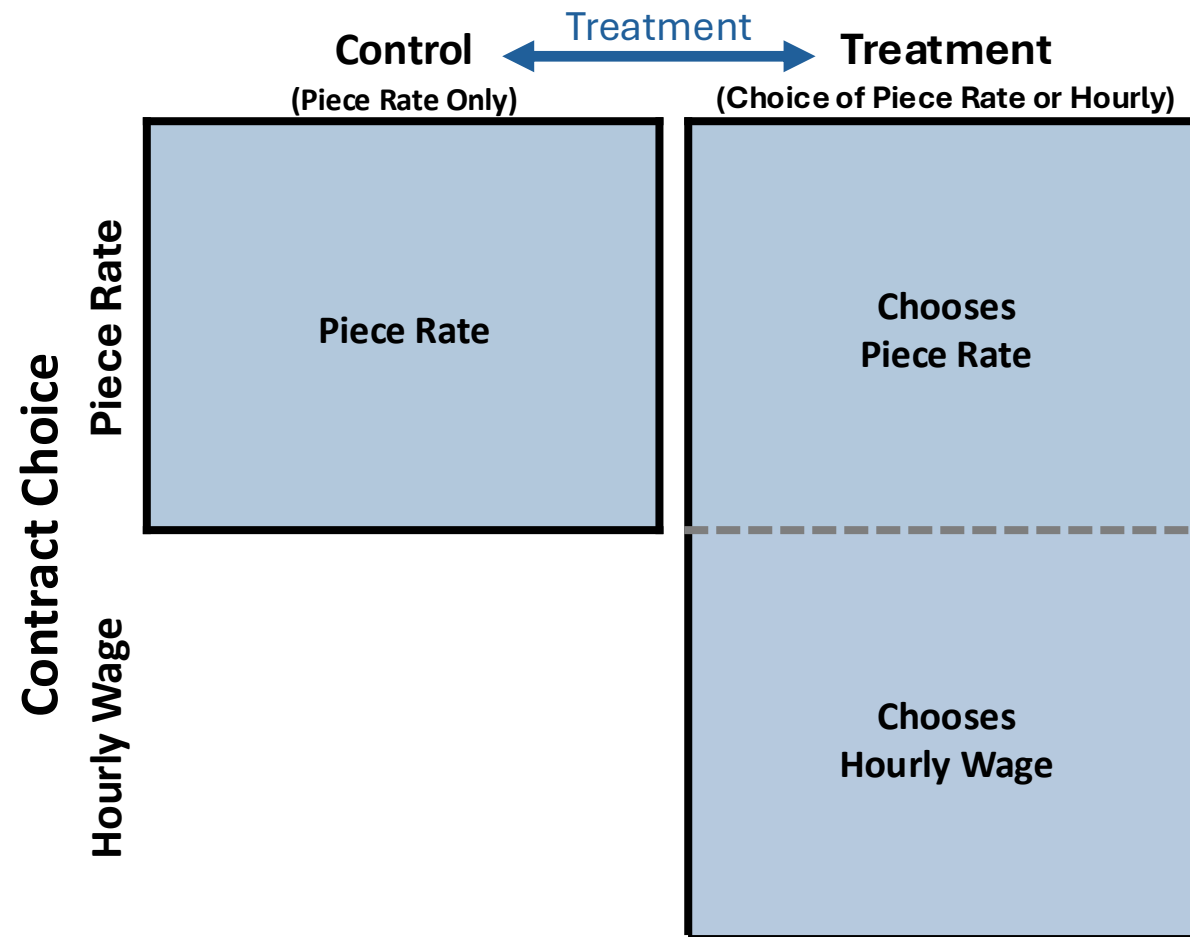


Experimental Design: Single Wage Offer

Comparing between workers facing different hourly offers identifies **treatment effect** of hourly wages:

$$\frac{E[Y_i|W_i = 1] - E[Y_i|W_i = 0]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{1i} - Y_{0i}|H_i = 1]}_{\text{Local Average Treatment Effect (LATE)}}$$

Moral Hazard

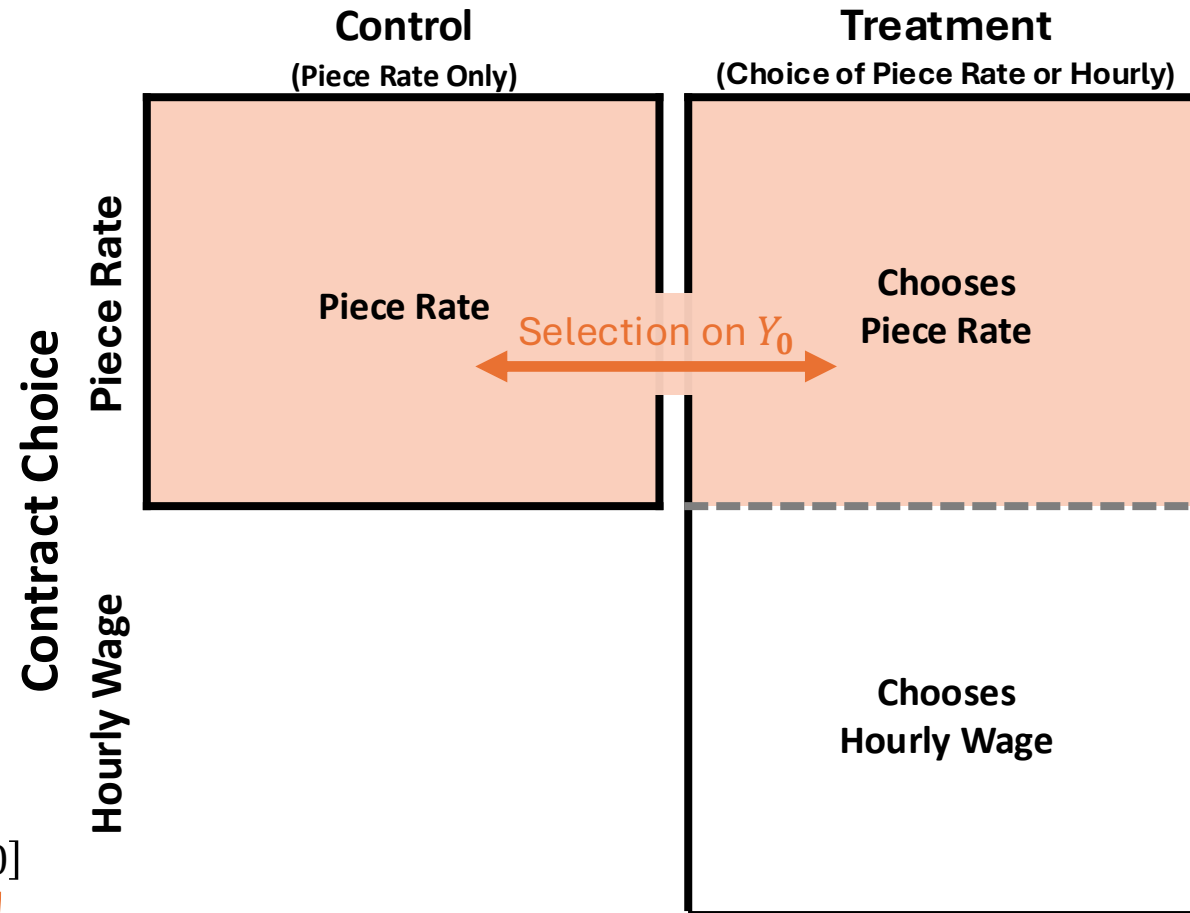


Experimental Design: Single Wage Offer

Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment***:

$$\frac{E[Y_i | W_i = 0] - E[Y_i | H_i = 0, W_i = 1]}{\Pr(H_i = 1 | W_i = 1)} = \underbrace{E[Y_{0i} | H_i = 1] - E[Y_{0i} | H_i = 0]}_{\text{Selection on } Y_0}$$

Adverse Selection



Experimental Design: Single Wage Offer

Comparing between workers facing different hourly offers identifies **treatment effect** of hourly wages:

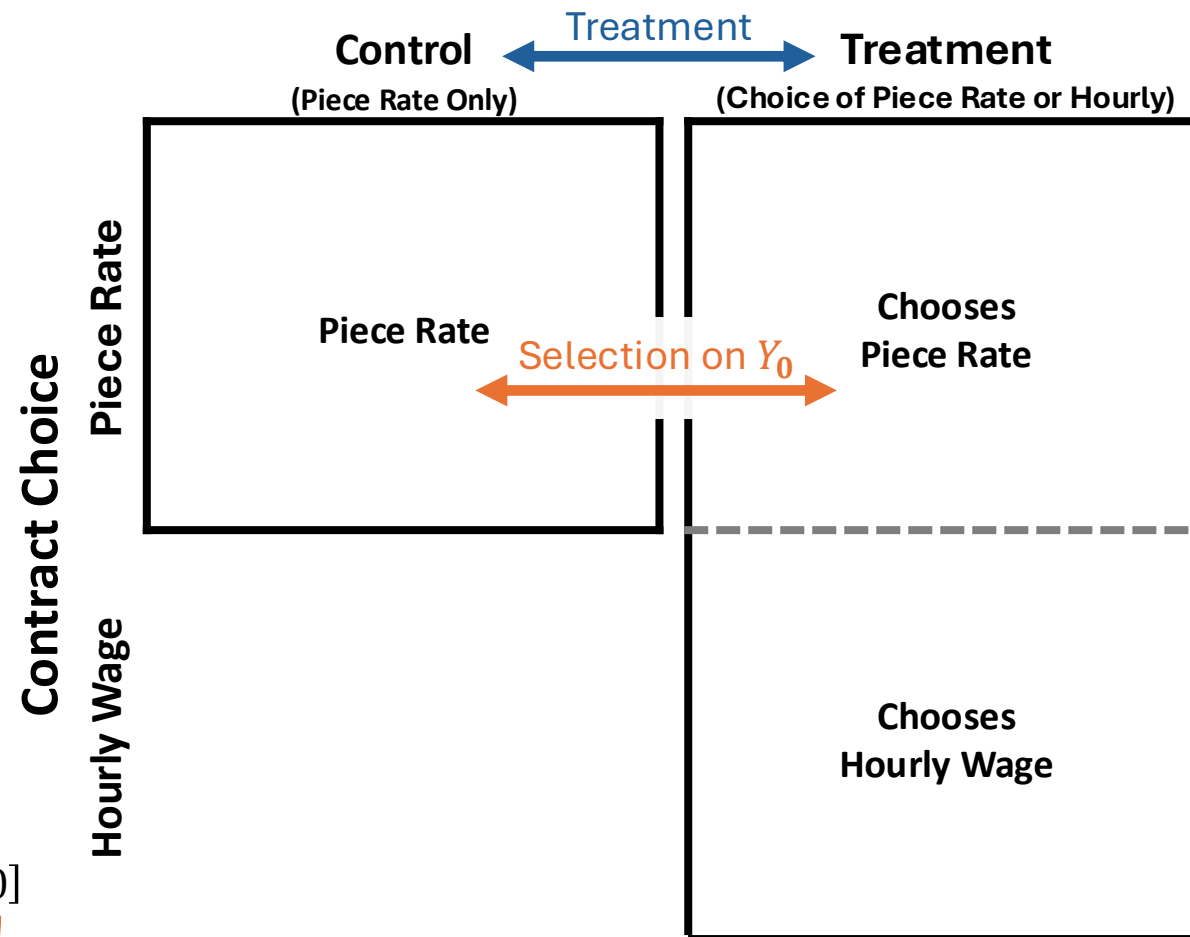
$$\frac{E[Y_i|W_i = 1] - E[Y_i|W_i = 0]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{1i} - Y_{0i}|H_i = 1]}_{\text{Local Average Treatment Effect (LATE)}}$$

Moral Hazard

Comparing piece-rate workers who faced different offer sets identifies **selection into treatment**:

$$\frac{E[Y_i|W_i = 0] - E[Y_i|H_i = 0, W_i = 1]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{0i}|H_i = 1] - E[Y_{0i}|H_i = 0]}_{\text{Selection on } Y_0}$$

Adverse Selection



Contract Choice
Piece Rate
Hourly Wage

Control

(Piece Rate Only)

Piece Rate

Treatment 1

(Choice of Piece Rate or Low Hourly)

Chooses
Piece Rate

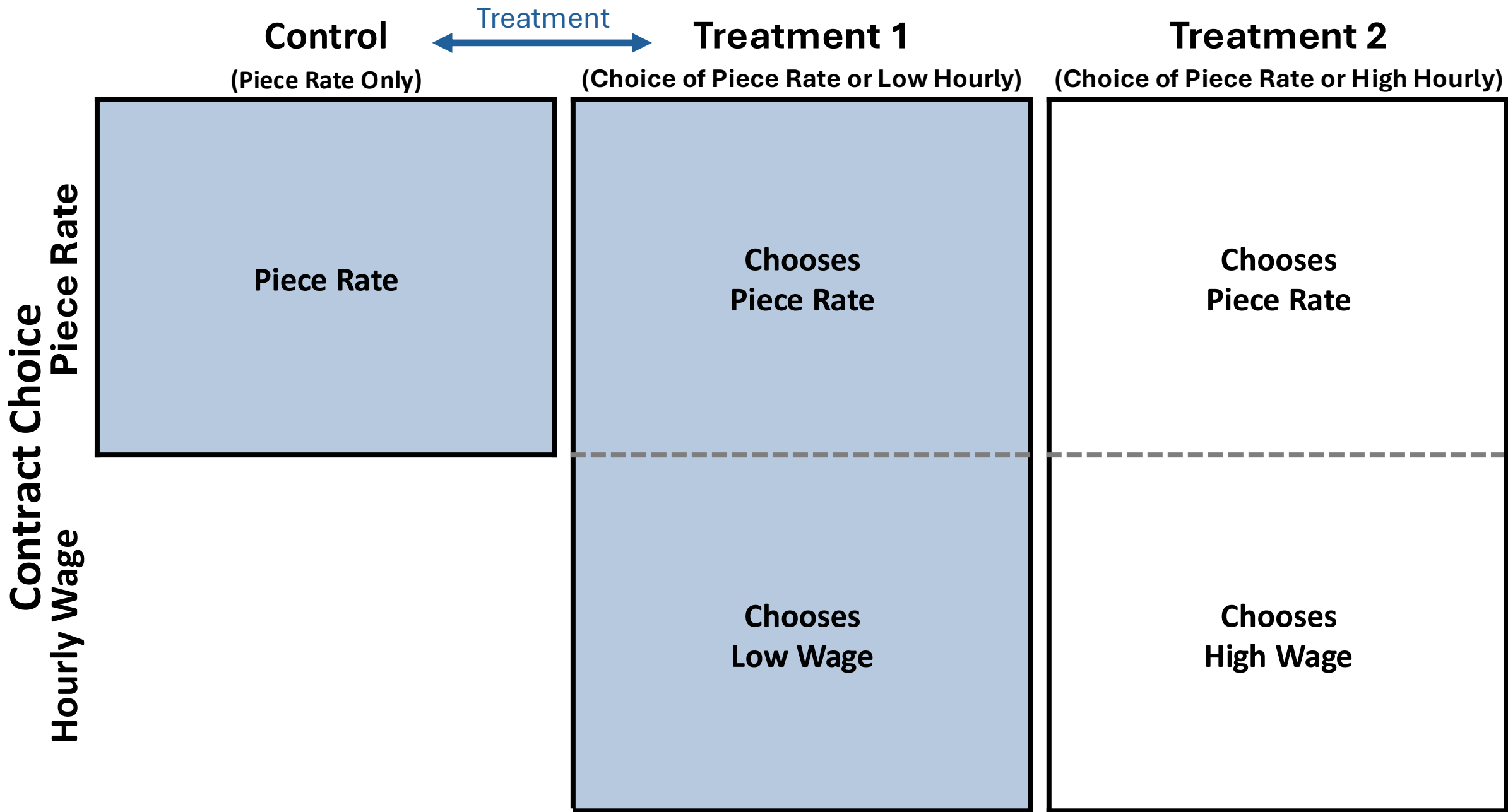
Chooses
Low Wage

Treatment 2

(Choice of Piece Rate or High Hourly)

Chooses
Piece Rate

Chooses
High Wage



Contract Choice
Piece Rate
Hourly Wage

Control

(Piece Rate Only)

Piece Rate

Treatment 1

(Choice of Piece Rate or Low Hourly)

Chooses
Piece Rate

Chooses
Low Wage

Treatment 2

(Choice of Piece Rate or High Hourly)

Chooses
Piece Rate

Chooses
High Wage

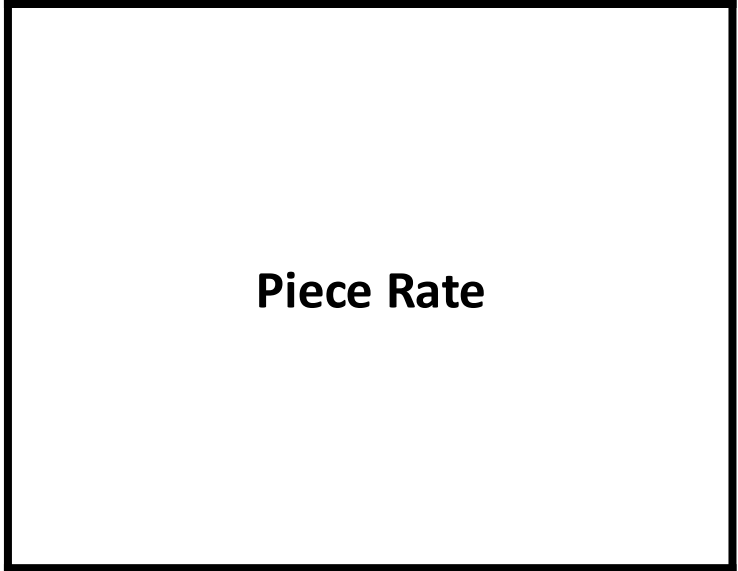
Selection on Y_0



Contract Choice
Piece Rate
Hourly Wage

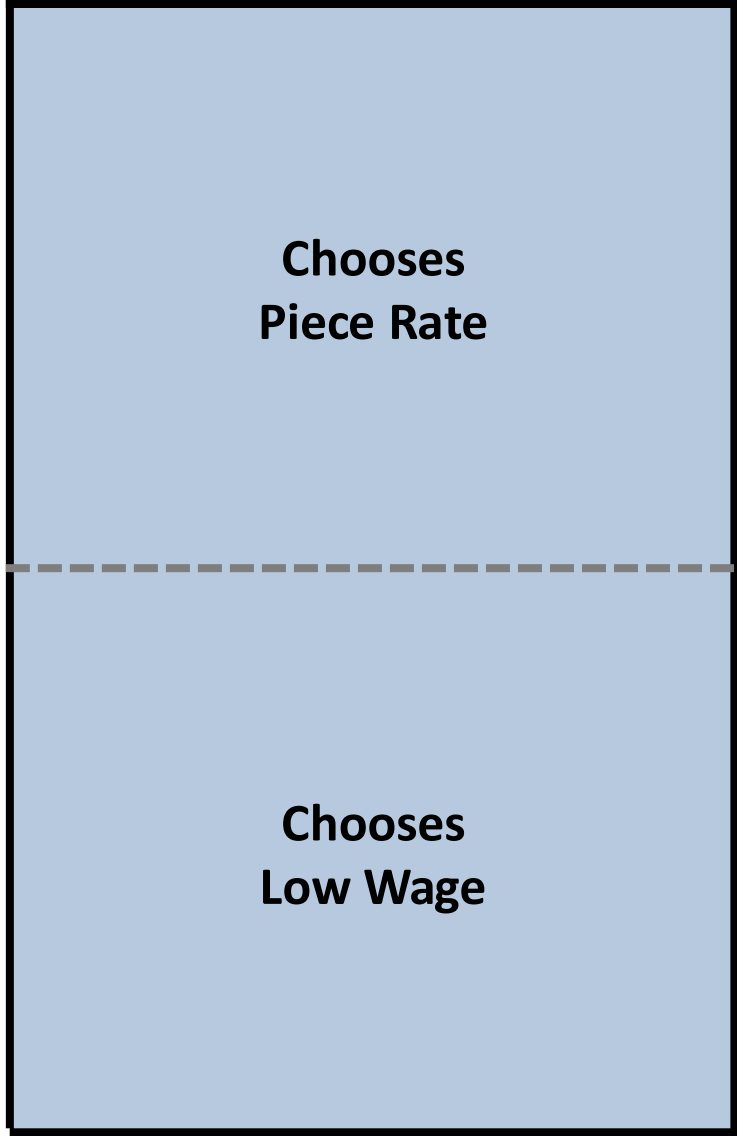
Control

(Piece Rate Only)



Treatment 1

(Choice of Piece Rate or Low Hourly)

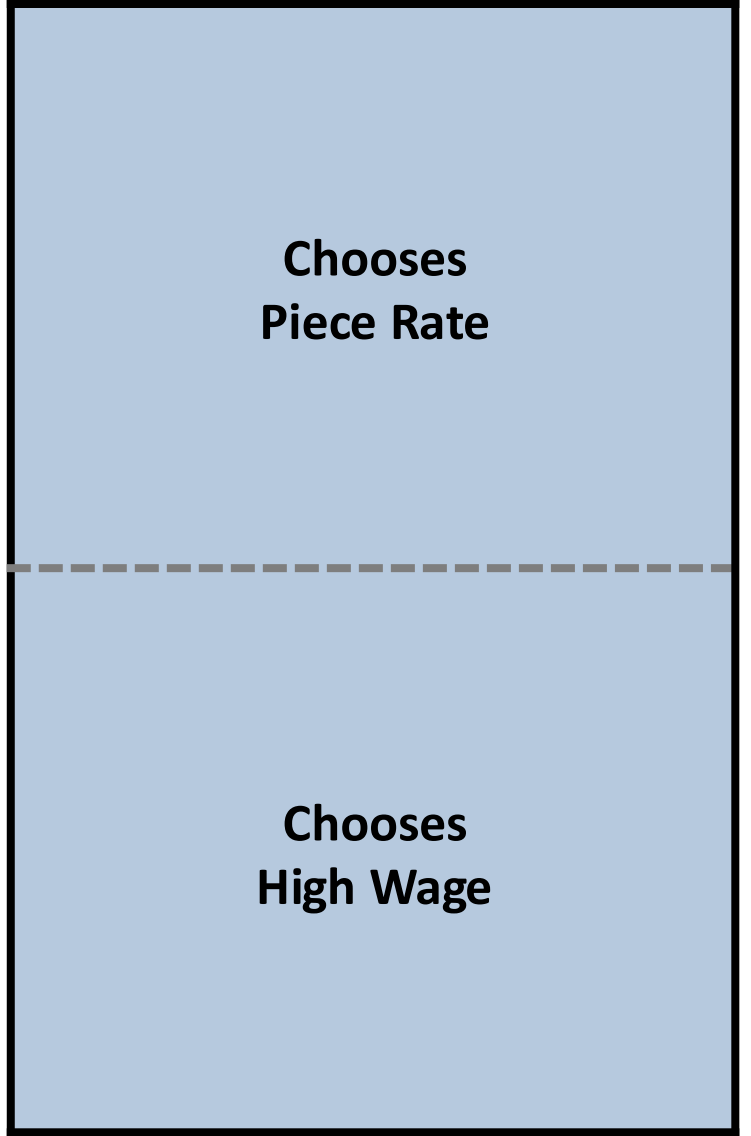


Treatment ?



Treatment 2

(Choice of Piece Rate or High Hourly)



Contract Choice
Piece Rate
Hourly Wage

Control

(Piece Rate Only)

Piece Rate

Treatment 1

(Choice of Piece Rate or Low Hourly)

Chooses
Piece Rate

Chooses
Low Wage

~~Treatment ?~~

Treatment 2

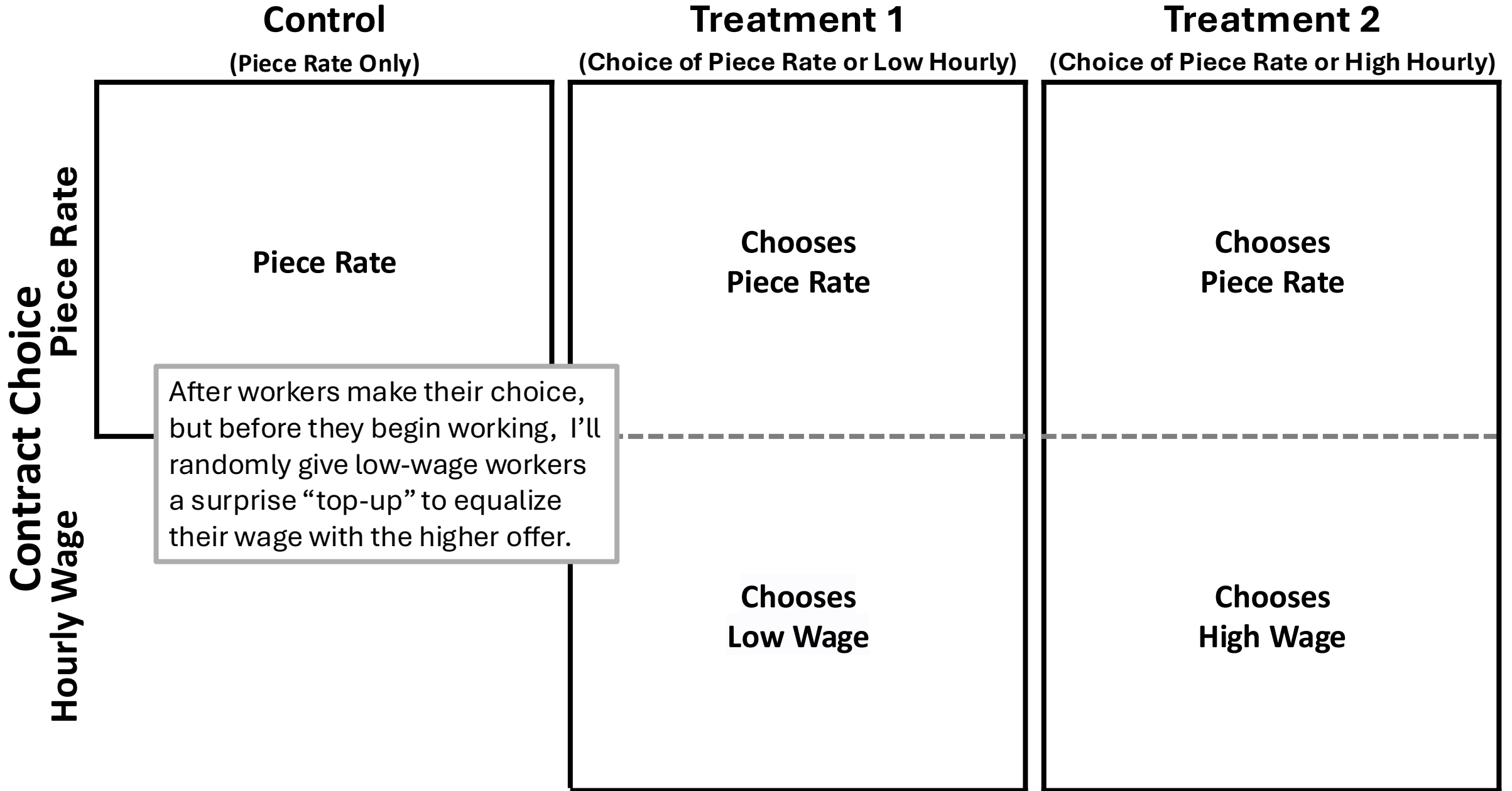
(Choice of Piece Rate or High Hourly)

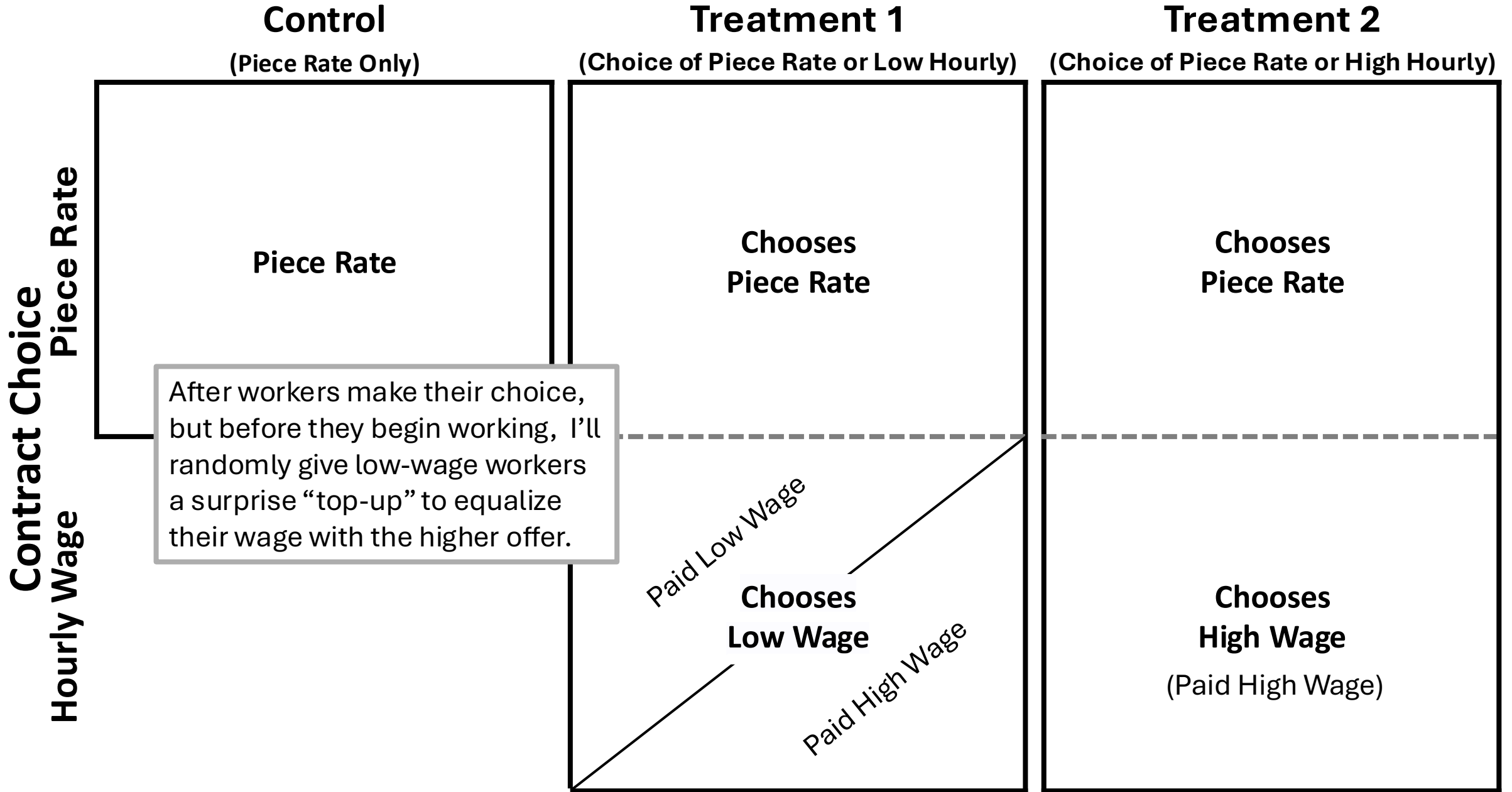
Chooses
Piece Rate

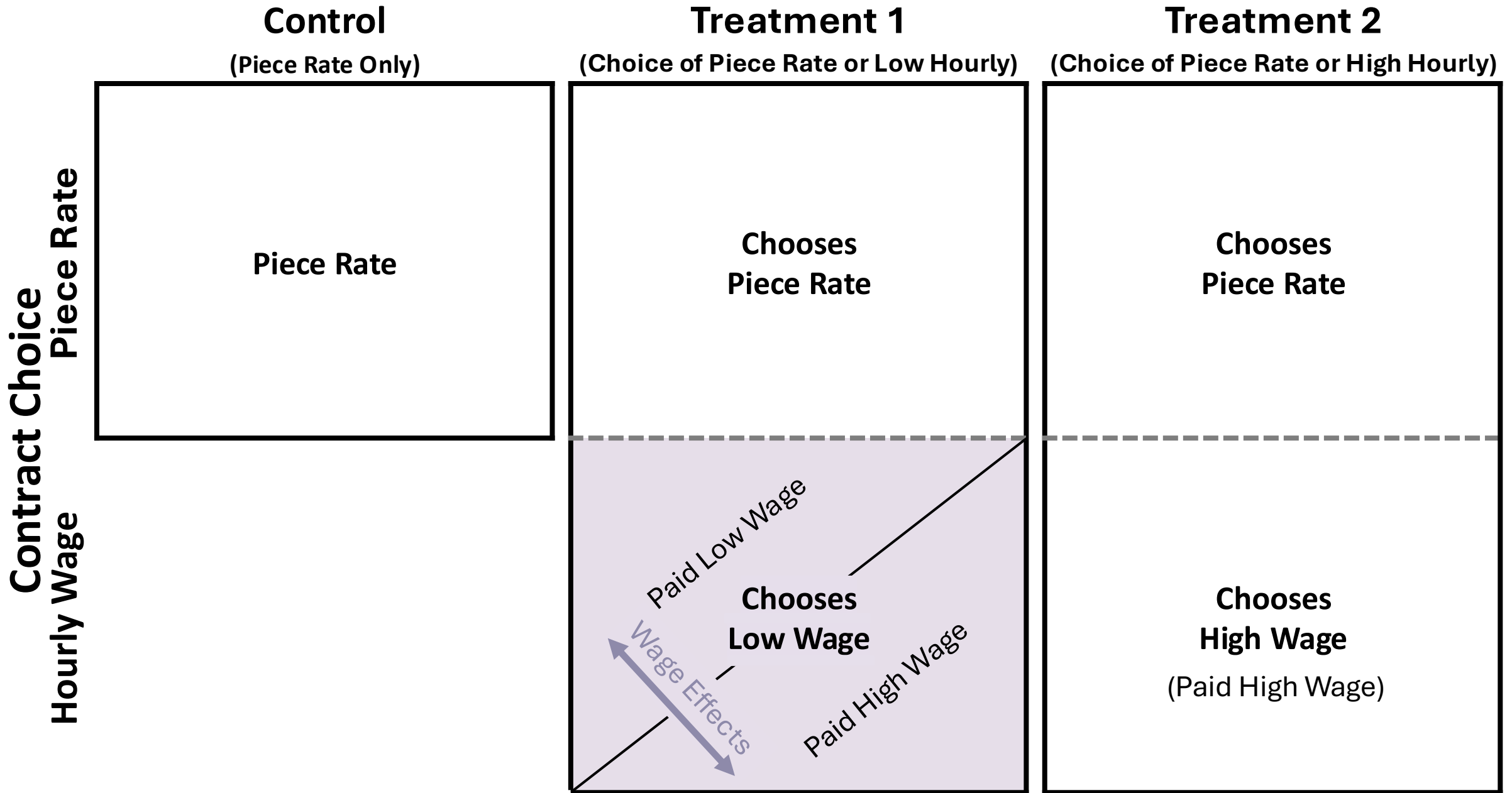
Chooses
High Wage

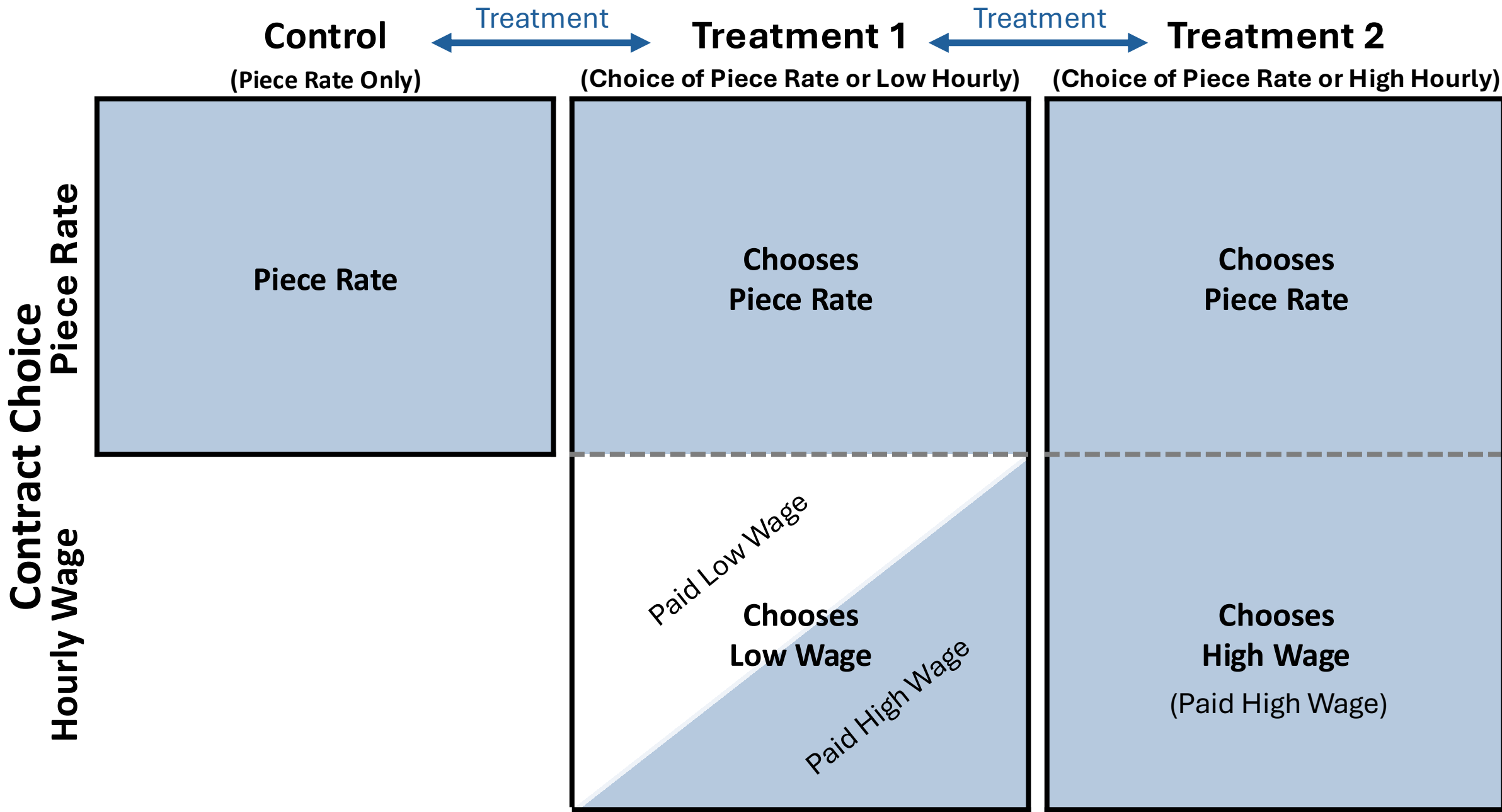
Not if $E[Y_{1i}|W_i = High] \neq E[Y_{1i}|W_i = Low]$
(i.e., Wage Effects)

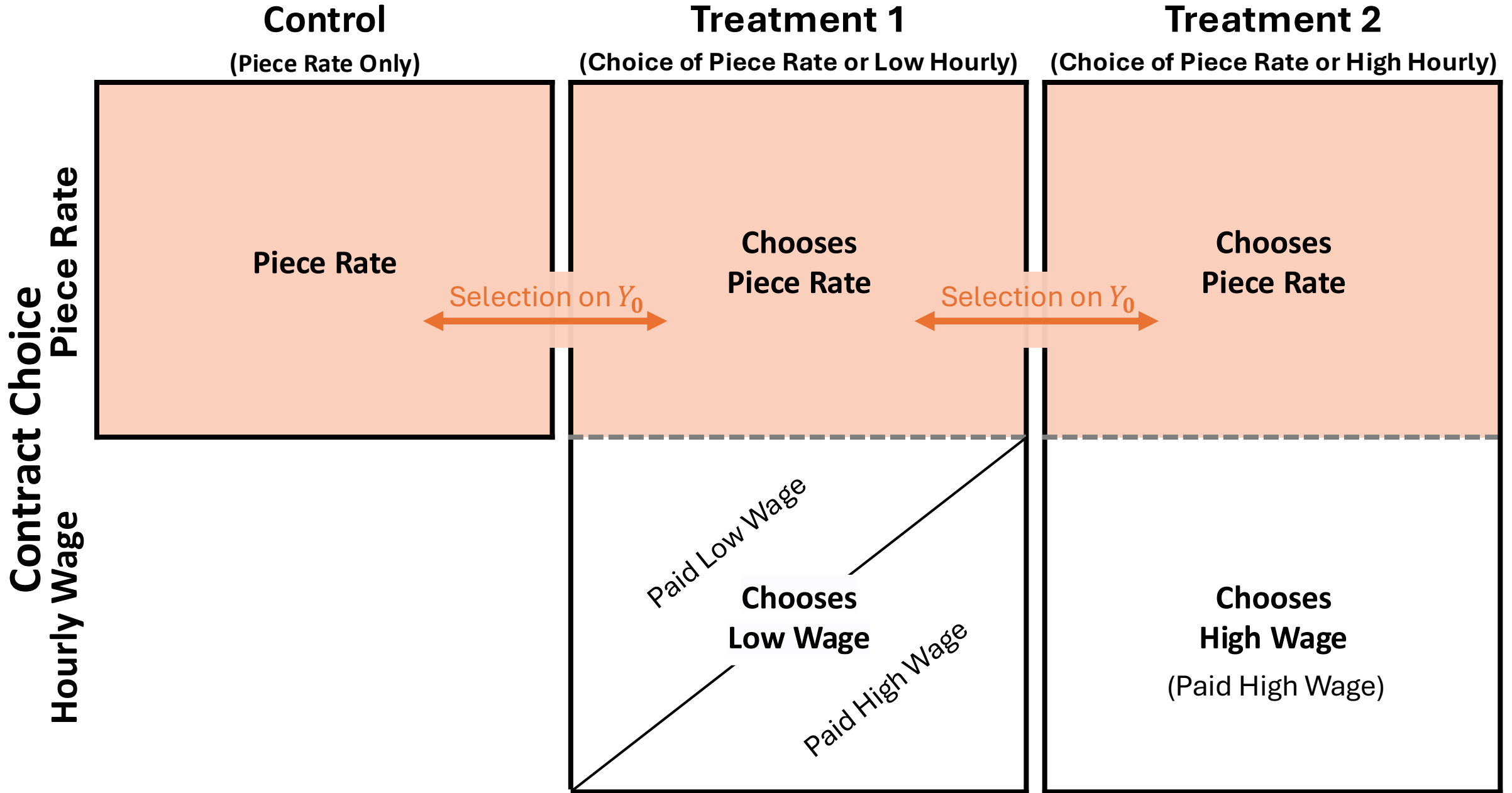
If higher wages increase
worker motivation, treatment
and selection estimates would
be biased by wage effects.

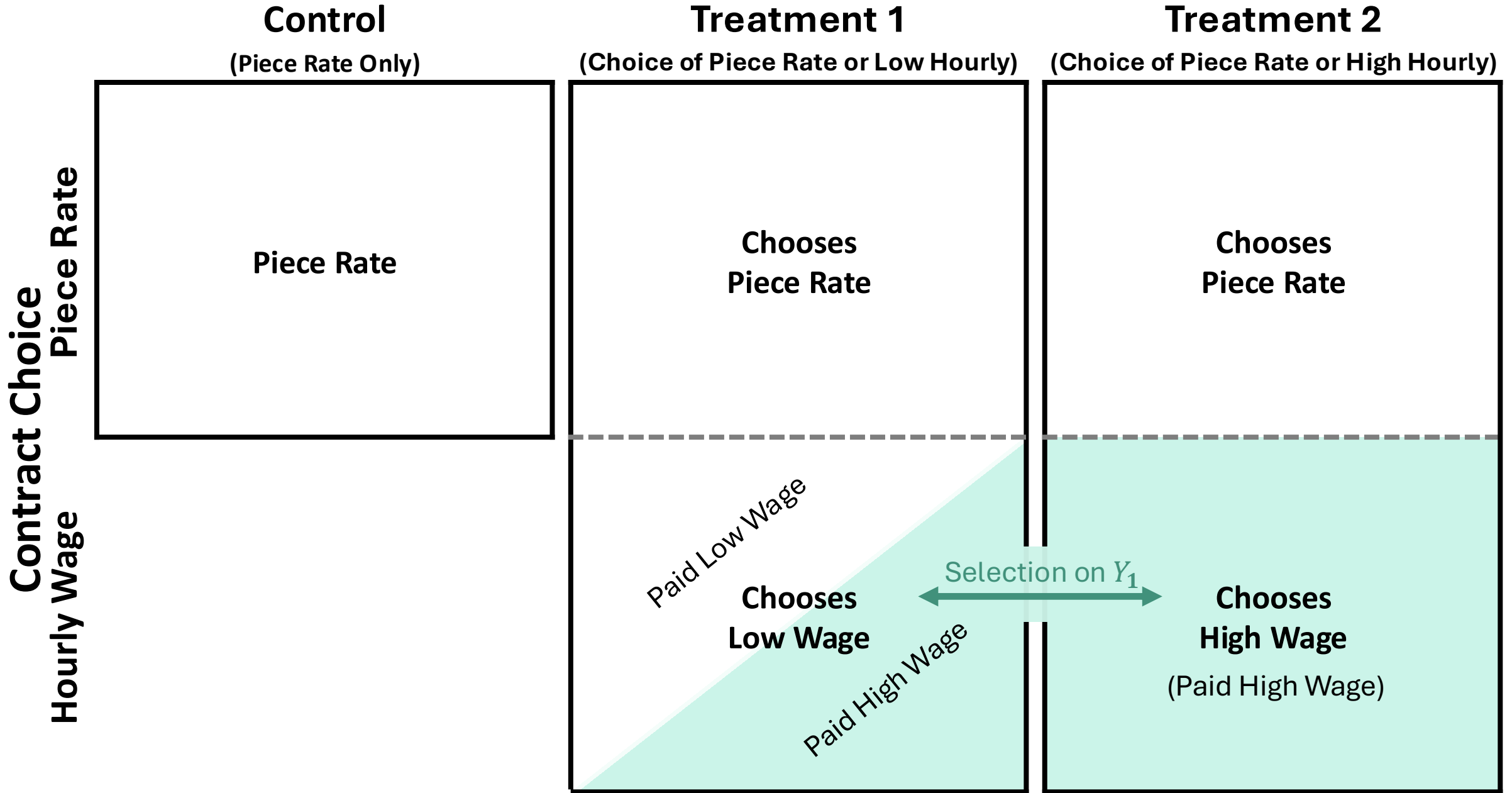


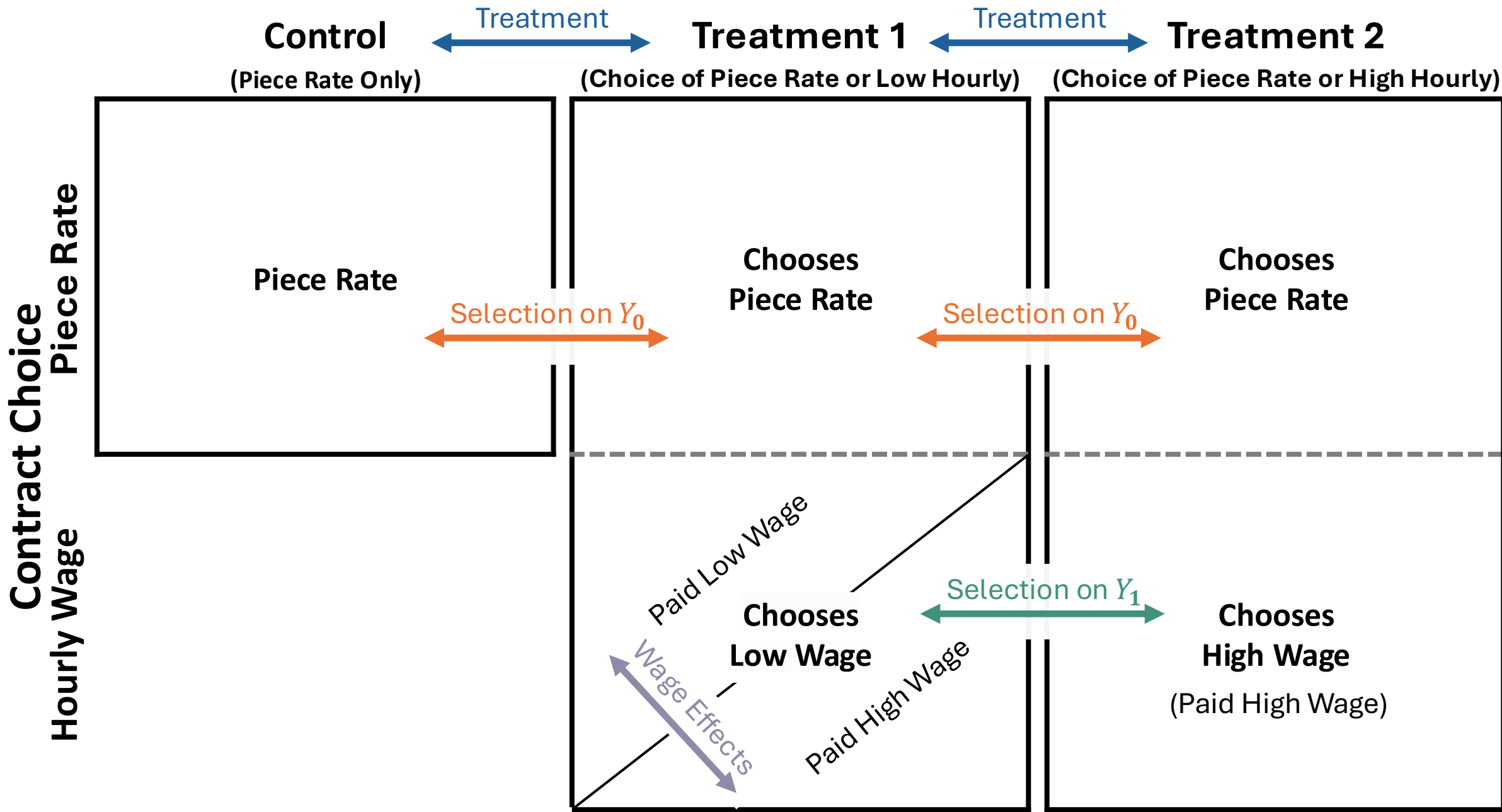












Experimental Procedure

Workers Recruited

Prolific job posting advertises a \$1.00 reward for a five-minute data-entry task, plus \$0.03 per correct entry.

- Posting screens for
 - Approval rating $\geq 98\%$
 - Number of approved tasks ≥ 10
 - Located within US
- In practice, most users who see the posting accept the job.
- Experiment takes place over ten waves of ≈ 300 workers

🇺🇸 \$1.00 • \$12.00/hr ⌚ 5 mins 👤 300 places 📝 Writing

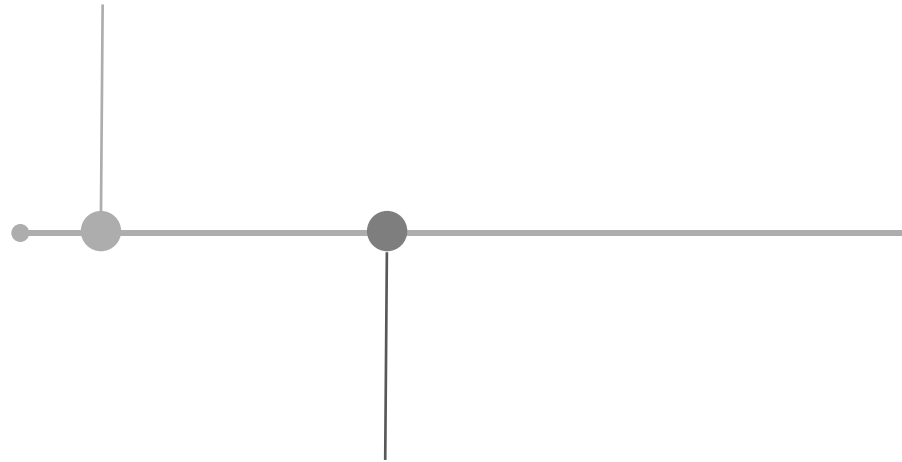
You will be shown a series of handwritten sentences for 5 minutes. Your task is to type each sentence into the corresponding text box.

*You can earn an additional \$0.03 in bonus compensation for each **correctly** typed sentence. Any bonus payments will be deposited within 24 hours of completion. You must reach the end of the 5-minute task to receive credit for your submission.*

Experimental Procedure

Workers Recruited

Prolific job posting advertises a \$1.00 reward for a five-minute data-entry task, plus \$0.03 per correct entry.



Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Hourly Wage Offer	Piece-Rate Offer	Number of Participants
No Hourly Offer	\$0.03 per sentence	302
\$1.20/hr	\$0.03 per sentence	300
\$1.80/hr	\$0.03 per sentence	101
\$2.40/hr	\$0.03 per sentence	103
\$3.00/hr	\$0.03 per sentence	304
\$3.60/hr	\$0.03 per sentence	100
\$4.20/hr	\$0.03 per sentence	99
\$4.80/hr	\$0.03 per sentence	101
\$5.40/hr	\$0.03 per sentence	101
\$6.00/hr	\$0.03 per sentence	305
\$7.20/hr	\$0.03 per sentence	100
\$8.40/hr	\$0.03 per sentence	102
\$9.60/hr	\$0.03 per sentence	101
\$10.80/hr	\$0.03 per sentence	100
\$12.00/hr	\$0.03 per sentence	305
\$15.00/hr	\$0.03 per sentence	100
\$18.00/hr	\$0.03 per sentence	102
\$21.00/hr	\$0.03 per sentence	304
<i>Total:</i> 3030		

Experimental Procedure

Workers Recruited

Prolific job posting advertises a \$1.00 reward for a five-minute data-entry task, plus \$0.03 per correct entry.

Choice of Compensation

Treated workers choose between a \$0.03 piece rate and hourly wage offers ranging from \$1.20 to \$21.00 per hour.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Before you begin the task, we'd like to offer you a choice of how to receive your bonus payment. Please select your preferred method of compensation from the options below:

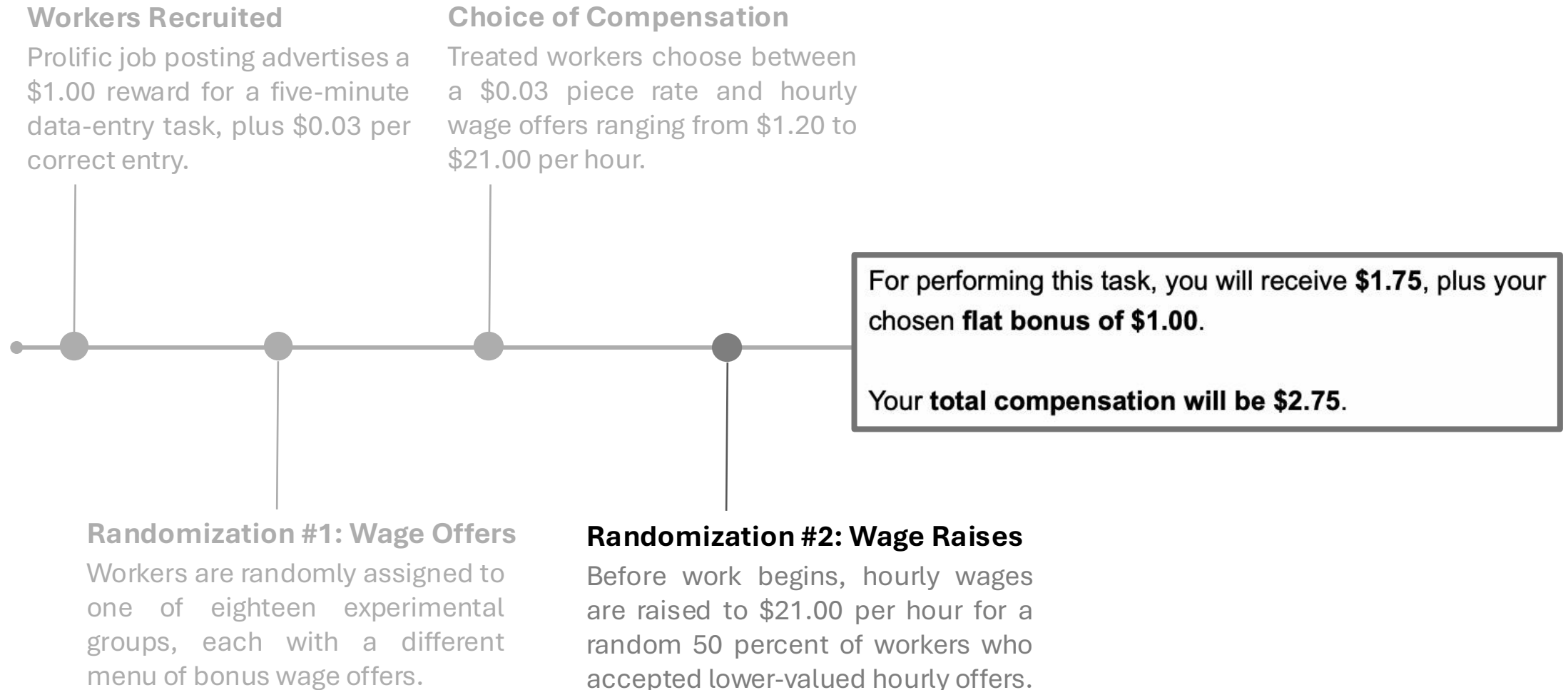
Get paid a flat bonus of \$1.00.



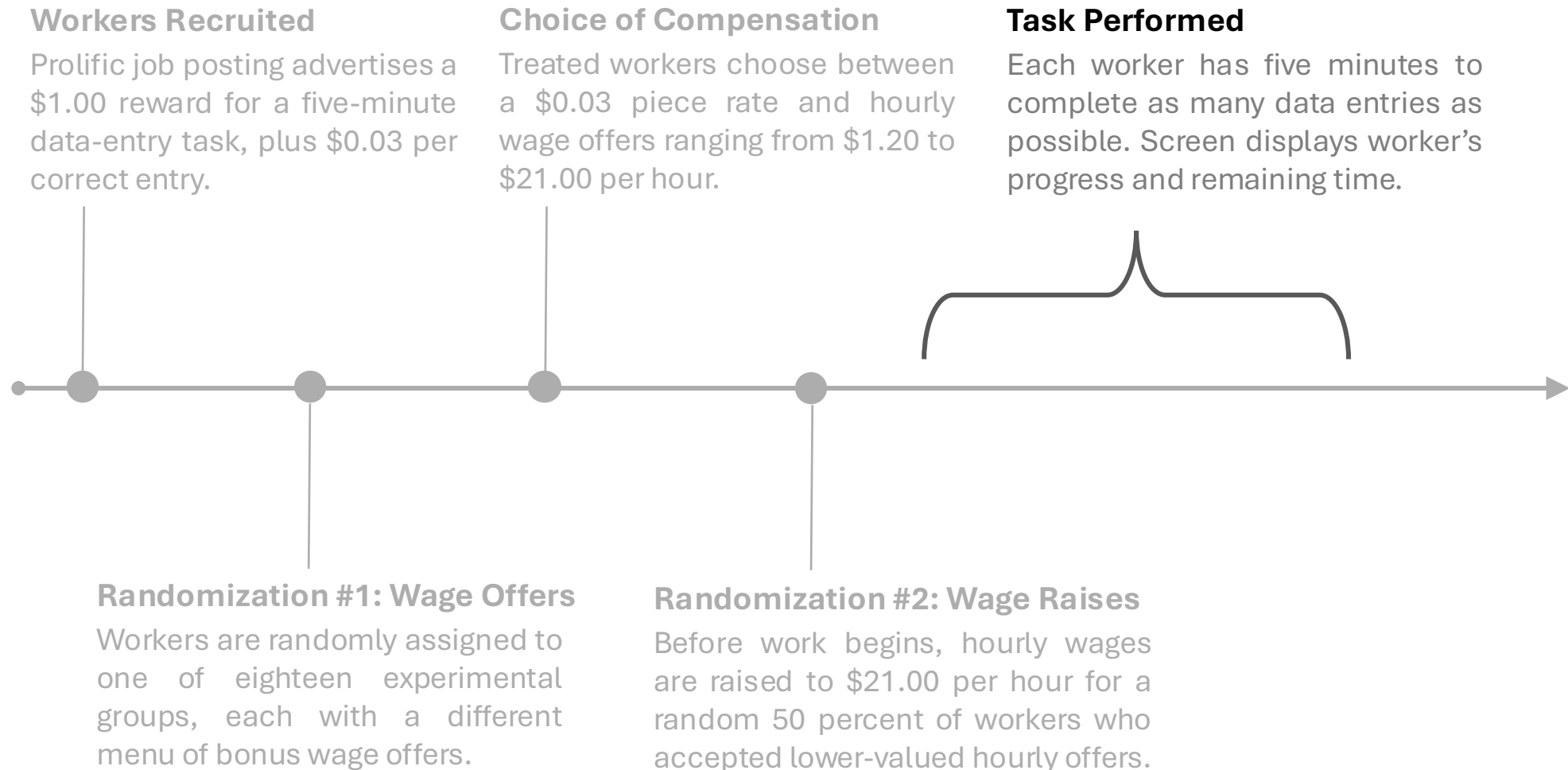
Get paid \$0.03 for each sentence you correctly complete.



Experimental Procedure



Experimental Procedure



You will be shown a series of handwritten sentences. On each page, your task is to type the sentence into the text box below.

Here is one example of a completed sentence:

The quick brown fox jumps over the lazy dog.

The quick brown fox jumps over the lazy dog.

Your answers should be as accurate as possible. Please be mindful of capitalization, spacing, and punctuation. When you've completed a sentence, click the "→" button to move on.

You will have 5 minutes to complete as many sentences as you can. You cannot start over, and you can only perform this task once.



Before you begin the task, we'd like to offer you a choice of how to earn your bonus payment. Please select your preferred bonus compensation from the options below:

Get paid a \$0.03 bonus for each sentence you correctly complete.



Get paid a flat bonus of \$1.00.



For performing this task, you will receive **\$1.00**, plus your chosen bonus of **\$0.03 per correct sentence**.

Click "Begin Task" to begin your 5-minute typing task.

Once you begin, the timer will start. Do not press the back button or refresh your browser—doing so may invalidate your results!

Begin Task

Score: 0
Earnings: \$1.00

a new leaf sprouted on the old plant.

A new leaf sprouted on the old p



Score: 8

Earnings: \$1.24

Time is up.

Thank you for participating!

You completed 31 sentences correctly.

Please click [here](#) to receive your \$1.93 payment.

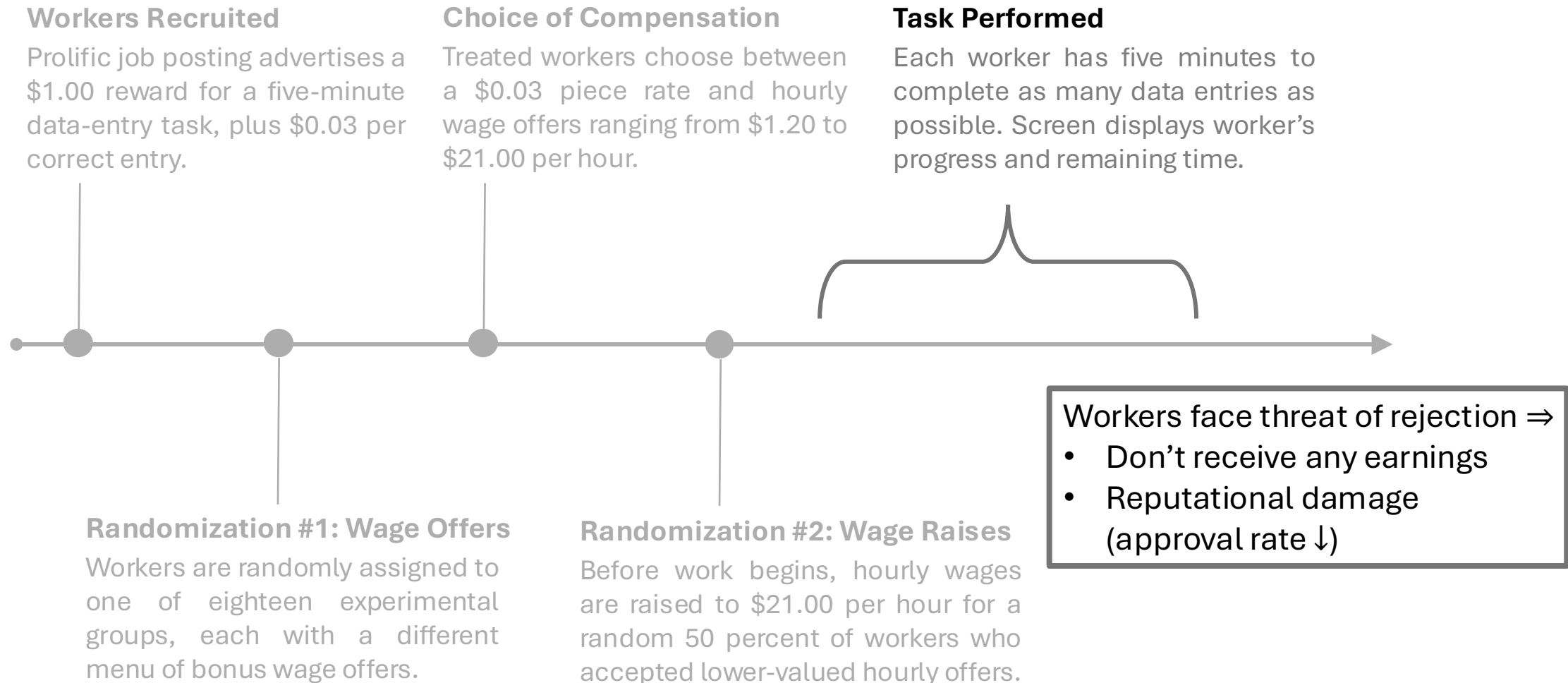
Score: 31

Earnings: \$1.93

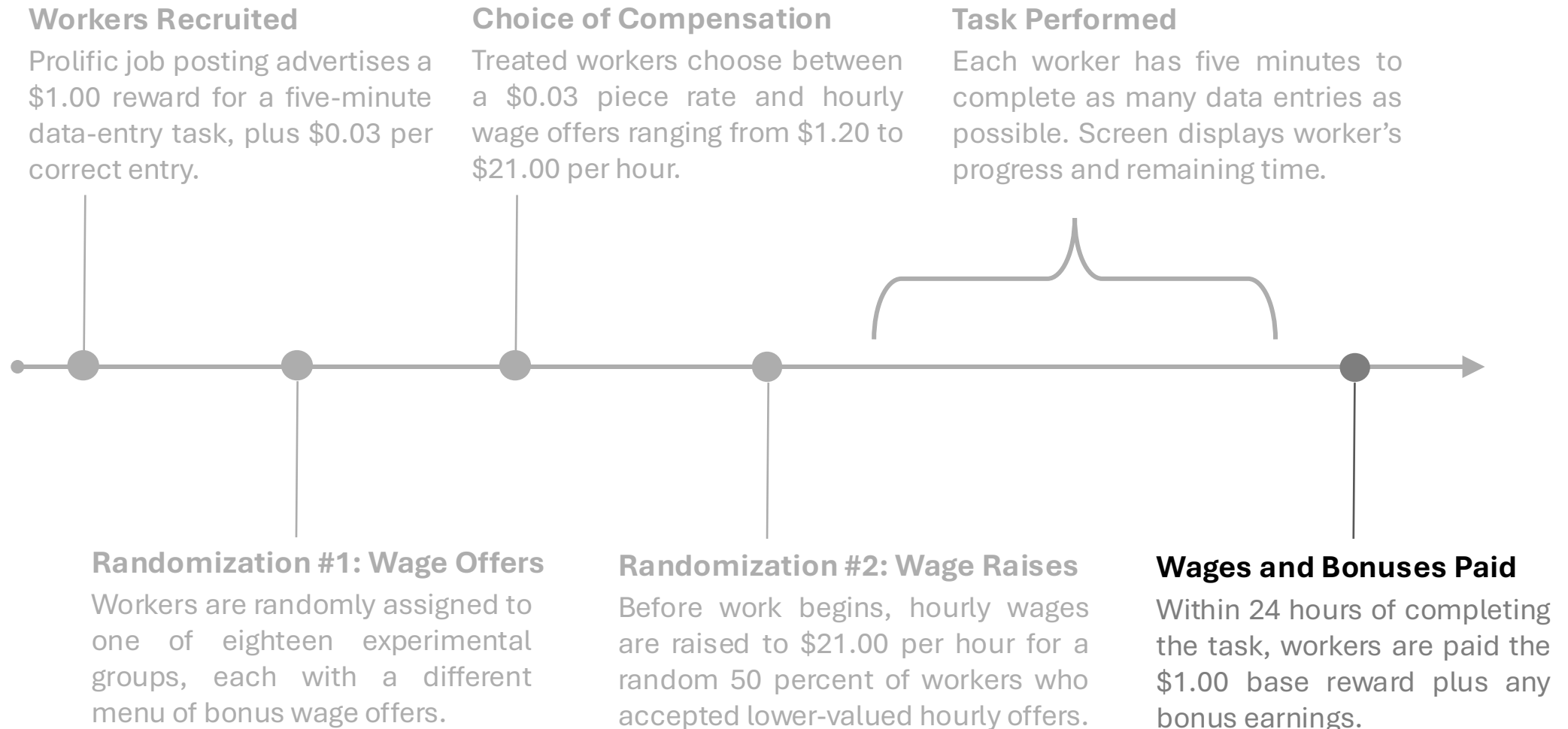


[Back](#)

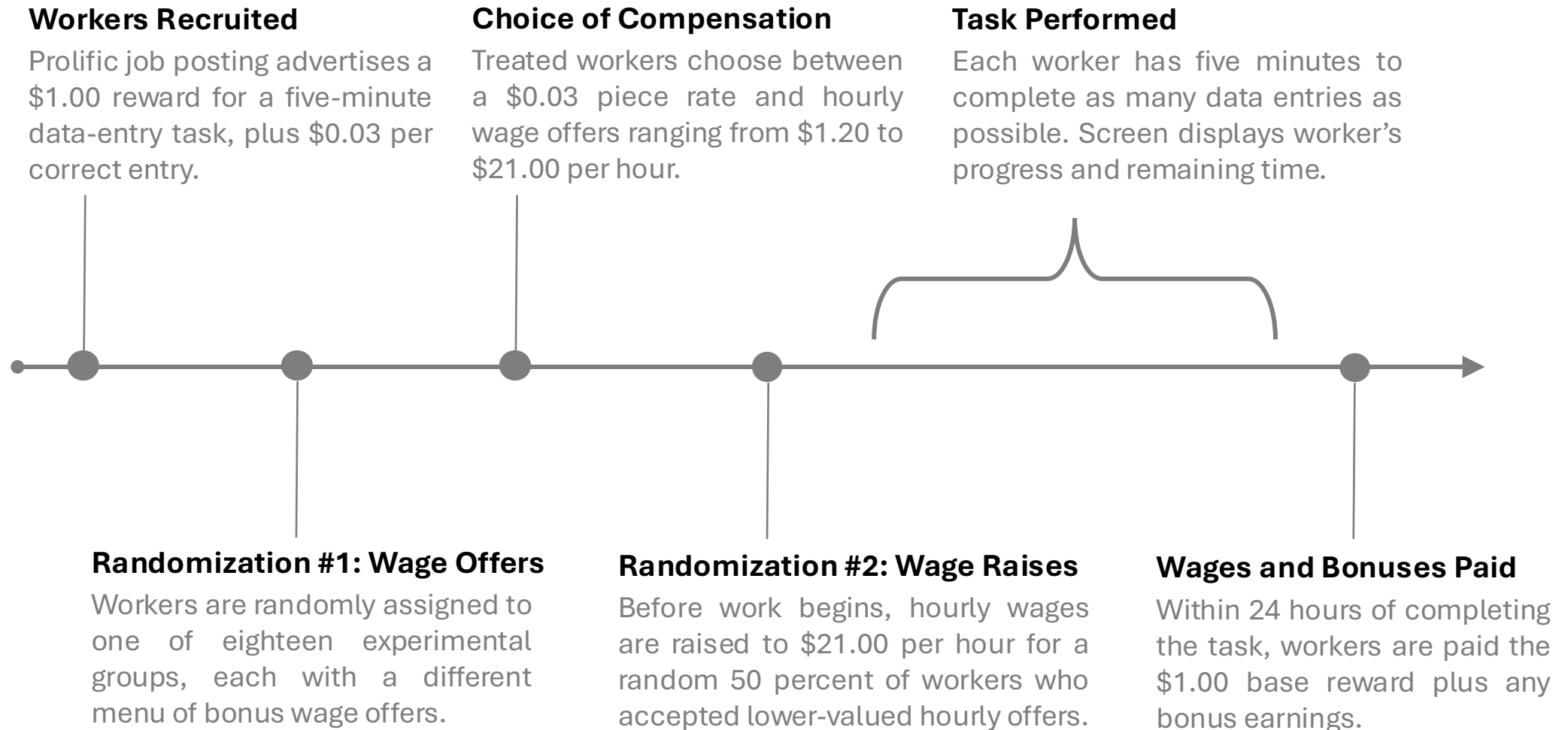
Experimental Procedure



Experimental Procedure



Experimental Procedure



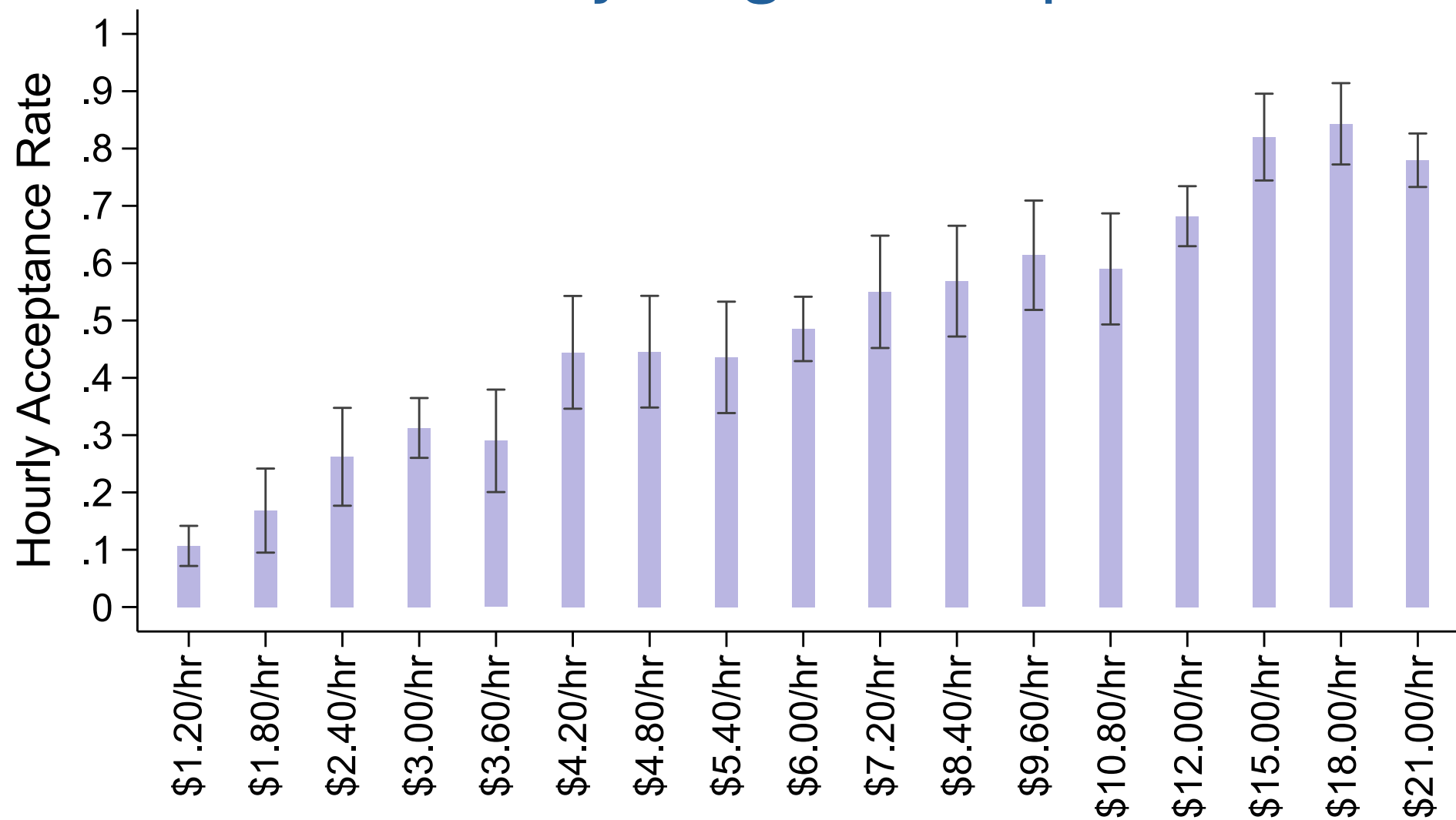
Key Advantages

- Sample is not selected into pre-existing employer or contract
 - e.g., sample of firm's existing employees would exclude freelancers
- Control of piece-rate outside option $\Rightarrow$ *market* supply curve
- Wide range of contract offers, not just those “under the lamppost”
 - i.e., options include potentially *unraveled* contracts that are not profitable
- Observe output for all workers—both *accepters* and *decliners*
 $\Rightarrow$ Can estimate selection on Y_0 and Y_1
- Experimental task is designed to be as generalizable as possible
 - 66% of US workers type on the job
 - Workers don't know it's an experiment
 - Task features tend to *attenuate* welfare effects relative to other gig markets
- Highly Replicable!

Summary Statistics

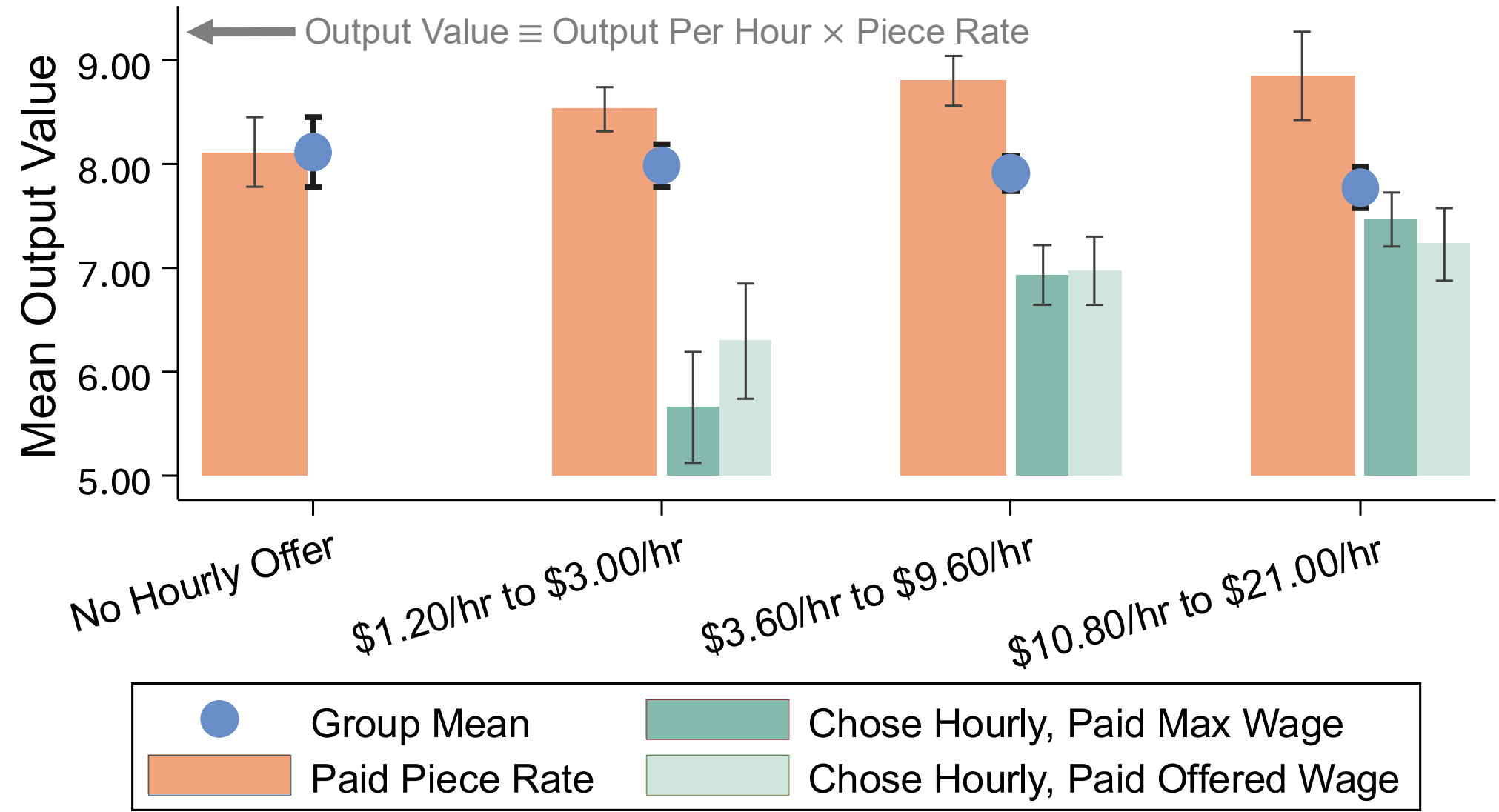
Category	Variable	Mean	SD
<i>Panel A: Task Performance</i>	Accepted Hourly Offer	0.438	0.496
	Completed Sentences	21.98	8.147
	Correct Sentences	17.79	9.360
	Output Value	7.913	2.933
	Finished	0.986	0.118
<i>Panel B: Demographics & Employment</i>	Age	37.23	12.18
	Female	0.643	0.479
	Minority	0.357	0.479
	Employed	0.685	0.465
	Student	0.187	0.390
	Number of Previous Tasks	1281.6	1746.4

Hourly Wage Take-up

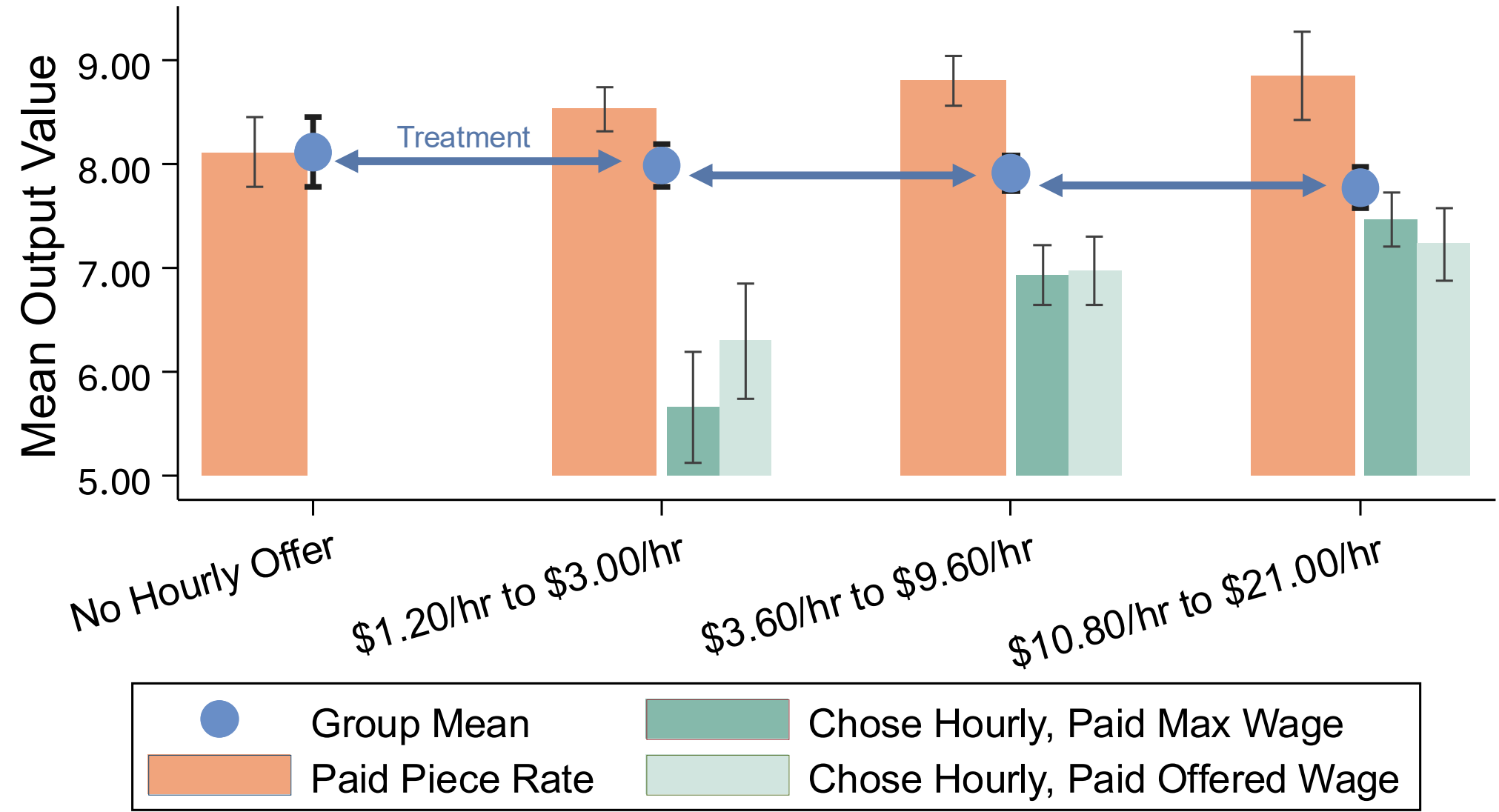


Logit Estimates

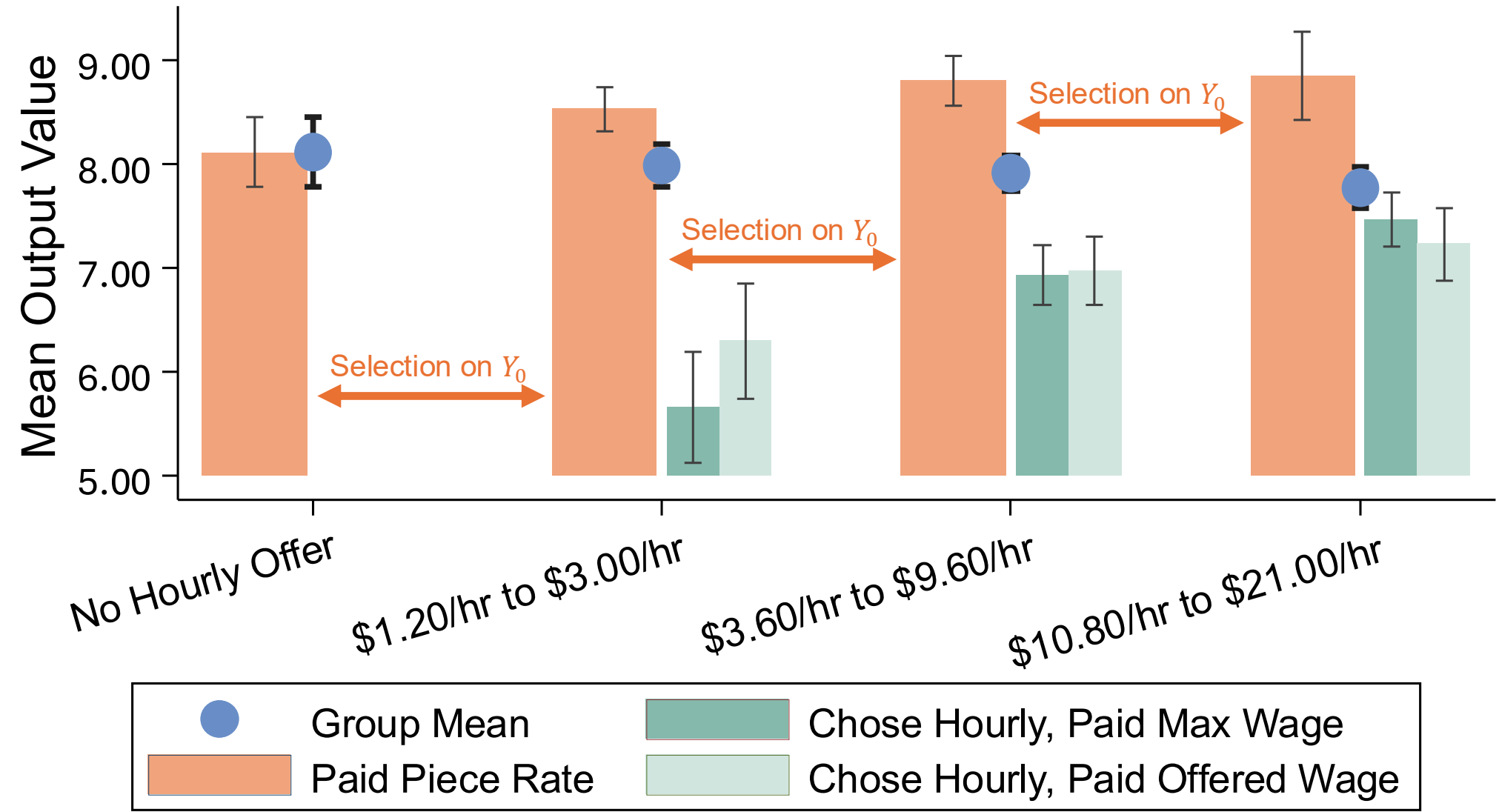
Worker Output by Offer & Contract



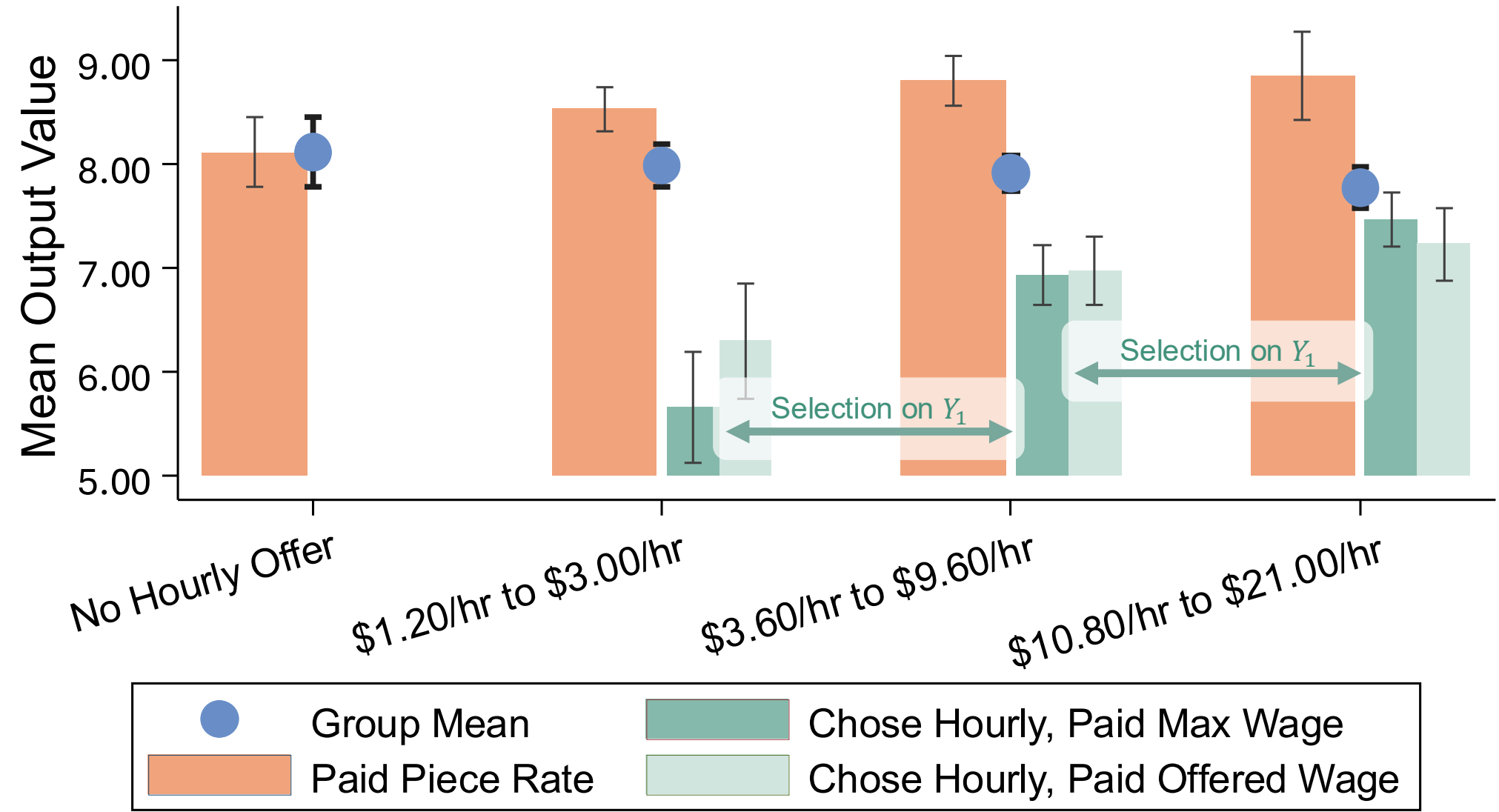
Worker Output by Offer & Contract



Worker Output by Offer & Contract



Worker Output by Offer & Contract



2SLS Estimates of Treatment Effects

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	−0.506** (0.206)	−0.500** (0.200)	−0.488** (0.200)	−0.366** (0.185)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.91	7.91	7.91	7.91
R-squared	0.037	0.080	0.096	0.232
N	3030	3030	3030	3030

$$\tilde{Y}_i = \delta H_i + \boldsymbol{\eta} \mathbf{X}_i + \epsilon_i$$

$$\tilde{Y}_i \equiv Y_i - \gamma H_i \times W_i^P$$

Selection by Log Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	−2.599*** (0.329)	−2.481*** (0.319)	−2.439*** (0.320)	−2.257*** (0.300)
Declined × Log Hourly Wage Offer	0.167* (0.0960)	0.193** (0.0932)	0.210** (0.0925)	0.229*** (0.0855)
Accepted × Log Hourly Wage Offer	0.621*** (0.116)	0.570*** (0.112)	0.568*** (0.113)	0.501*** (0.104)
Accepted × Log Effective Hourly Wage	−0.0608 (0.122)	−0.0443 (0.118)	−0.0444 (0.118)	−0.000133 (0.110)
Control Group (No Hourly Offer)	−0.341 (0.225)	−0.303 (0.219)	−0.250 (0.219)	−0.236 (0.201)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.082	0.123	0.139	0.273
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \xi X_i + \epsilon_i$$

Selection by Log Wage Offer

As wage offer increases...

output of
decliners ↑

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	−2.599*** (0.329)	−2.481*** (0.319)	−2.439*** (0.320)	−2.257*** (0.300)
Declined × Log Hourly Wage Offer	0.167* (0.0960)	0.193** (0.0932)	0.210** (0.0925)	0.229*** (0.0855)
Accepted × Log Hourly Wage Offer	0.621*** (0.116)	0.570*** (0.112)	0.568*** (0.113)	0.501*** (0.104)
Accepted × Log Effective Hourly Wage	−0.0608 (0.122)	−0.0443 (0.118)	−0.0444 (0.118)	−0.000133 (0.110)
Control Group (No Hourly Offer)	−0.341 (0.225)	−0.303 (0.219)	−0.250 (0.219)	−0.236 (0.201)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.082	0.123	0.139	0.273
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \xi X_i + \epsilon_i$$

Selection by Log Wage Offer

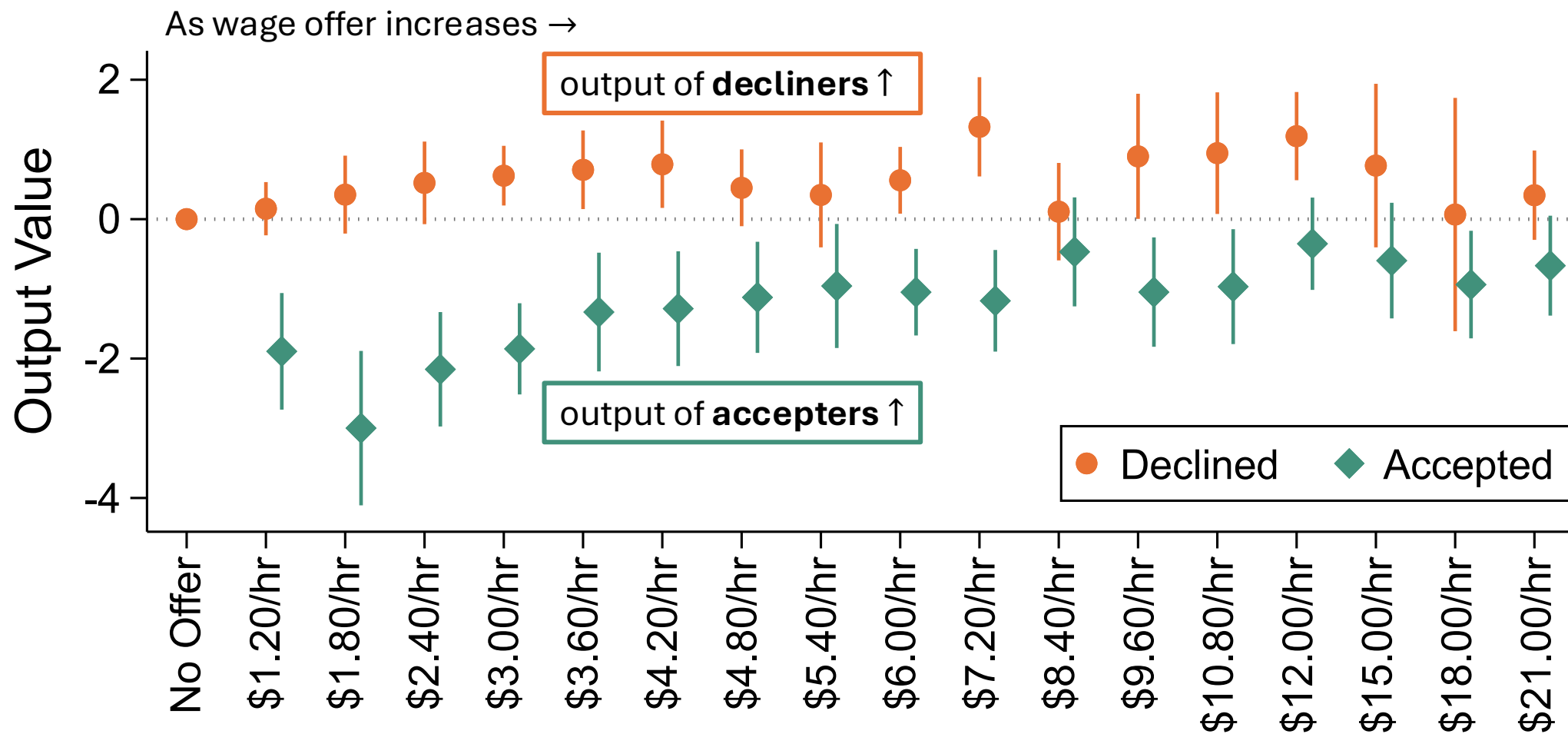
As wage offer increases...

output of
accepters ↑

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	−2.599*** (0.329)	−2.481*** (0.319)	−2.439*** (0.320)	−2.257*** (0.300)
Declined × Log Hourly Wage Offer	0.167* (0.0960)	0.193** (0.0932)	0.210** (0.0925)	0.229*** (0.0855)
Accepted × Log Hourly Wage Offer	0.621*** (0.116)	0.570*** (0.112)	0.568*** (0.113)	0.501*** (0.104)
Accepted × Log Effective Hourly Wage	−0.0608 (0.122)	−0.0443 (0.118)	−0.0444 (0.118)	−0.000133 (0.110)
Control Group (No Hourly Offer)	−0.341 (0.225)	−0.303 (0.219)	−0.250 (0.219)	−0.236 (0.201)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.082	0.123	0.139	0.273
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \xi X_i + \epsilon_i$$

Selection by Log Wage Offer



$$Y_i = \sum_{w \in W} [(\beta_0^w (1 - H_i) + \beta_1^w H_i) \times \mathbf{1}\{W_i = w\}] + \gamma H_i \times W_i^P + \xi X_i + \epsilon_i$$

Summary so far...

Let $\bar{w}_i \equiv$ **hourly reservation wage** (the lowest hourly wage i would accept)

Estimates of average treatment and selection into fixed wage, $w \Rightarrow$

Evidence of Moral Hazard

Hourly wages reduce average output among workers who accept.

$$E[Y_{1i} - Y_{0i} | \bar{w}_i \leq w] < 0$$

Treatment Effect

Evidence of Adverse Selection

Low-productivity workers are more likely to accept low hourly wage offers.

$$\frac{\partial}{\partial w} E[Y_{0i} | \bar{w}_i > w] > 0$$

Selection on Y_0

$$\frac{\partial}{\partial w} E[Y_{1i} | \bar{w}_i \leq w] > 0$$

Selection on Y_1

What are the implications for... Market equilibrium? Social welfare? Public policy?

Model shows how answers depend on the marginal worker ($\bar{w}_i = w$)

$$E[Y_{1i} - Y_{0i} | \bar{w}_i = w]$$

$$E[Y_{0i} | \bar{w}_i = w]$$

$$E[Y_{1i} | \bar{w}_i = w]$$

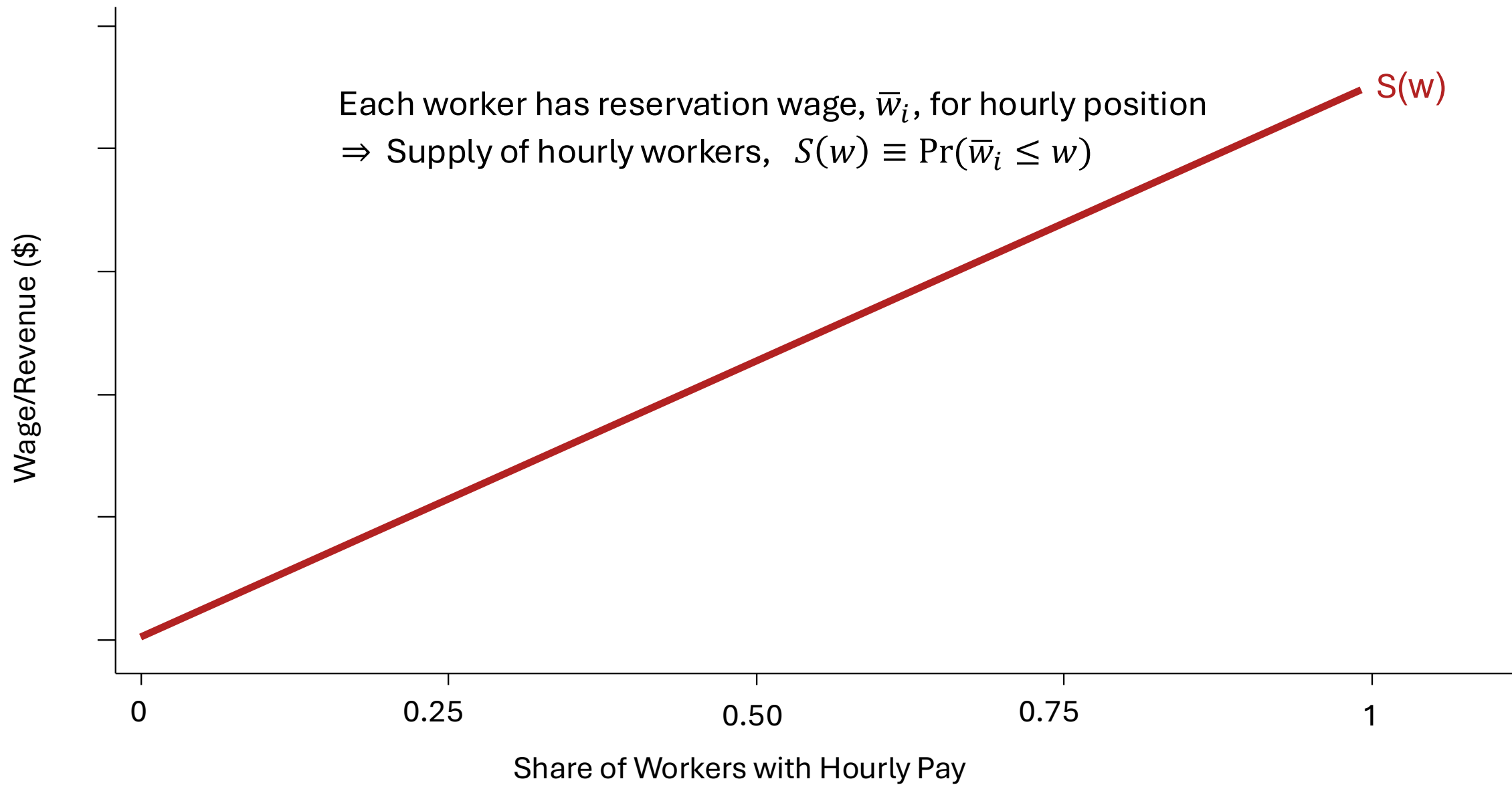
Identified with Marginal Treatment Effects (MTE)

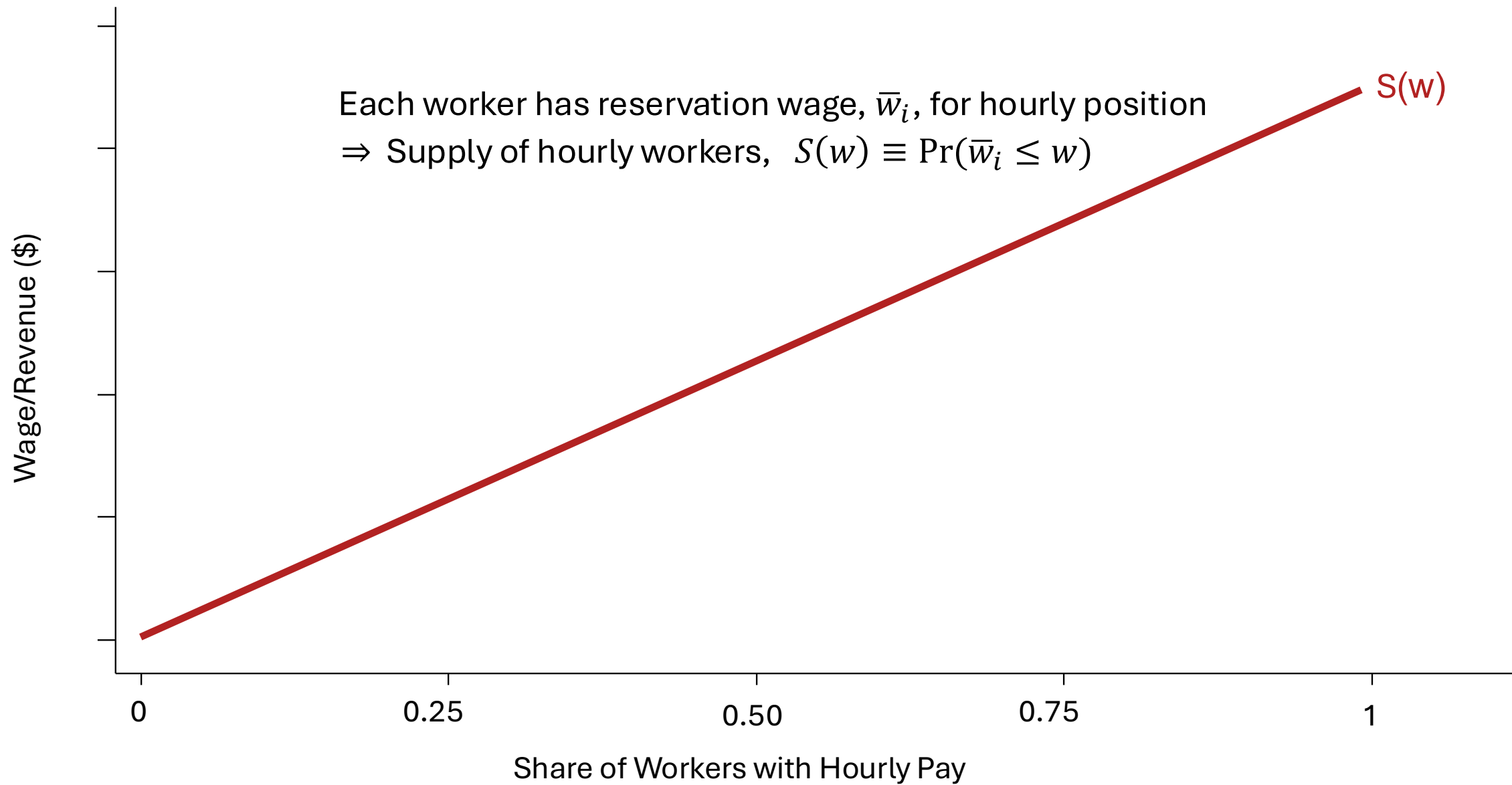
Outline

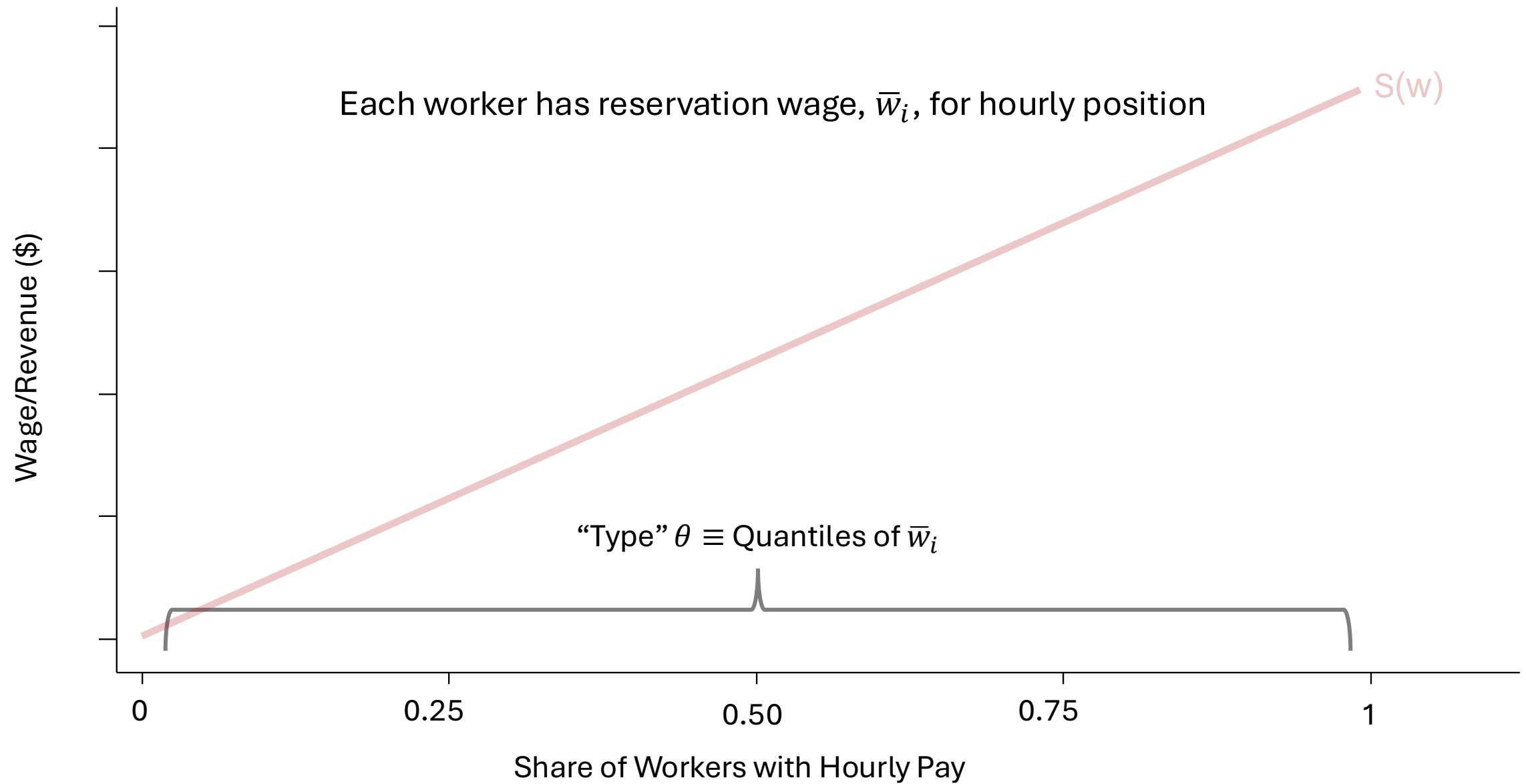
- 1 Experimental Design and Results
- 2 Model of Asymmetric Information
- 3 Estimates of Marginal Value and Welfare Loss
- 4 Optimal Fixed-Wage Subsidy and Piece-Rate Tax
- 5 Extensions

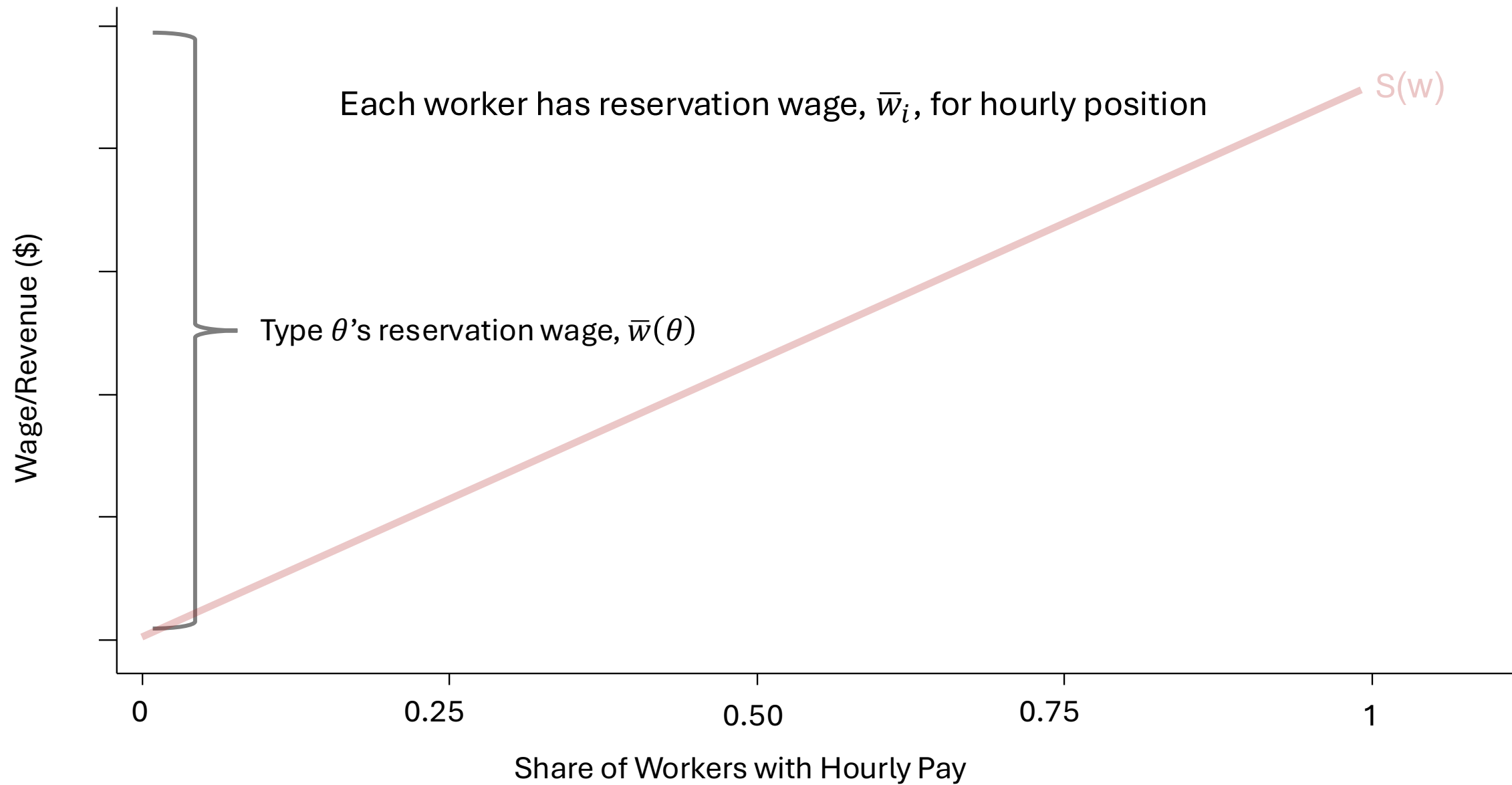
Model

- Population of observationally equivalent (pre-screened) workers
- Each worker produces hourly output $q_i = f(\zeta_i, e_i, v_i)$
 - ζ_i : (unobserved) worker characteristics
 - e_i : individual effort, potentially influenced by the contract (moral hazard)
 - v_i : random noise
- Value of worker i 's output is $Y_i \equiv pq_i$, where $p \equiv$ price per unit q
- Firms have two options:
 1. Buy output at per-unit market price p (e.g., piece rate, freelance hire)
 2. Offer worker *ex-ante* hourly wage, w , in exchange for claim on realized q_i

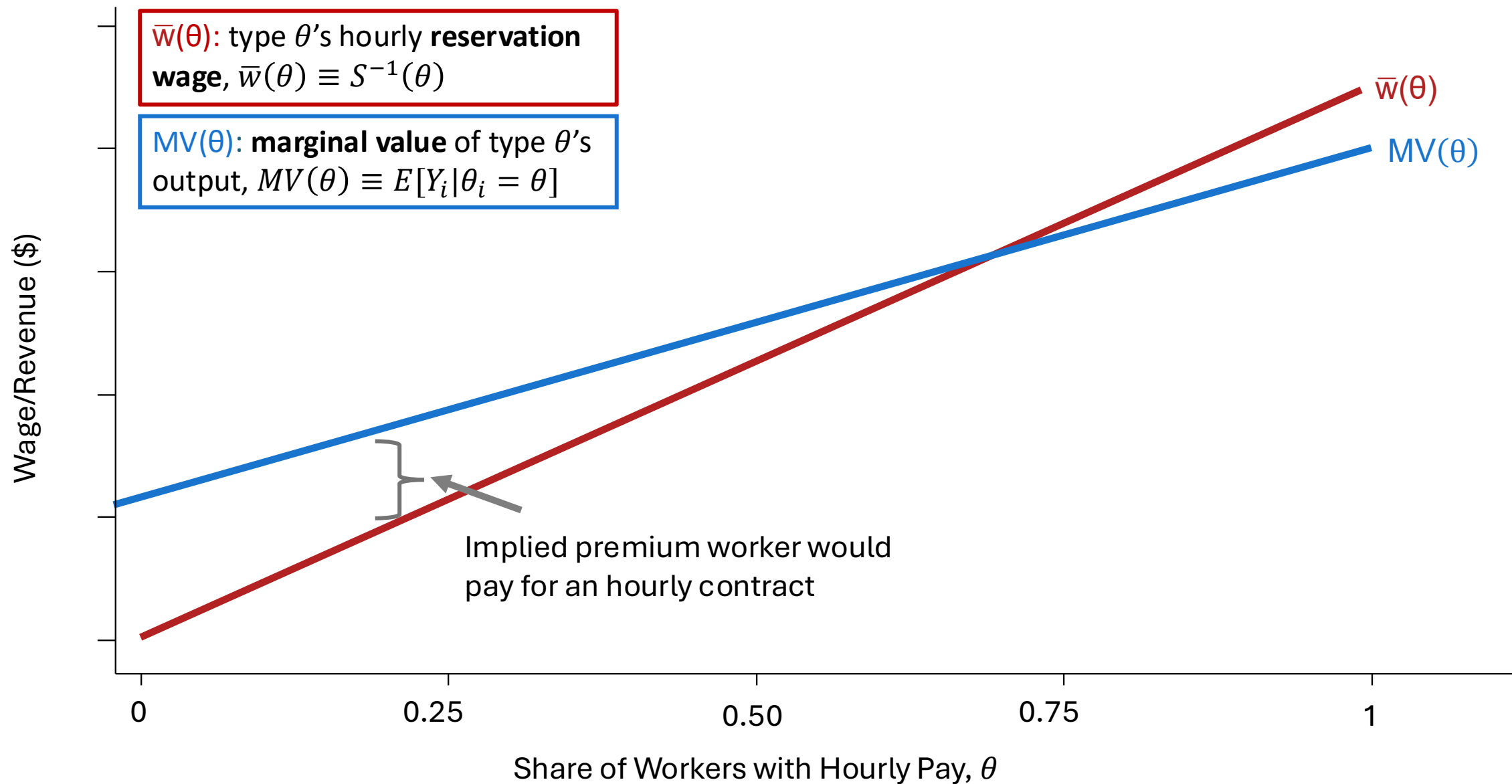


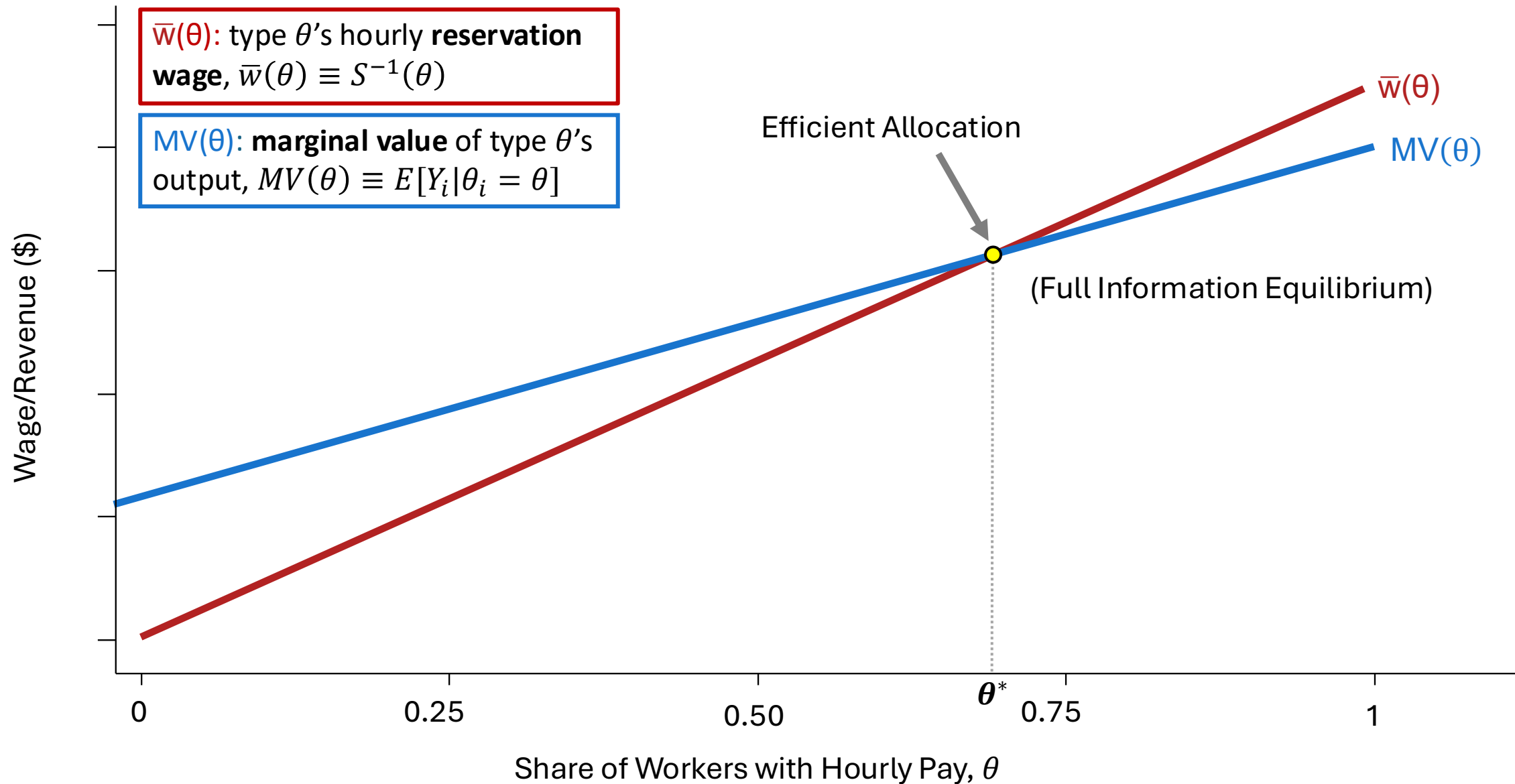


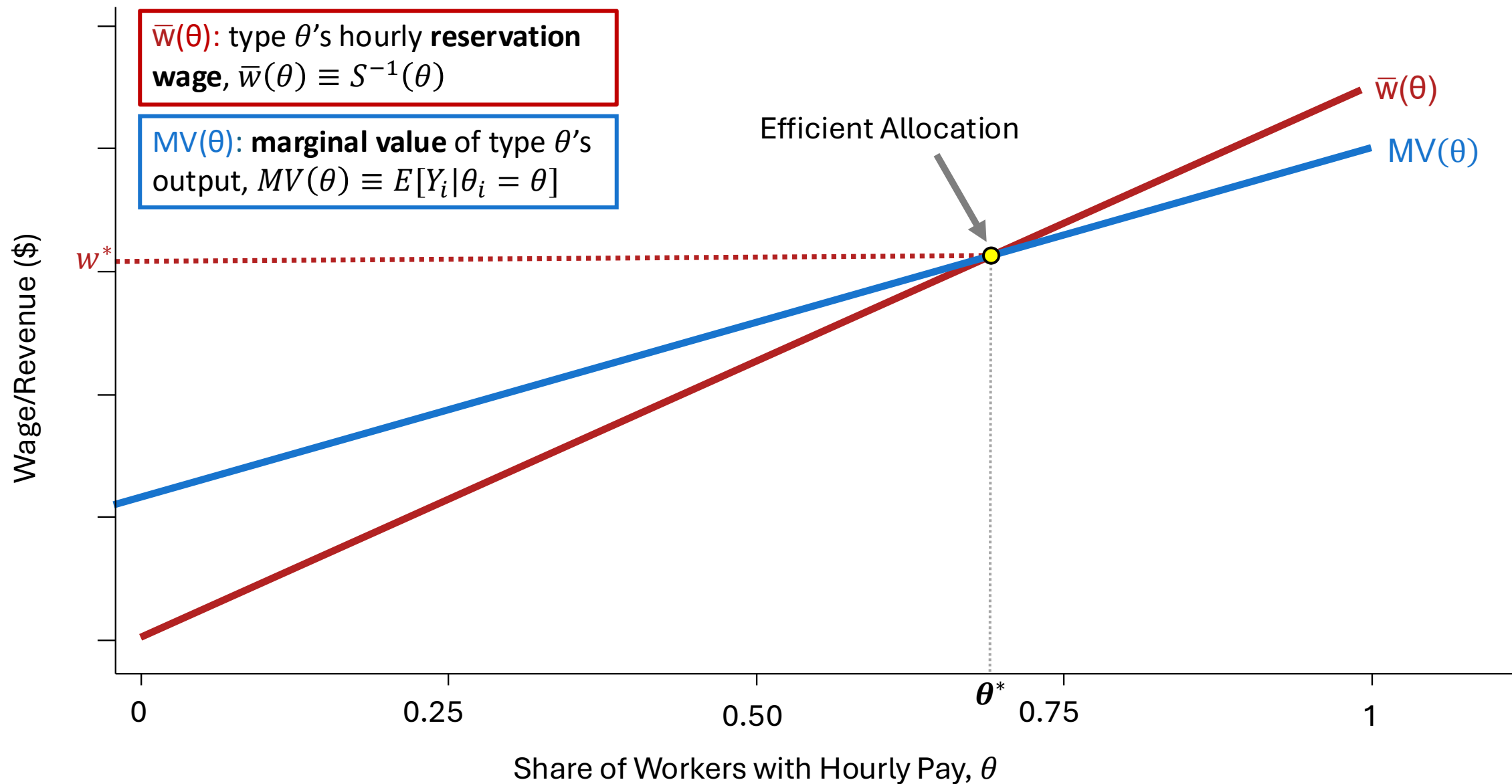


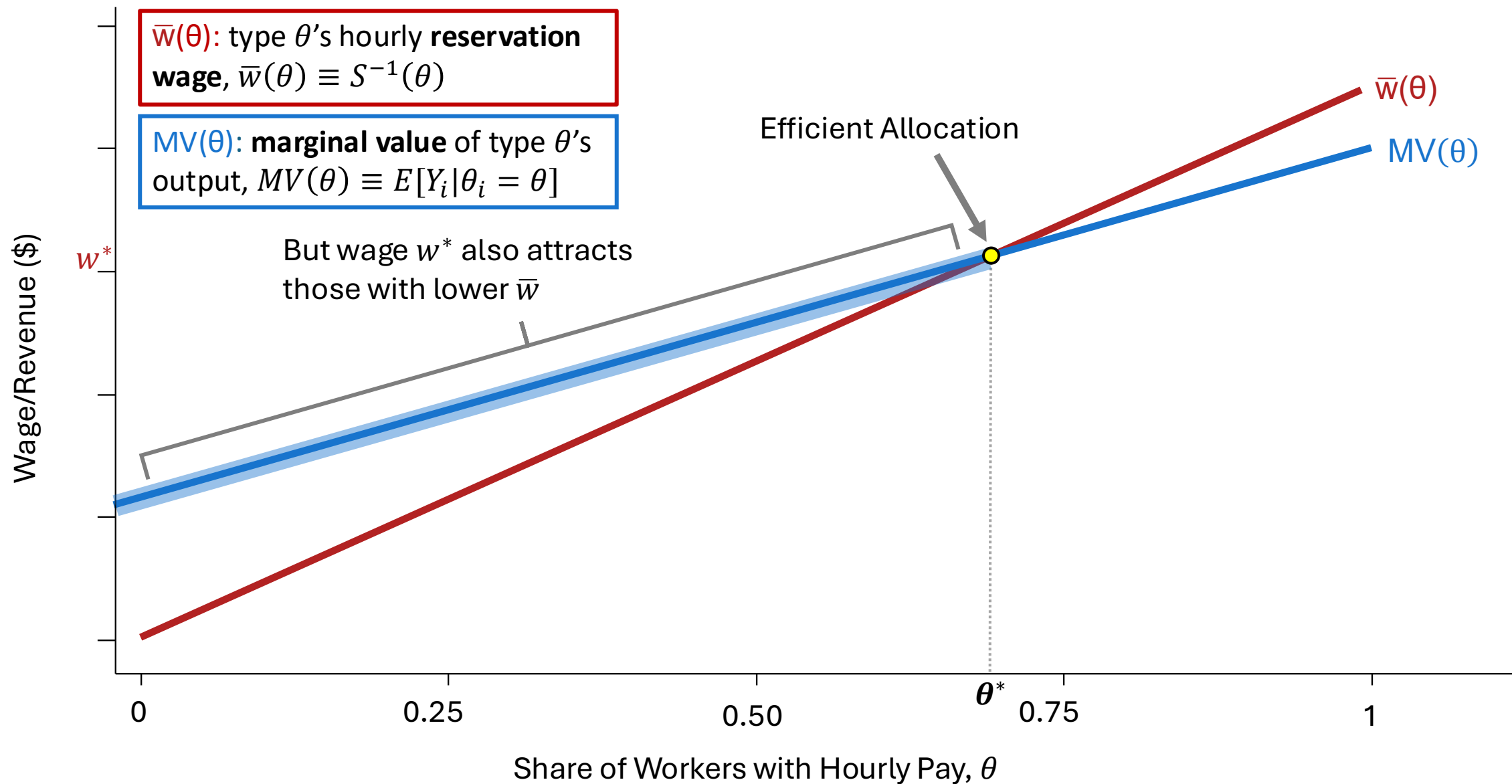


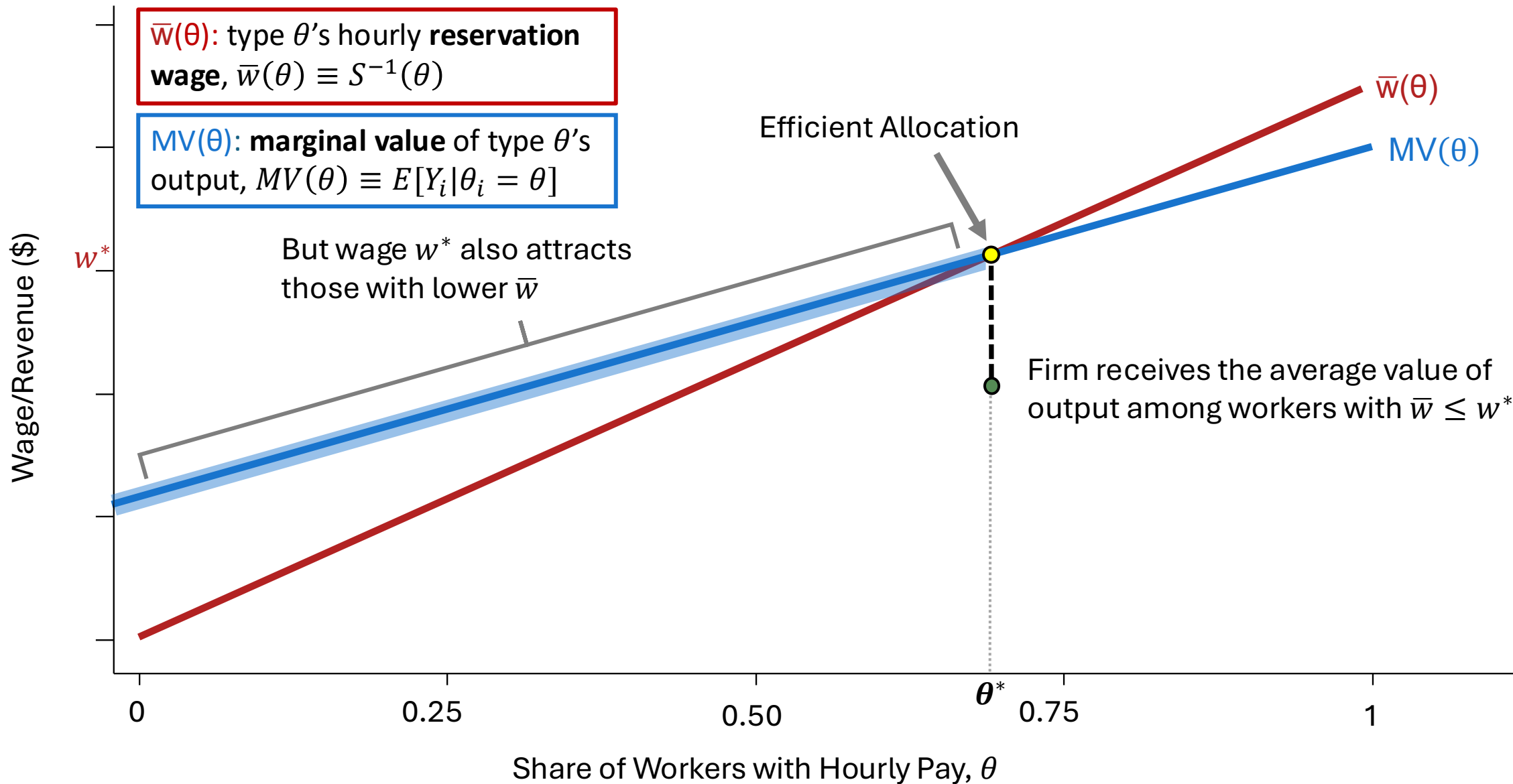


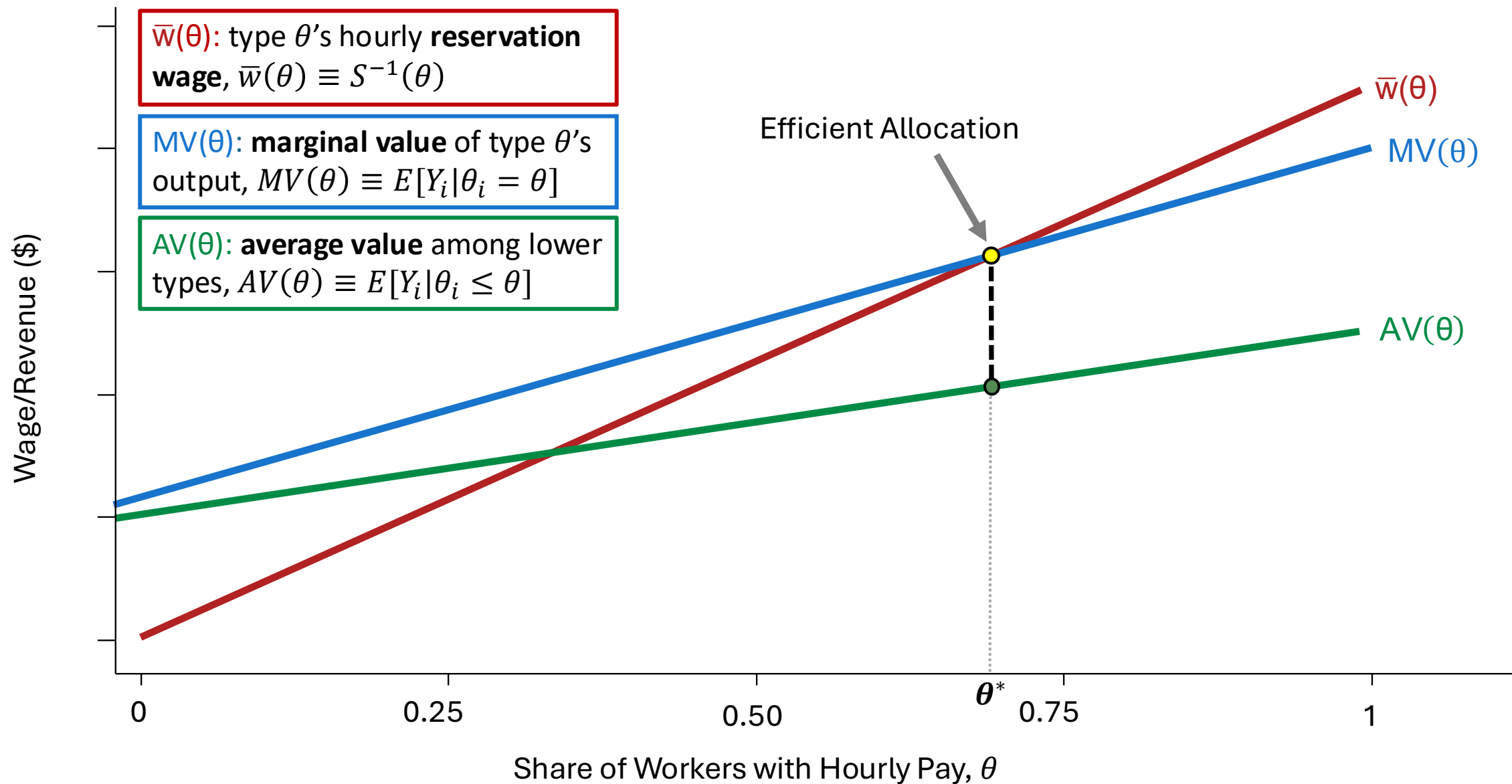


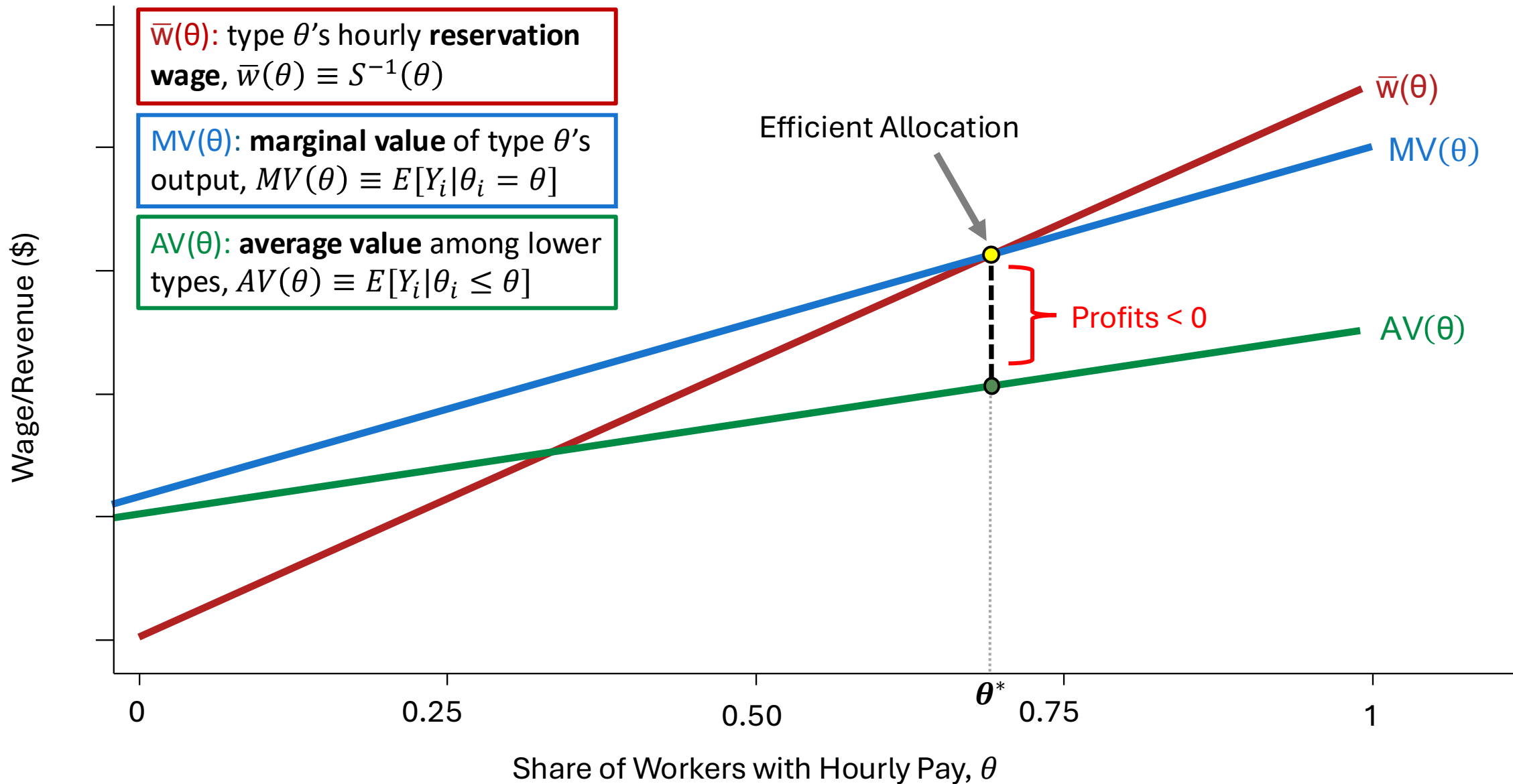


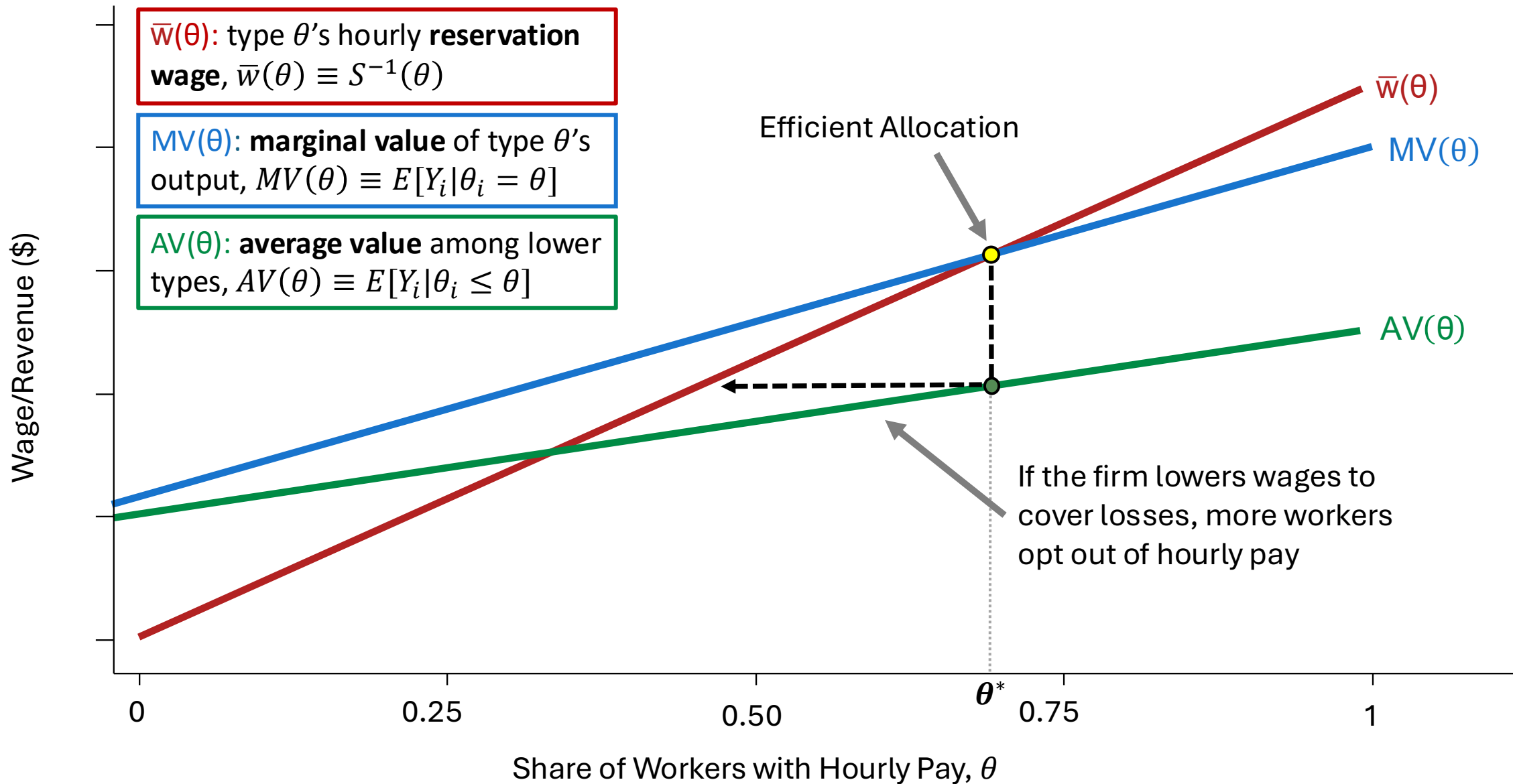


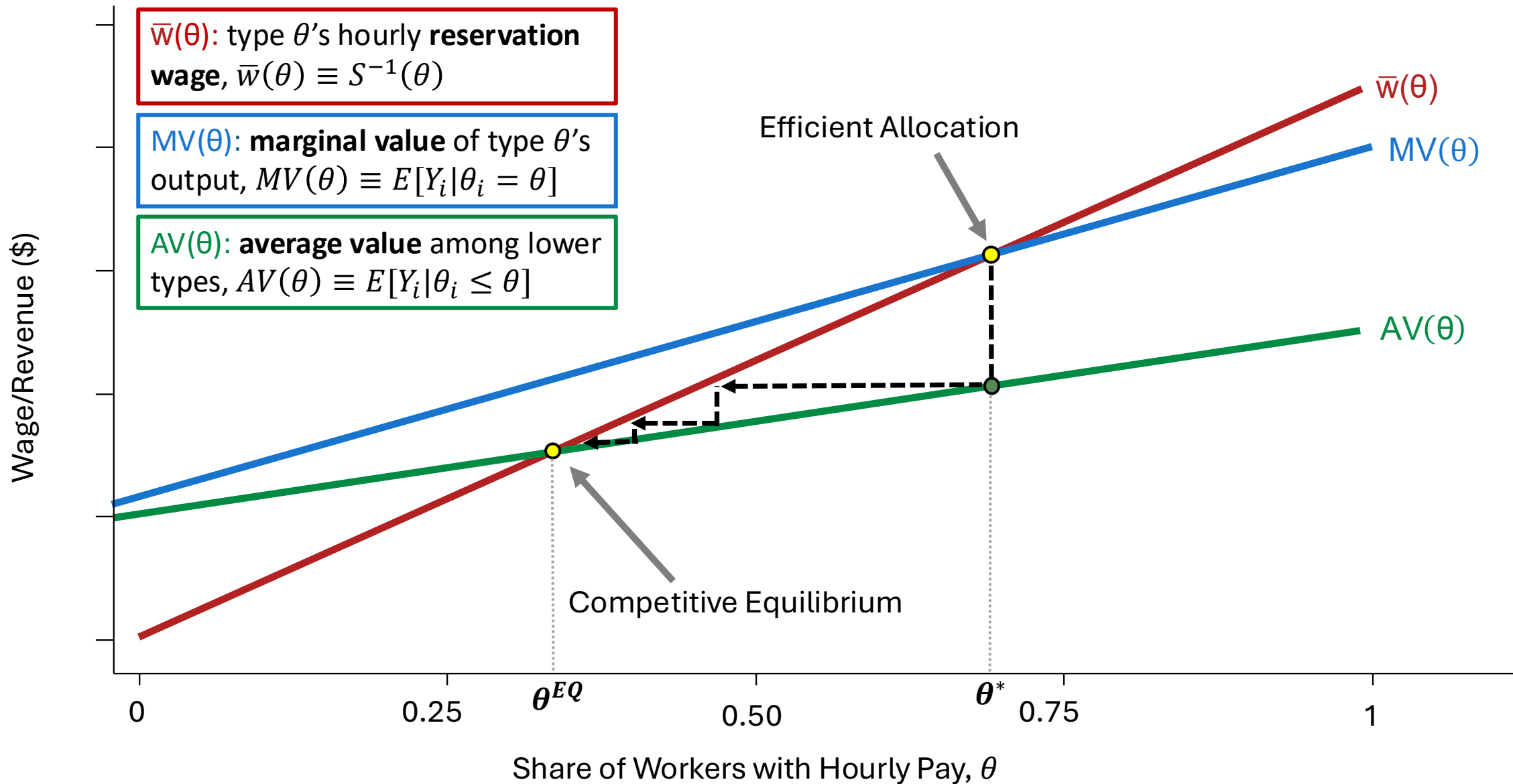


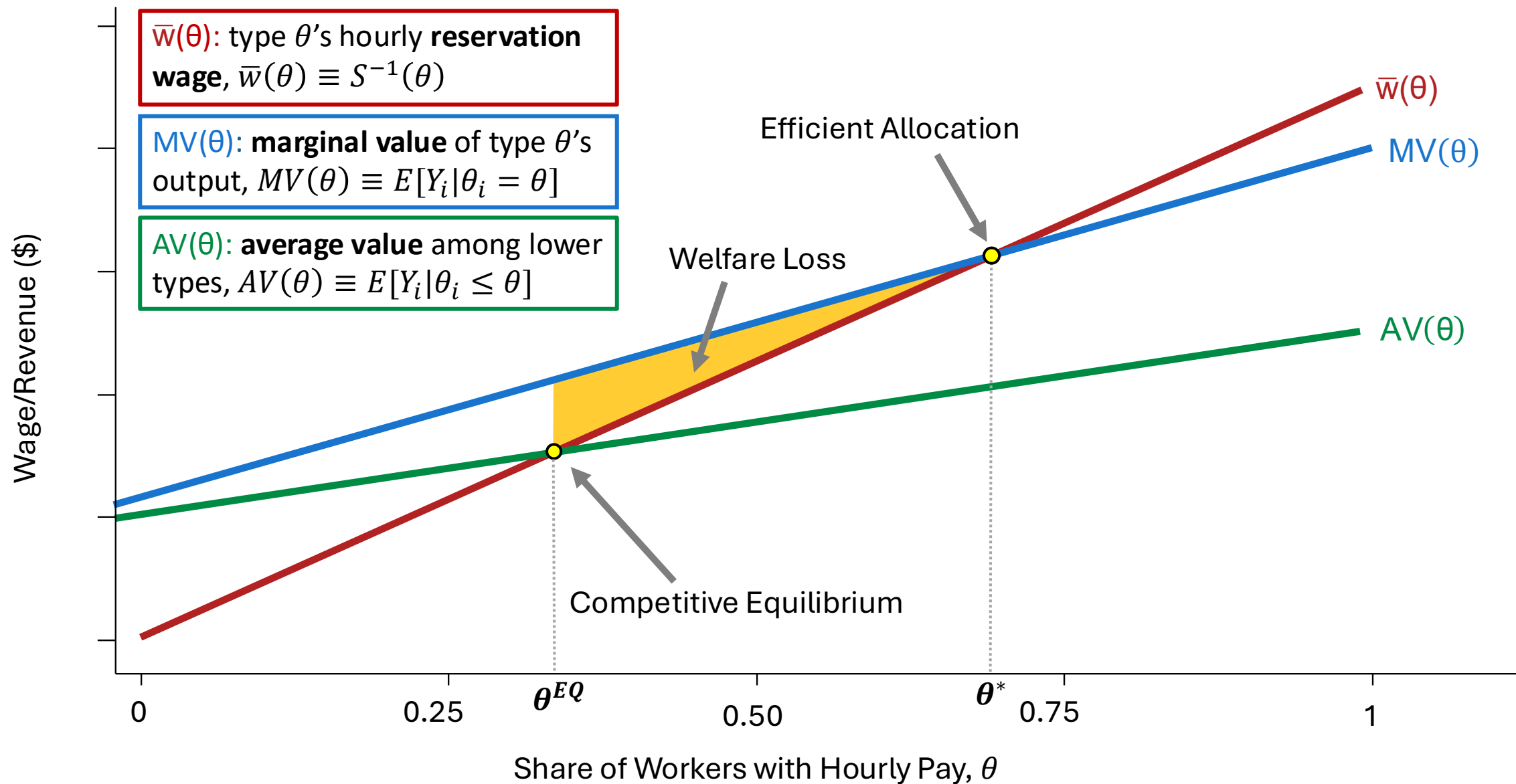








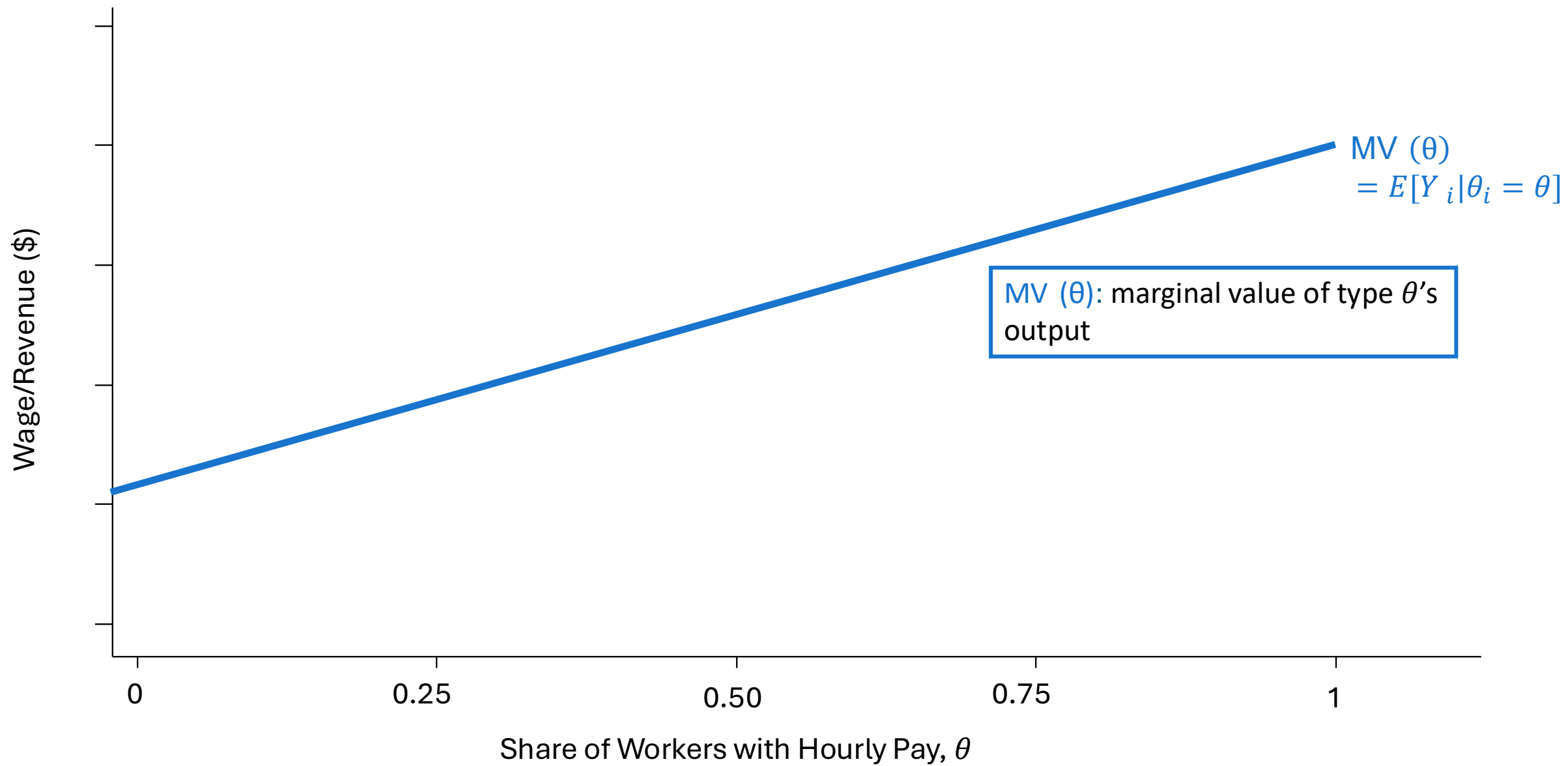


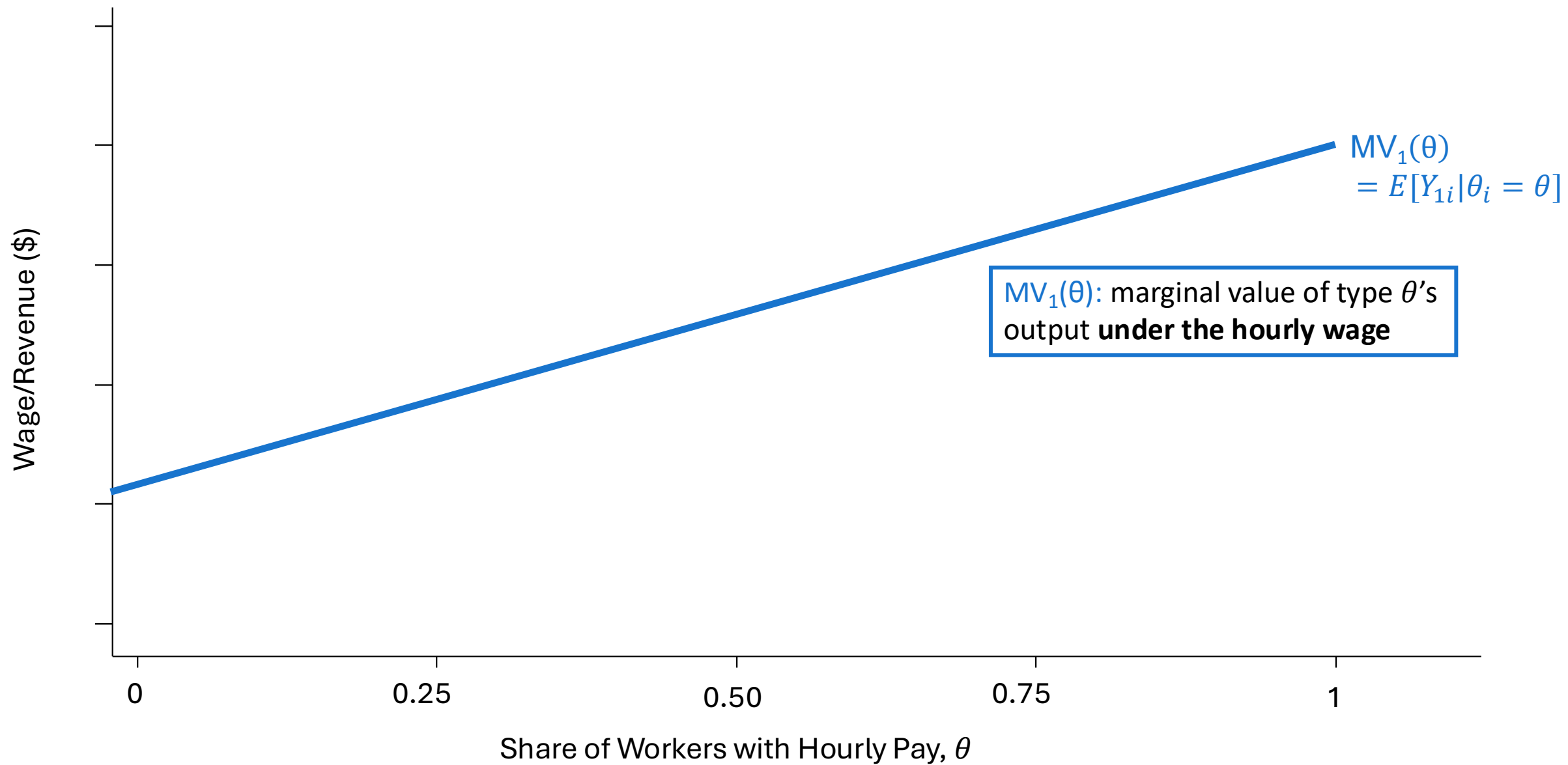


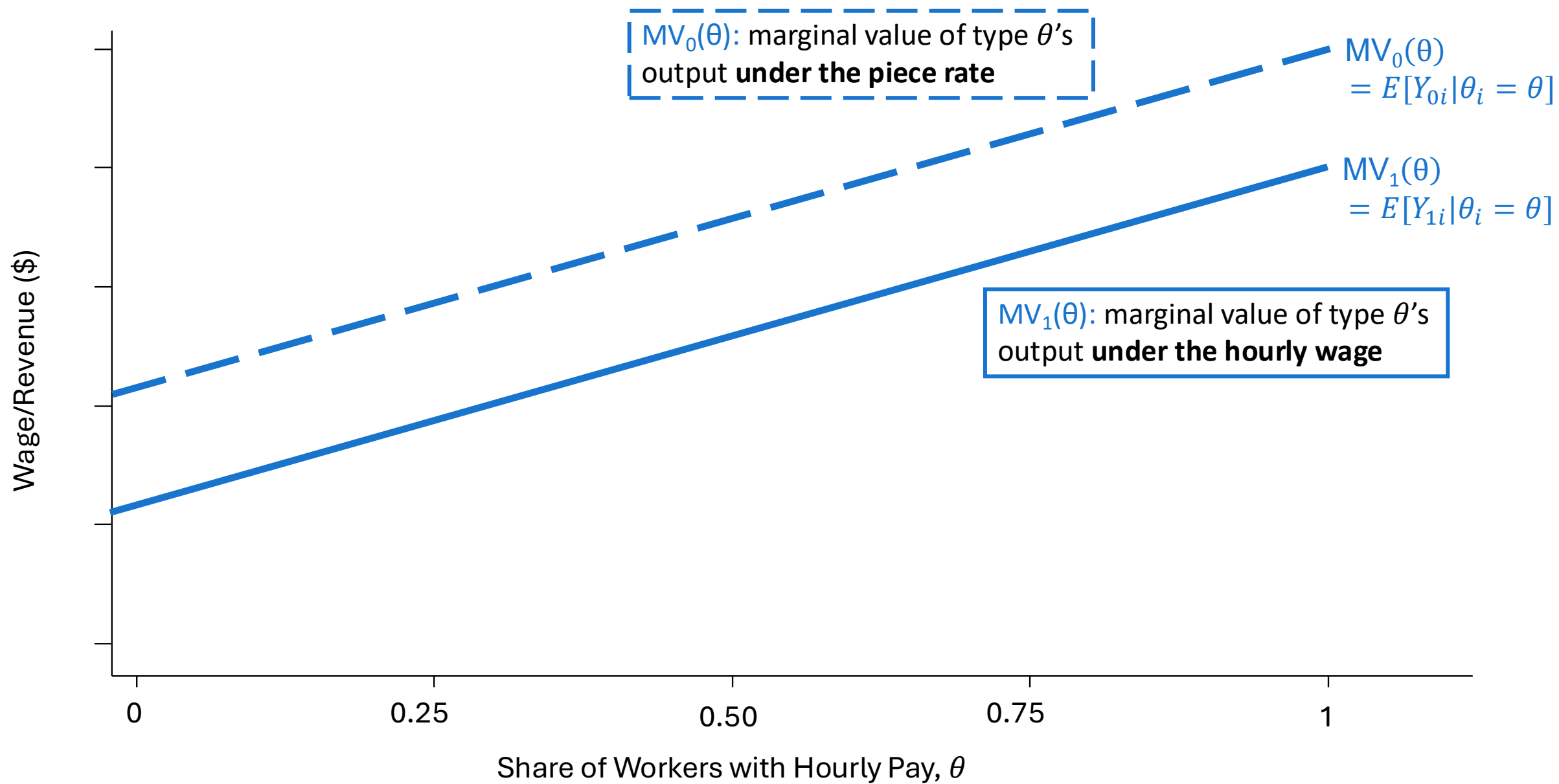
Incorporating Moral Hazard

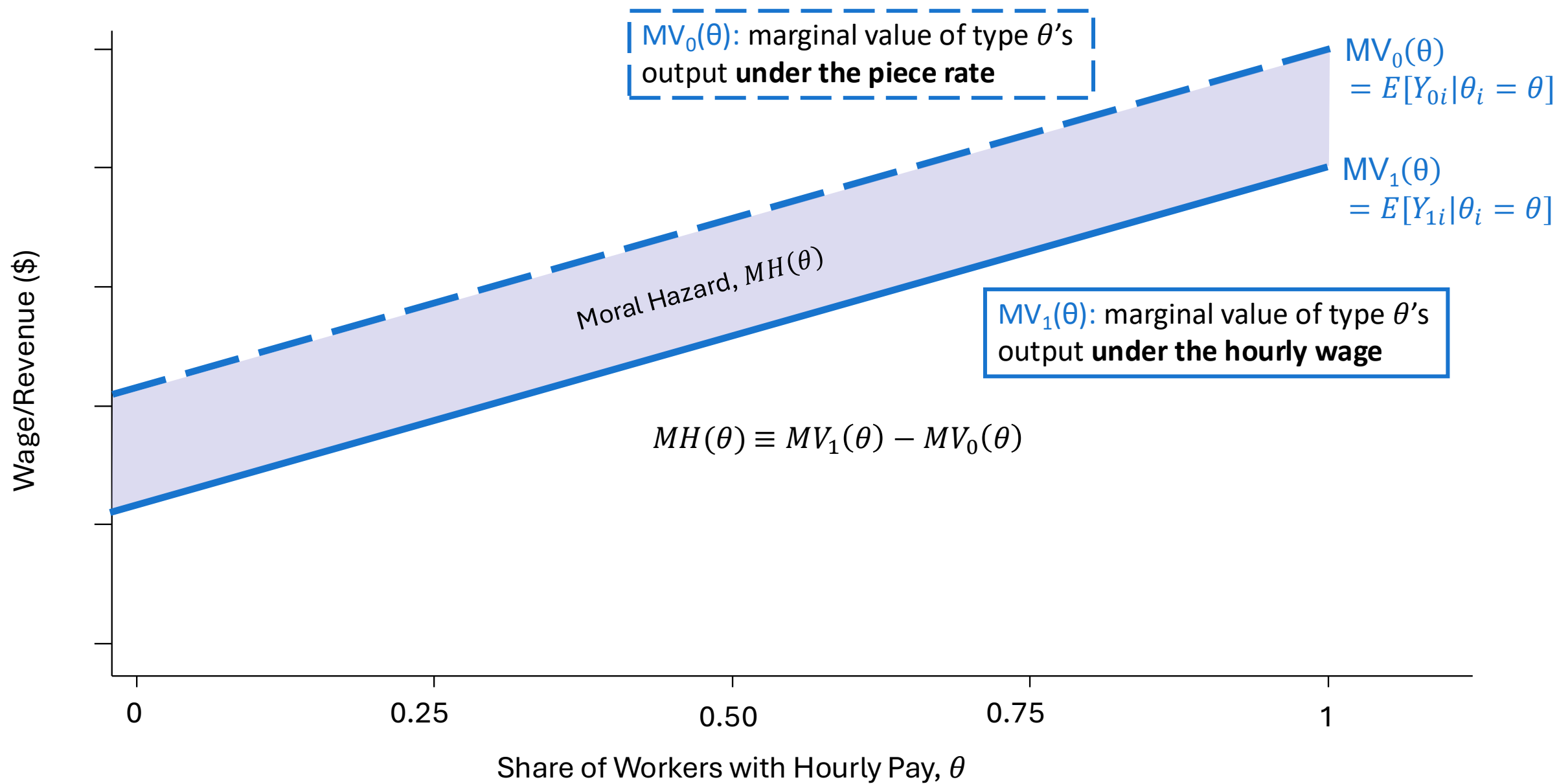
- Where does moral hazard fit in? Consider potential outputs
 - Y_{1i} : Potential output value under hourly wage
 - Y_{0i} : Potential output value under piece rate
$$Y_{1i} = Y_{0i} + MH_i$$
- Firms care about Y_{1i} , but they don't care about Y_{0i}
 - Piece-rate workers sell their output at a constant price per unit, so their productivity has no effect on firm profits
- $MV(\theta)$ & $AV(\theta)$ are defined *conditional* on accepting the hourly wage
$$MV(\theta) \equiv E[Y_i | \theta_i = \theta] = E[Y_{1i} | \theta_i = \theta]$$
$$AV(\theta) \equiv E[Y_i | \theta_i \leq \theta] = E[Y_{1i} | \theta_i \leq \theta]$$

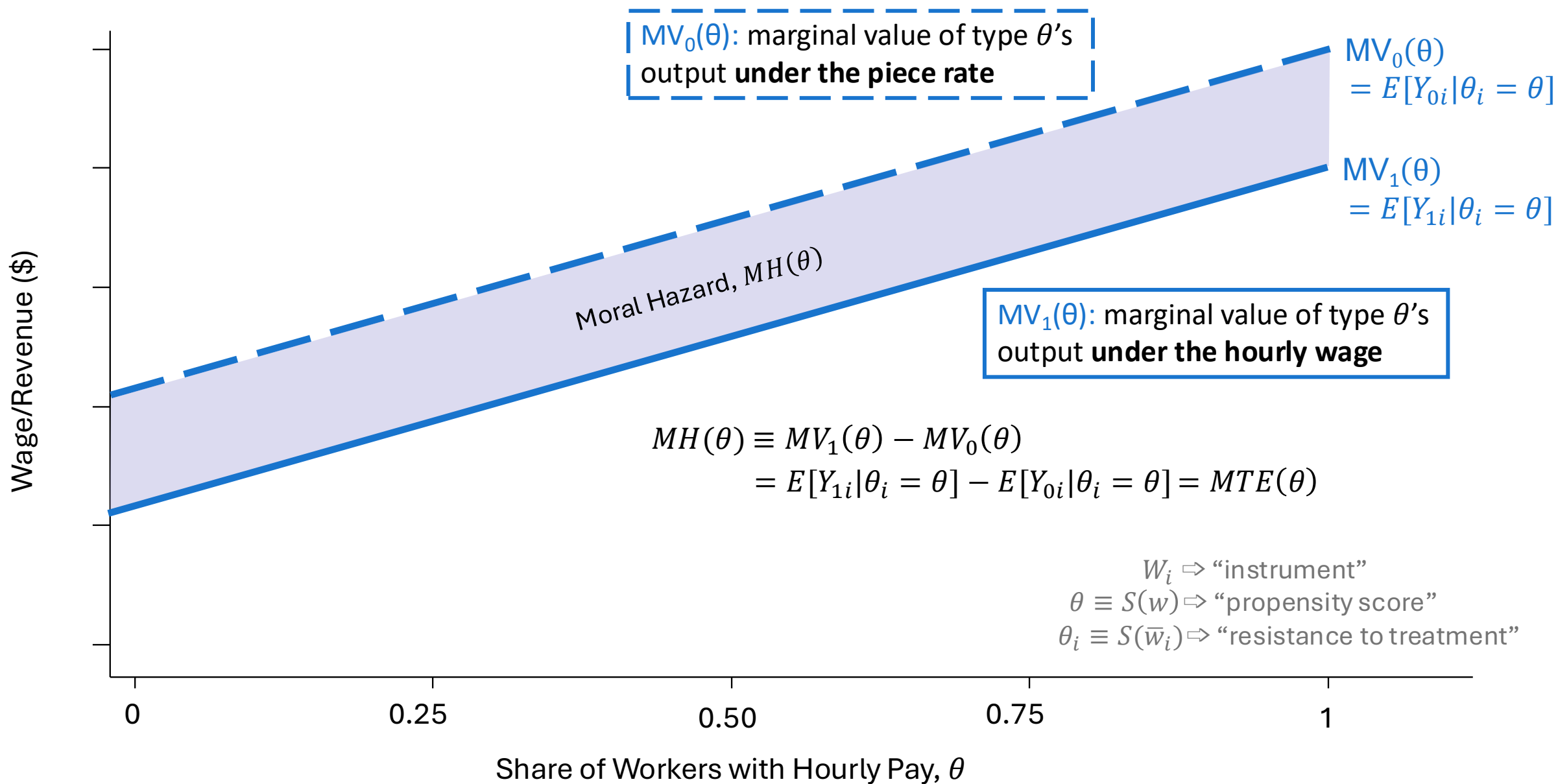
⇒ Equilibrium is inclusive of workers' moral hazard response to the hourly wage
- Still want to separately identify moral hazard
 - Firms might mitigate MH with “partial insurance” (e.g., sales commission, tips)
 - MH needed to measure welfare impact of policies promoting fixed-wages
 - Trade-off between insurance value and fiscal externality ($MH \Rightarrow$ tax revenue ↓)

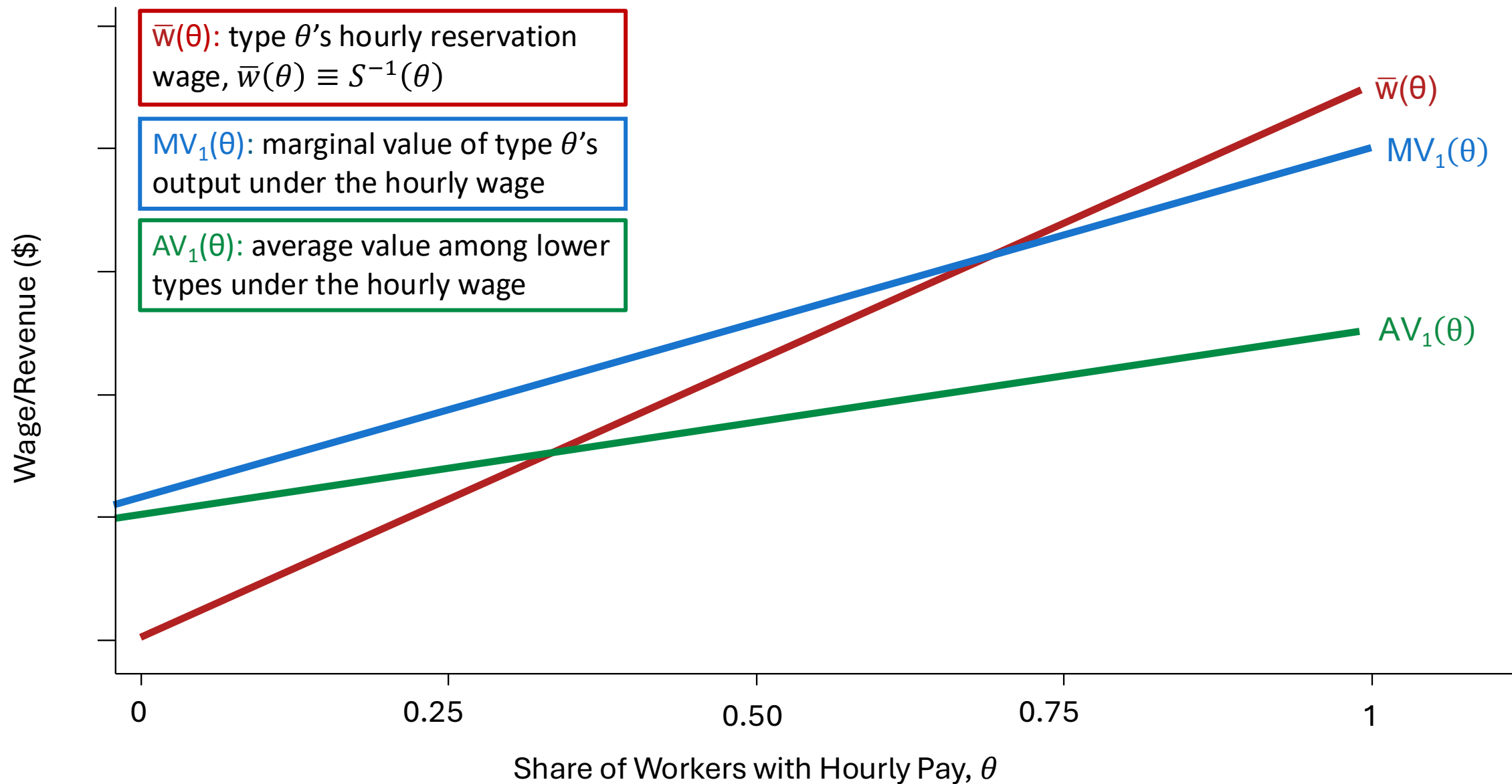








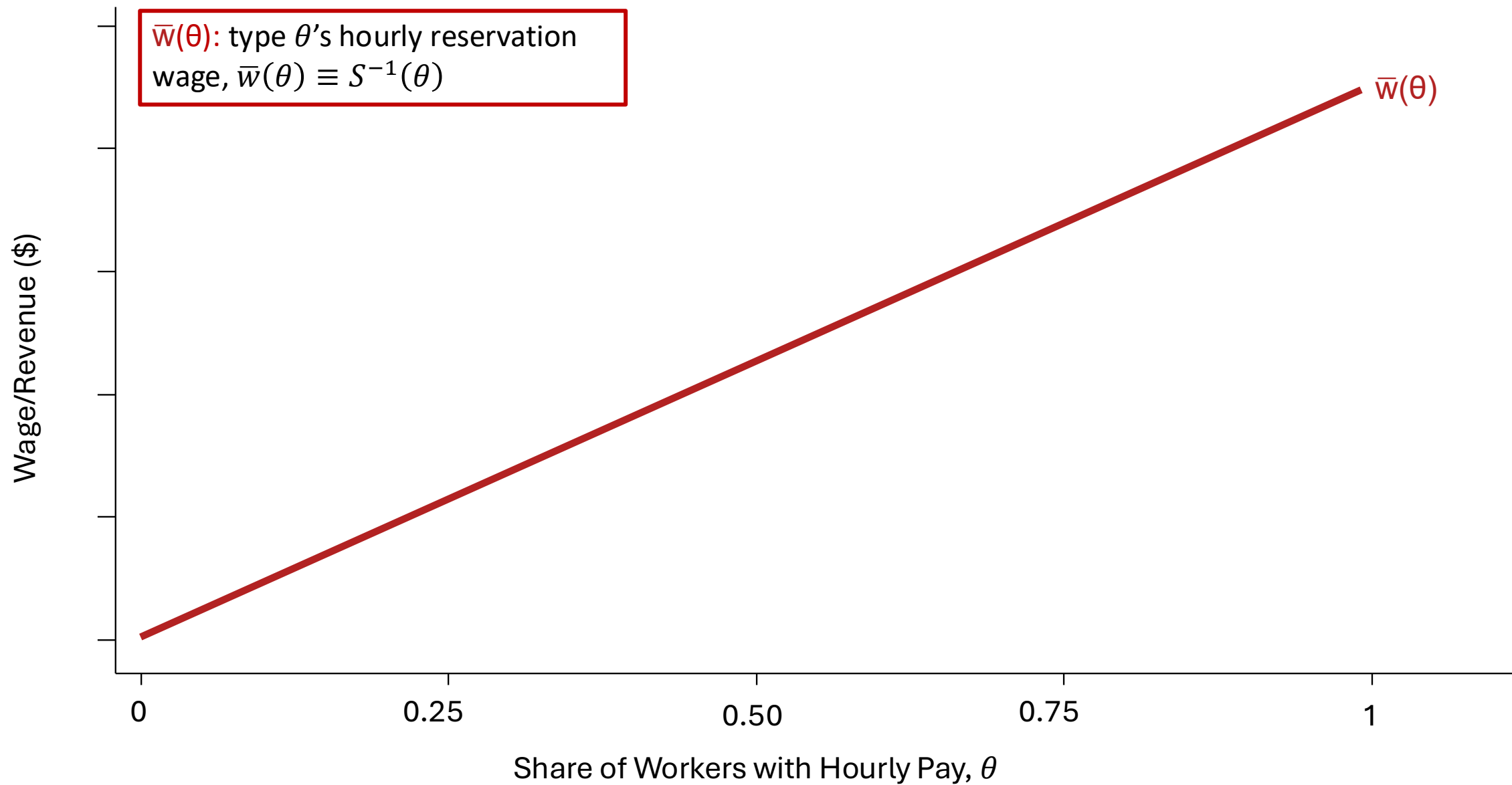


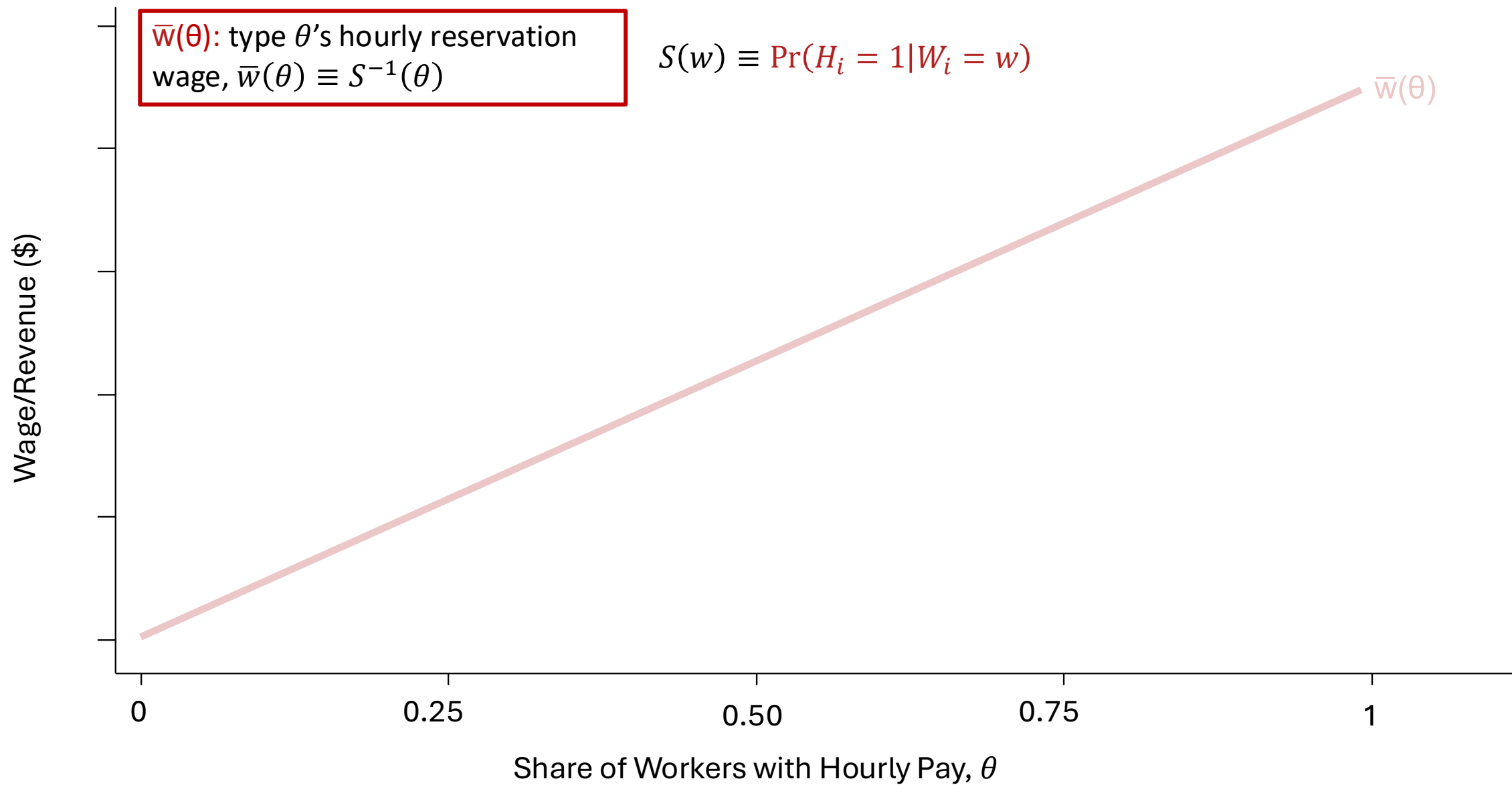


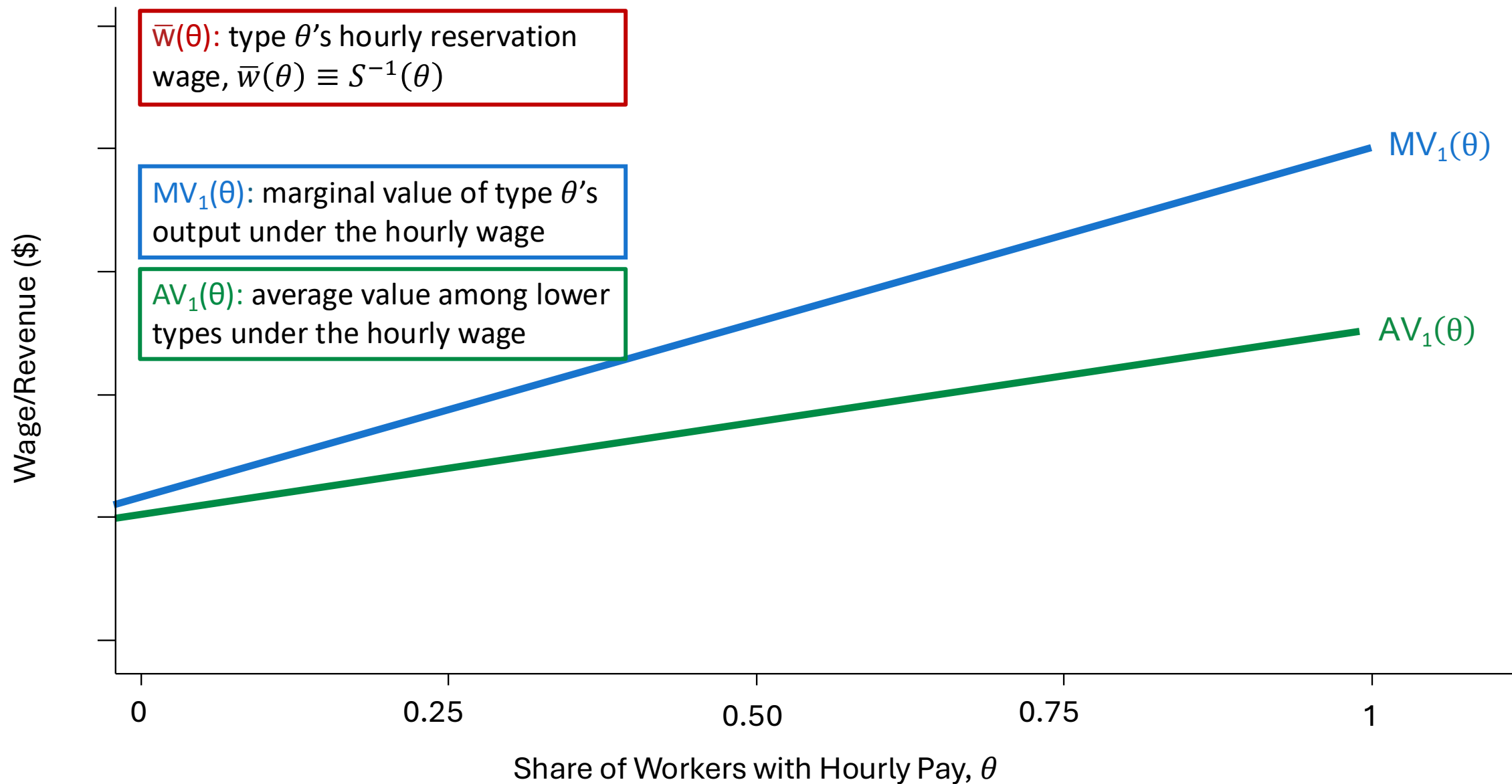


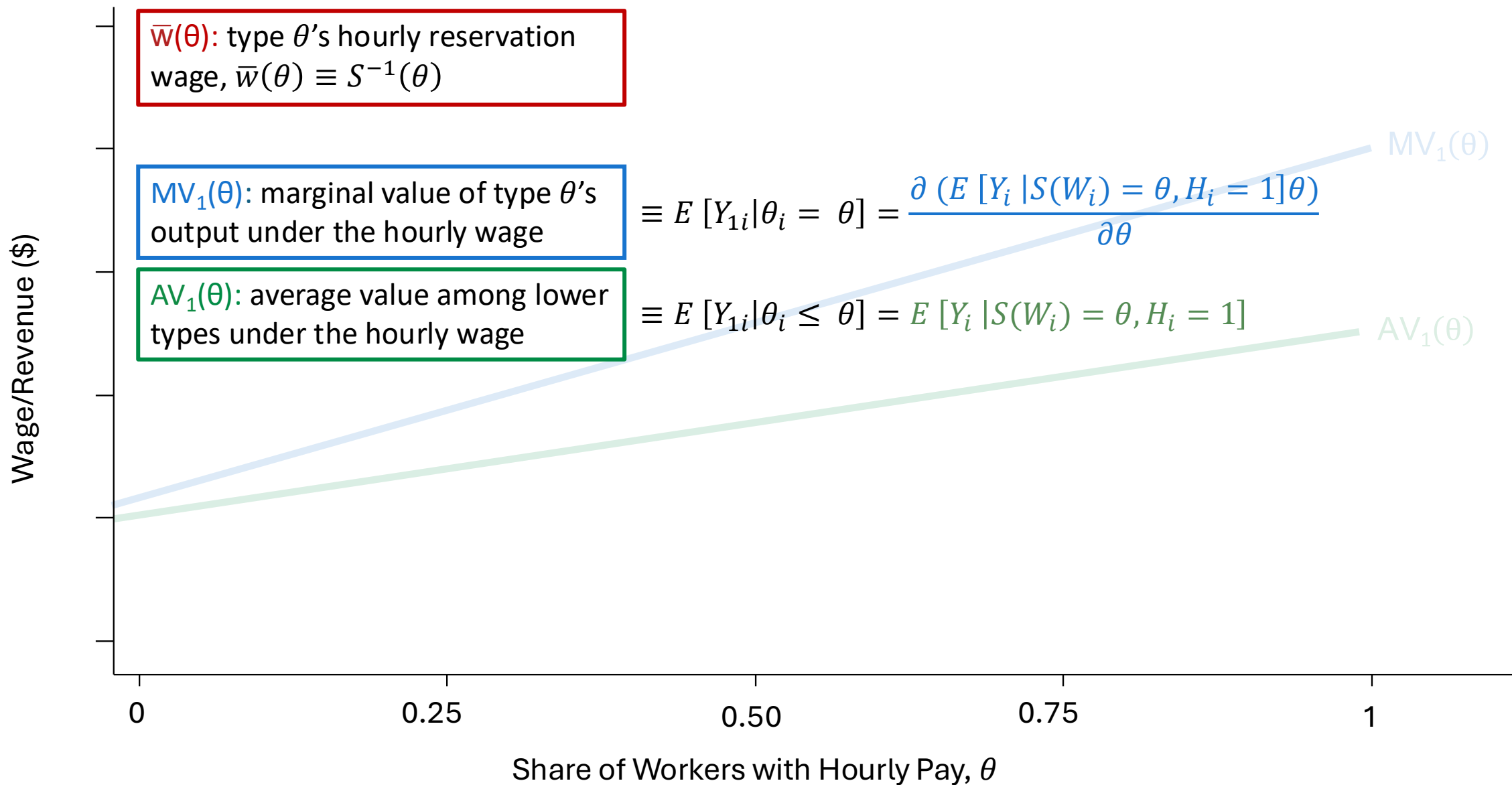
Outline

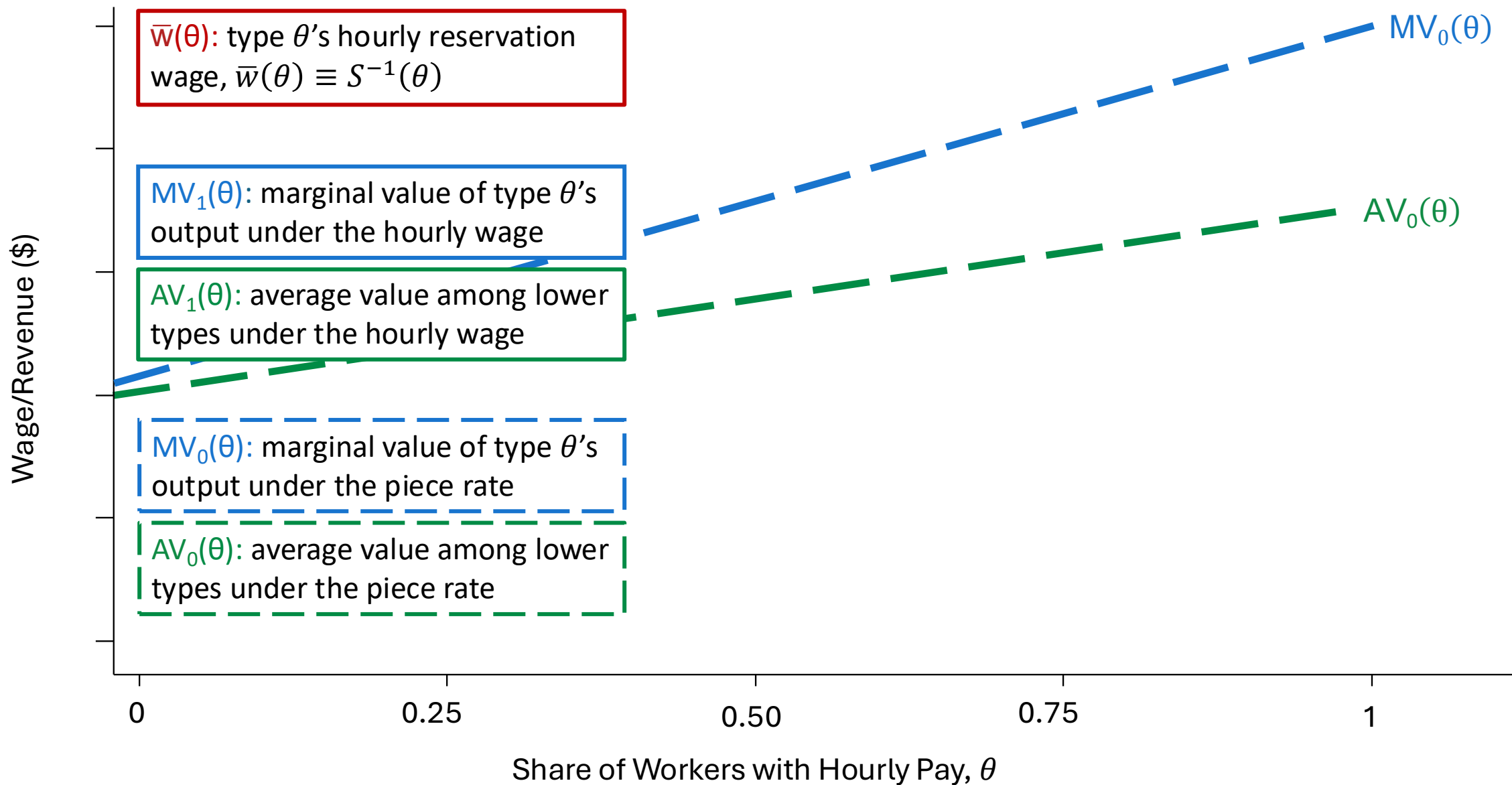
- 1 Experimental Design and Results
- 2 Model of Asymmetric Information
- 3 Estimates of Marginal Value and Welfare Loss
- 4 Optimal Fixed-Wage Subsidy and Piece-Rate Tax
- 5 Extensions

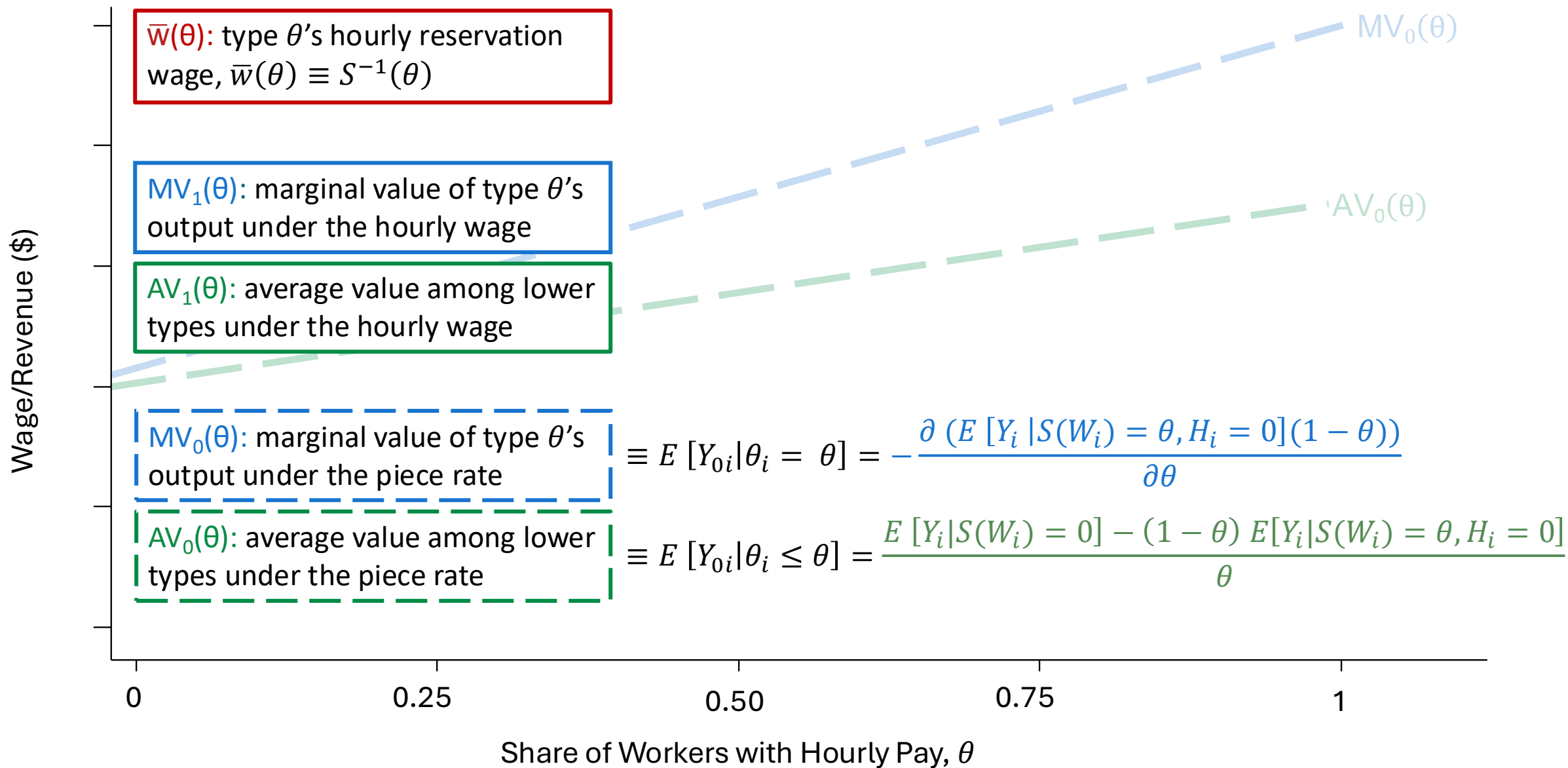












$\bar{w}(\theta)$: type θ 's hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$

$$S(w) \equiv \Pr(H_i = 1 | W_i = w)$$

$MV_1(\theta)$: marginal value of type θ 's output under the hourly wage

$$\equiv E[Y_{1i} | \theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1] \theta)}{\partial \theta}$$

$AV_1(\theta)$: average value among lower types under the hourly wage

$$\equiv E[Y_{1i} | \theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

$MV_0(\theta)$: marginal value of type θ 's output under the piece rate

$$\equiv E[Y_{0i} | \theta_i = \theta] = - \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0](1 - \theta))}{\partial \theta}$$

$AV_0(\theta)$: average value among lower types under the piece rate

$$\equiv E[Y_{0i} | \theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1 - \theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Estimation

(Carneiro et al. 2011)

1. Estimate $\Pr(H_i = 1|W_i = w)$
(logit regression)

$$S(w) \equiv \Pr(H_i = 1|W_i = w)$$

$$MV_1(\theta) \equiv E[Y_{1i}|\theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1]\theta)}{\partial \theta}$$

$$AV_1(\theta) \equiv E[Y_{1i}|\theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

$$MV_0(\theta) \equiv E[Y_{0i}|\theta_i = \theta] = -\frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0](1 - \theta))}{\partial \theta}$$

$$AV_0(\theta) \equiv E[Y_{0i}|\theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1 - \theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Estimation

(Carneiro et al. 2011)

1. Estimate $\Pr(H_i = 1|W_i = w)$
(logit regression)

2. Separately estimate

$$E[Y_i | S(W_i) = \theta, H_i = 1]$$

&

$$E[Y_i | S(W_i) = \theta, H_i = 0]$$

(local polynomial regression)

$$S(w) \equiv \Pr(H_i = 1|W_i = w)$$

$$MV_1(\theta) \equiv E[Y_{1i} | \theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1]\theta)}{\partial \theta}$$

$$AV_1(\theta) \equiv E[Y_{1i} | \theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

$$MV_0(\theta) \equiv E[Y_{0i} | \theta_i = \theta] = -\frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0](1 - \theta))}{\partial \theta}$$

$$AV_0(\theta) \equiv E[Y_{0i} | \theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1 - \theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Estimation

(Carneiro et al. 2011)

1. Estimate $\Pr(H_i = 1|W_i = w)$
(logit regression)

2. Separately estimate

$$E[Y_i | S(W_i) = \theta, H_i = 1]$$

&

$$E[Y_i | S(W_i) = \theta, H_i = 0]$$

(local polynomial regression)

3. Differentiate with respect to θ

$$S(w) \equiv \Pr(H_i = 1|W_i = w)$$

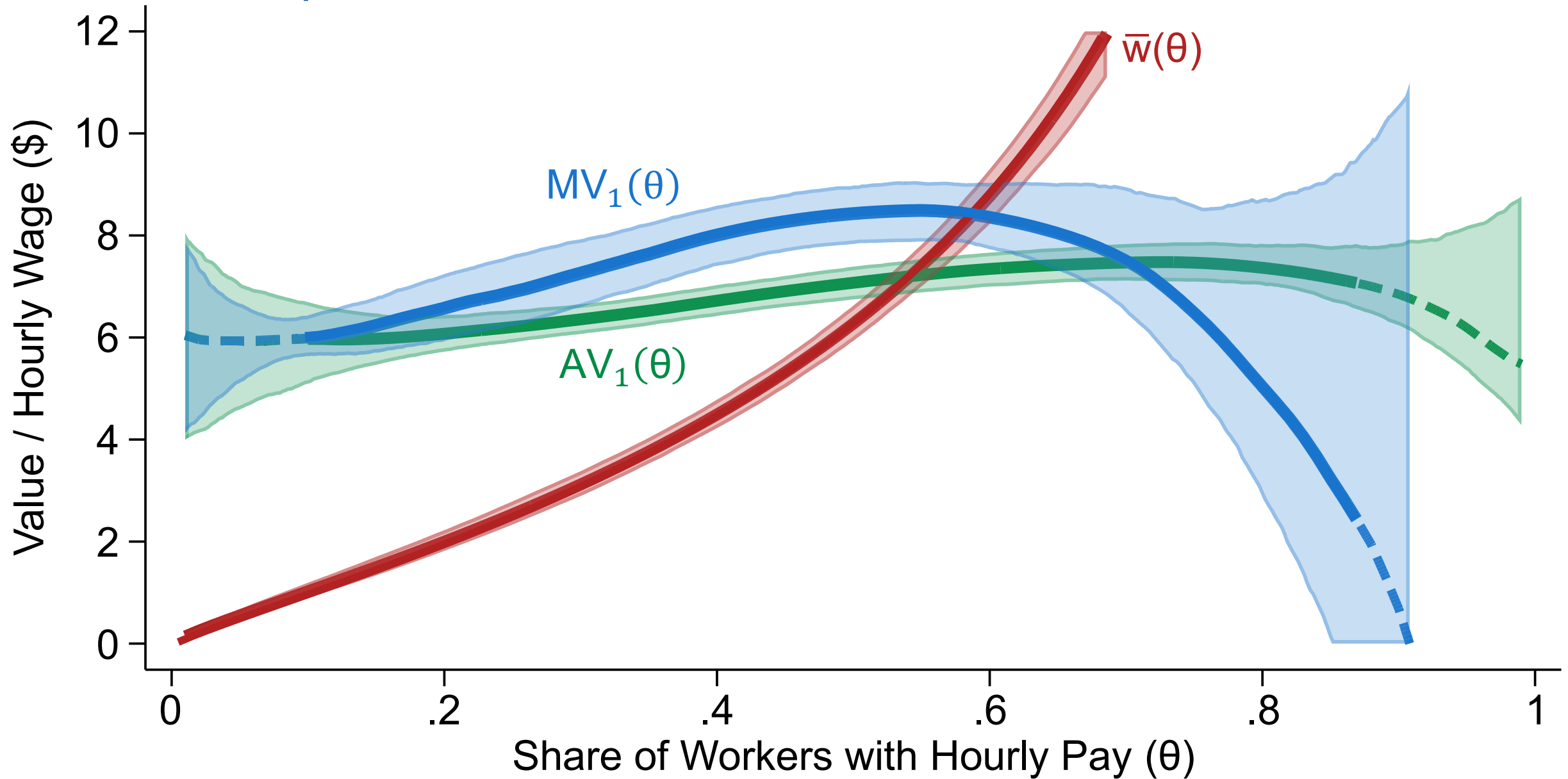
$$MV_1(\theta) \equiv E[Y_{1i} | \theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1])}{\partial \theta}$$

$$AV_1(\theta) \equiv E[Y_{1i} | \theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

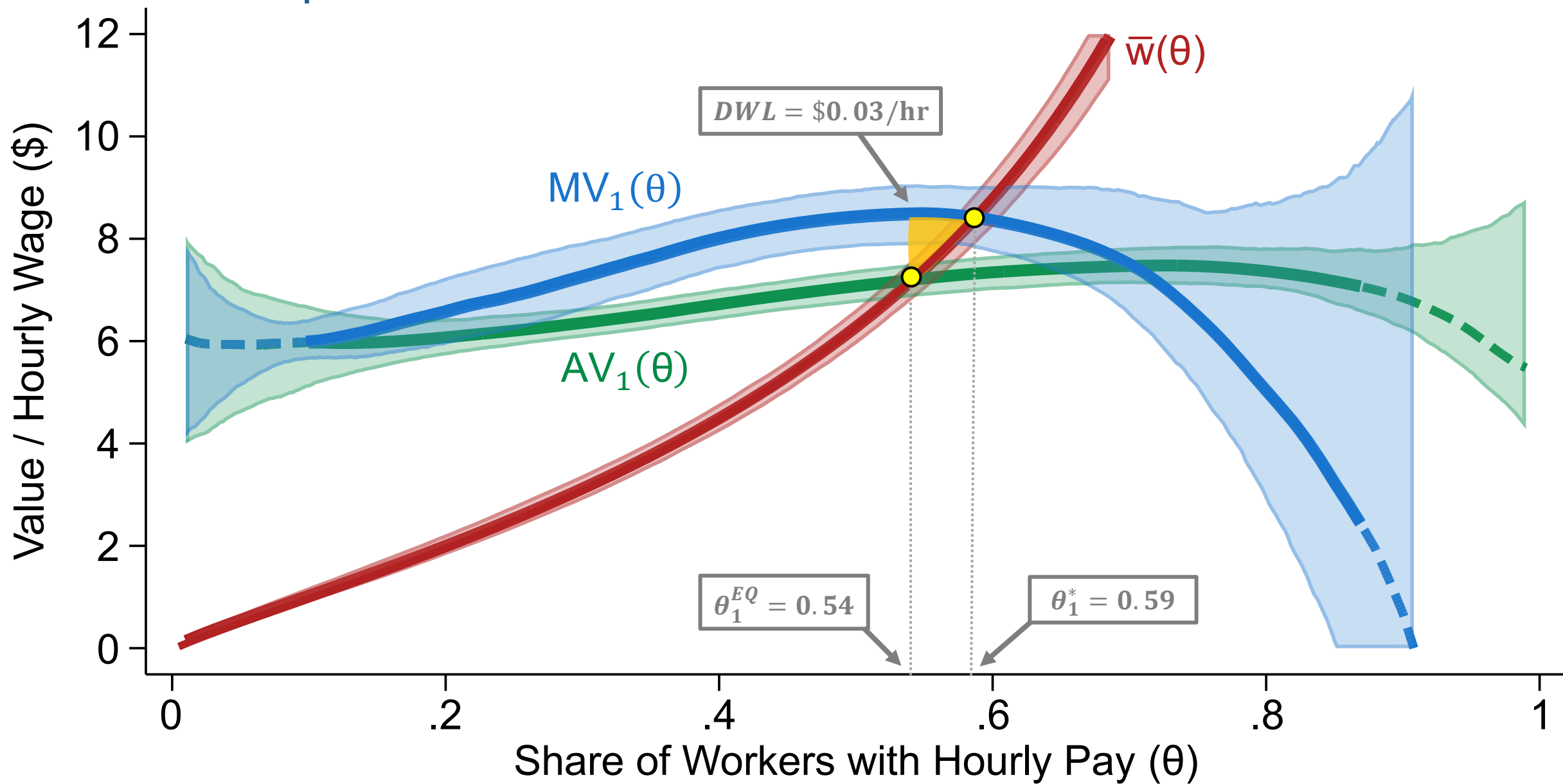
$$MV_0(\theta) \equiv E[Y_{0i} | \theta_i = \theta] = - \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0])}{\partial \theta} (1 - \theta)$$

$$AV_0(\theta) \equiv E[Y_{0i} | \theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1 - \theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

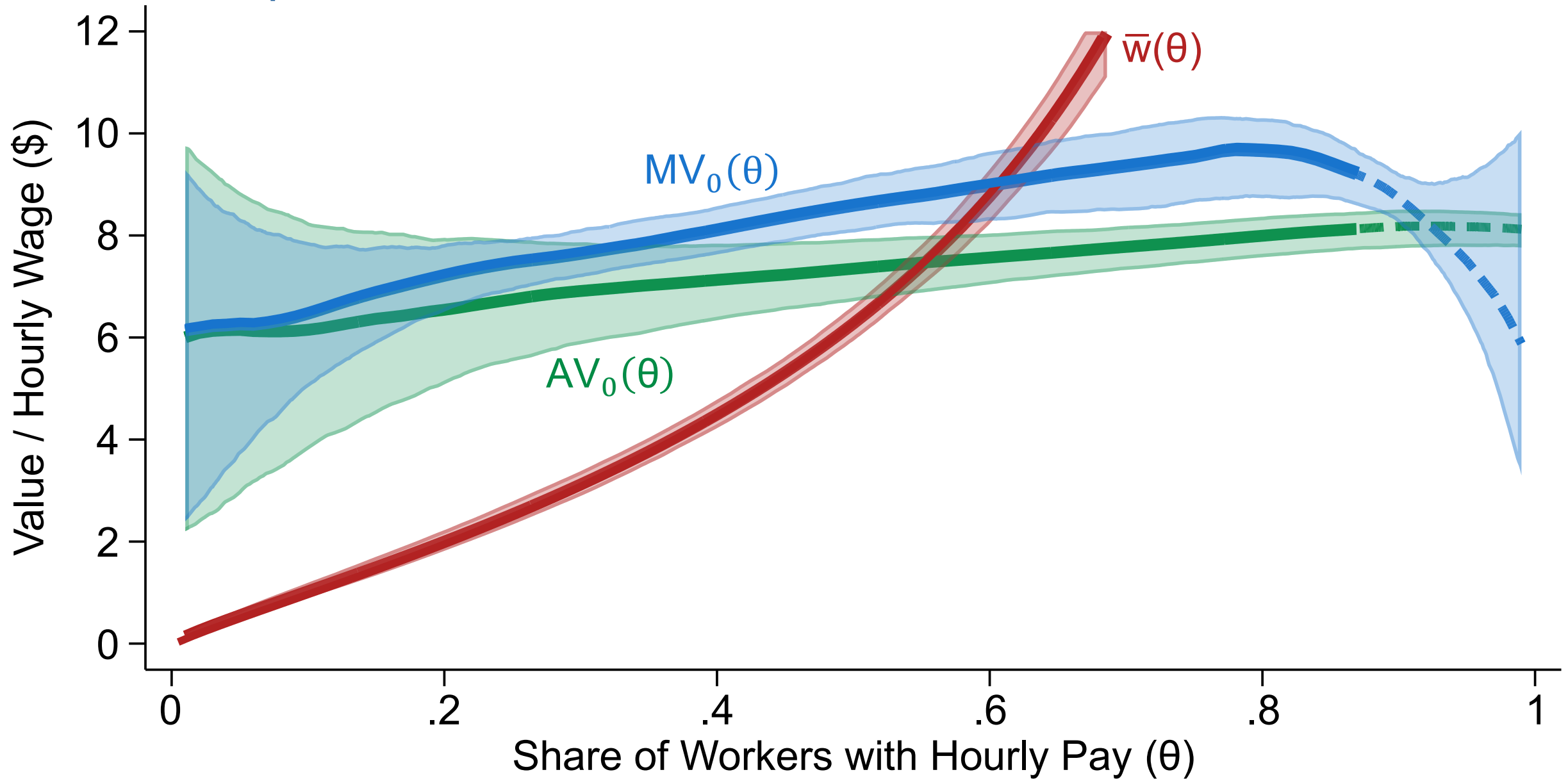
Equilibrium with Adverse Selection and Moral Hazard



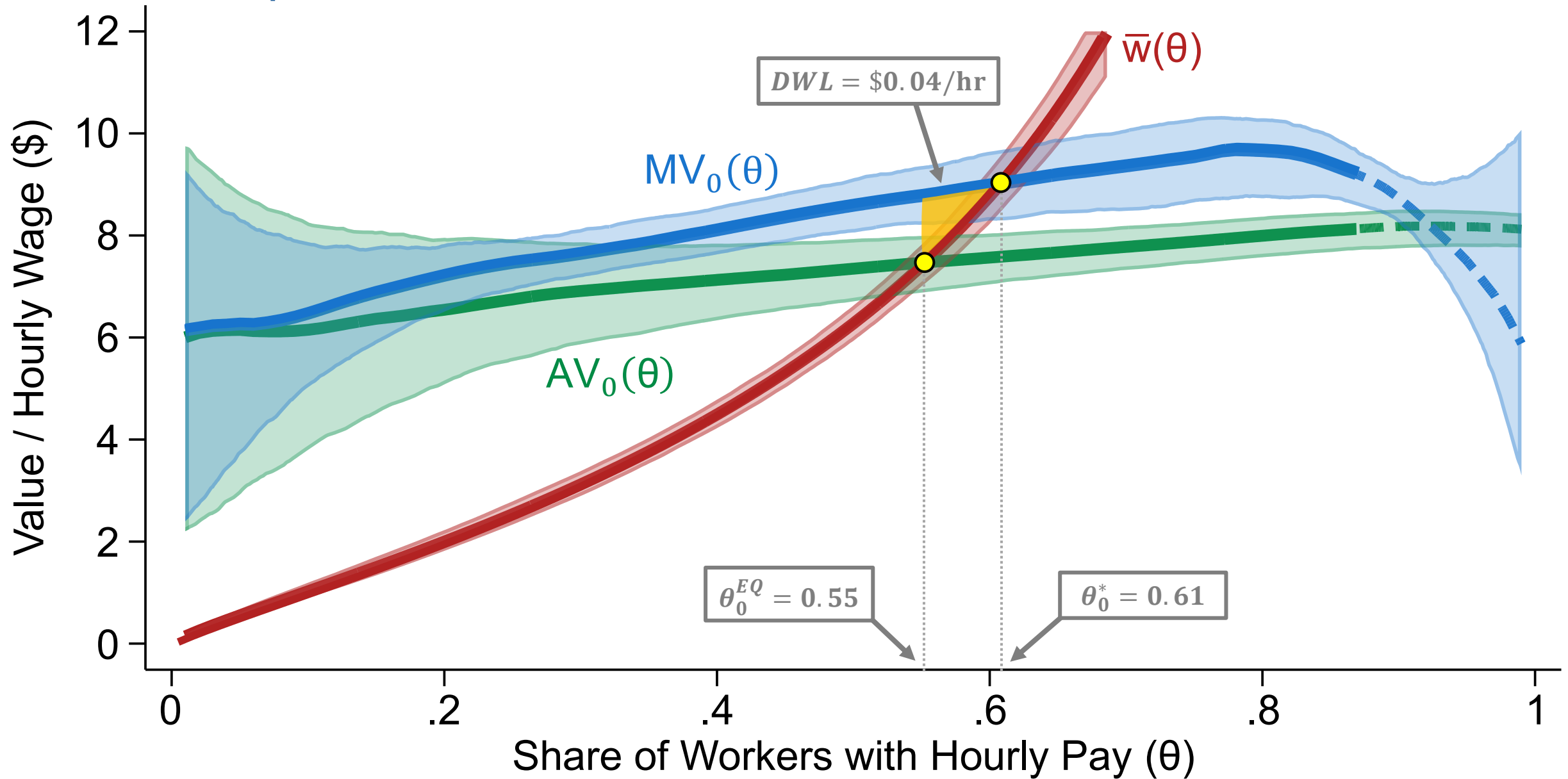
Equilibrium with Adverse Selection and Moral Hazard



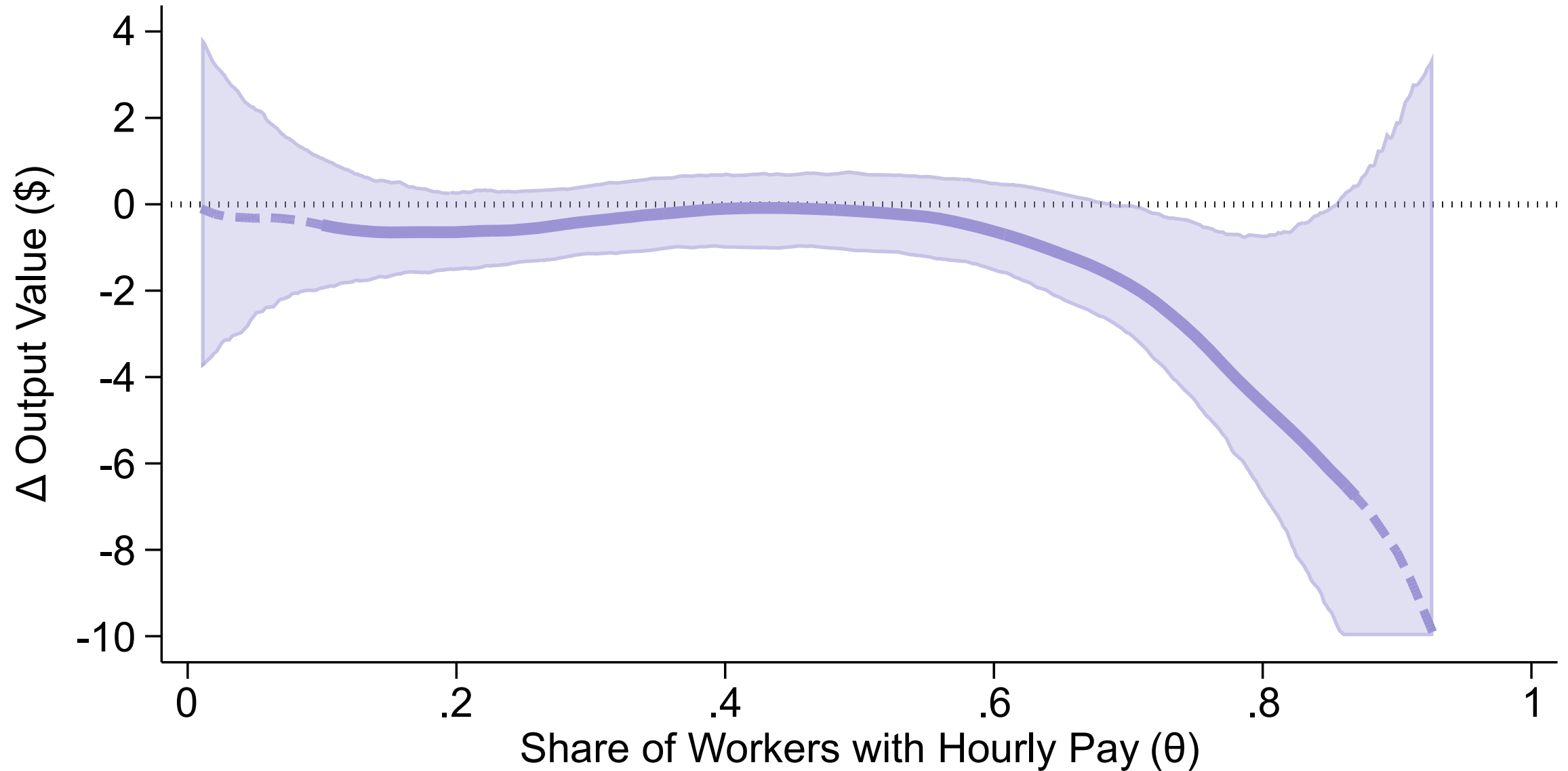
Equilibrium with Adverse Selection and no Moral Hazard



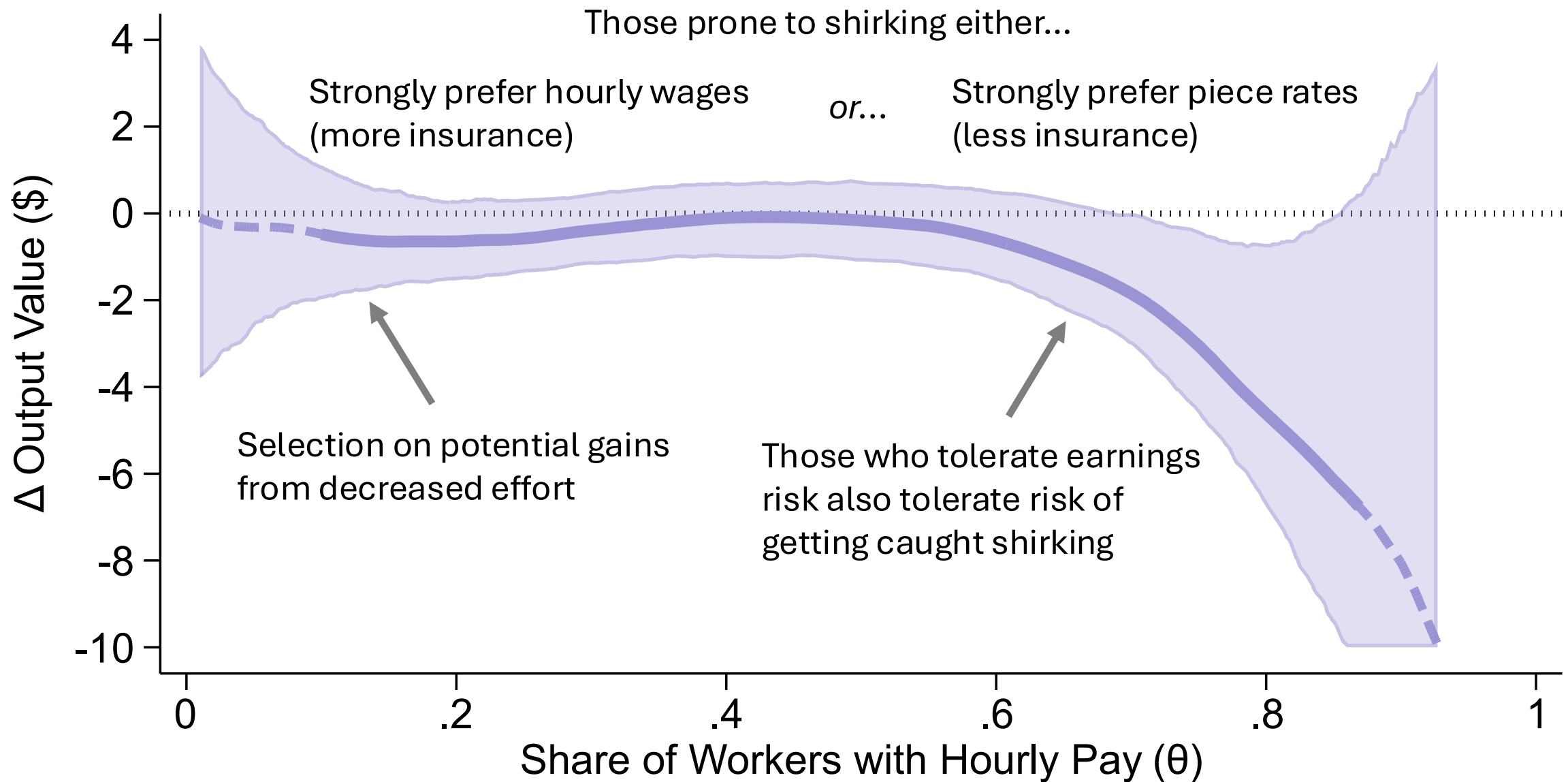
Equilibrium with Adverse Selection and no Moral Hazard



Moral Hazard: Estimates of Marginal Treatment Effects



Moral Hazard: Estimates of Marginal Treatment Effects



Outline

- ① **Experimental Design and Results**
- ② **Model of Asymmetric Information**
- ③ **Estimates of Marginal Value and Welfare Loss**
- ④ **Optimal Fixed-Wage Subsidy and Piece-Rate Tax**
- ⑤ **Extensions**

Policy Solutions

- Welfare loss from adverse selection suggests role for government: promote insurance-like provisions in labor contracts.
- I calculate the **Marginal Value of Public Funds (MVPF)** for two policies
 1. Fixed-wage subsidies
 - e.g., deductions for portion of hourly payments
 2. Piece-rate tax
 - e.g., eliminate tax credits on tipped earnings
- Results may inform broad set of labor policies, even if estimates are specific to data-entry setting

$$MVPF = \frac{Welfare\ Benefit}{Net\ Govt\ Expenditure}$$

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{Welfare\ Benefit}{Net\ Govt\ Expenditure}$$

Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

Net Govt Expenditure: The aggregate amount spent including fiscal externalities

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\textit{Transfer} + \textit{Insurance Value}}{\textit{Net Govt Expenditure}}$$

Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

- *Transfer* from government to hourly workers
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Expenditure: The aggregate amount spent including fiscal externalities

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\text{Transfer} + \text{Insurance Value}}{\text{Transfer} + \text{Tax Loss from MH}}$$

Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

- *Transfer* from government to hourly workers
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Expenditure: The aggregate amount spent including fiscal externalities

- *Transfer* from government to hourly workers
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\delta\theta^\delta + \int_{\theta^{EQ}}^{\theta^\delta} (MV_1(\theta) - \bar{w}(\theta))d\theta}{\delta\theta^\delta - \int_{\theta^{EQ}}^{\theta^\delta} \tau MH(\theta)d\theta}$$

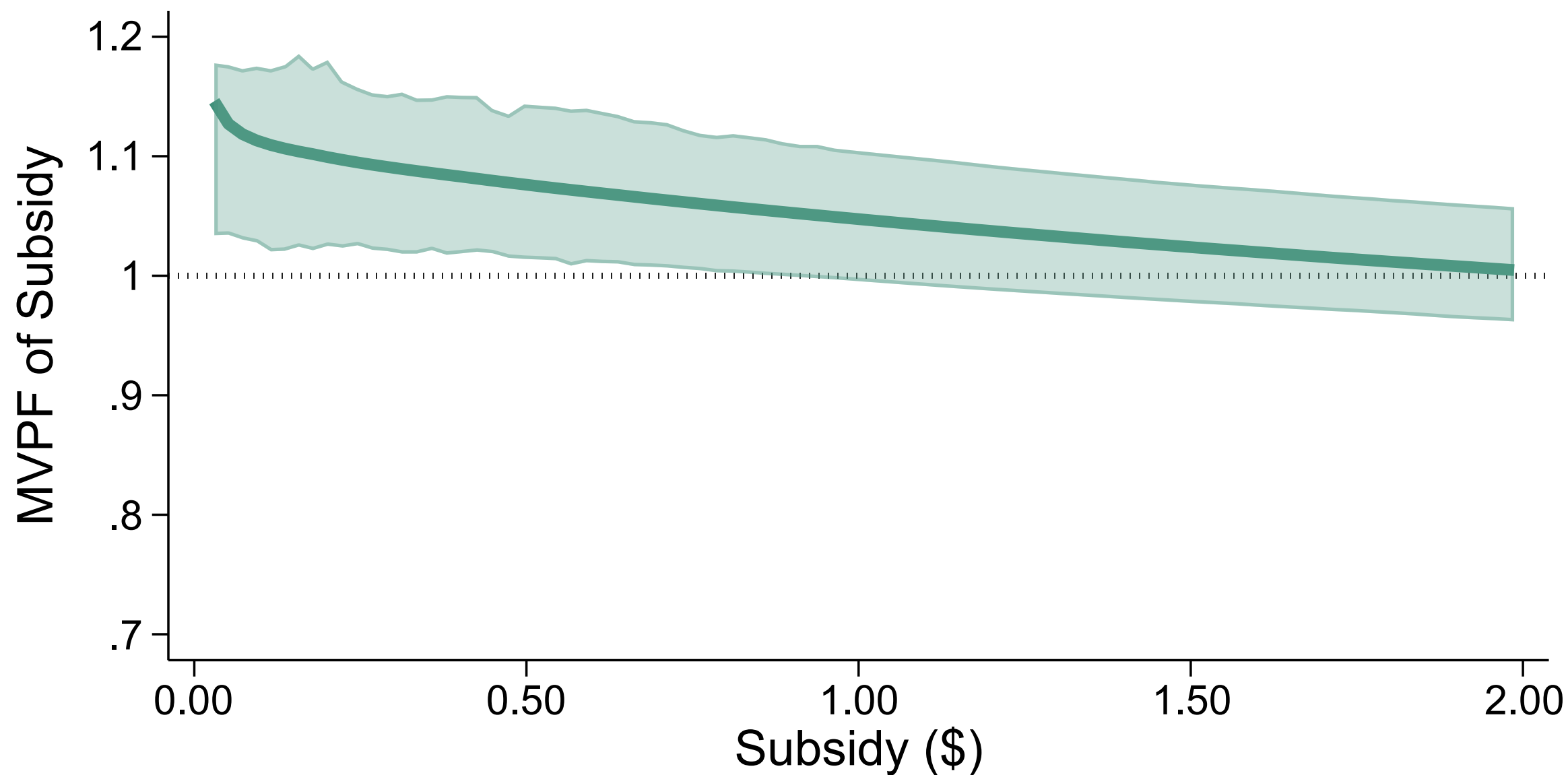
Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

- *Transfer* from government to hourly workers
- *Insurance Value* to workers who move from piece-rate to hourly pay

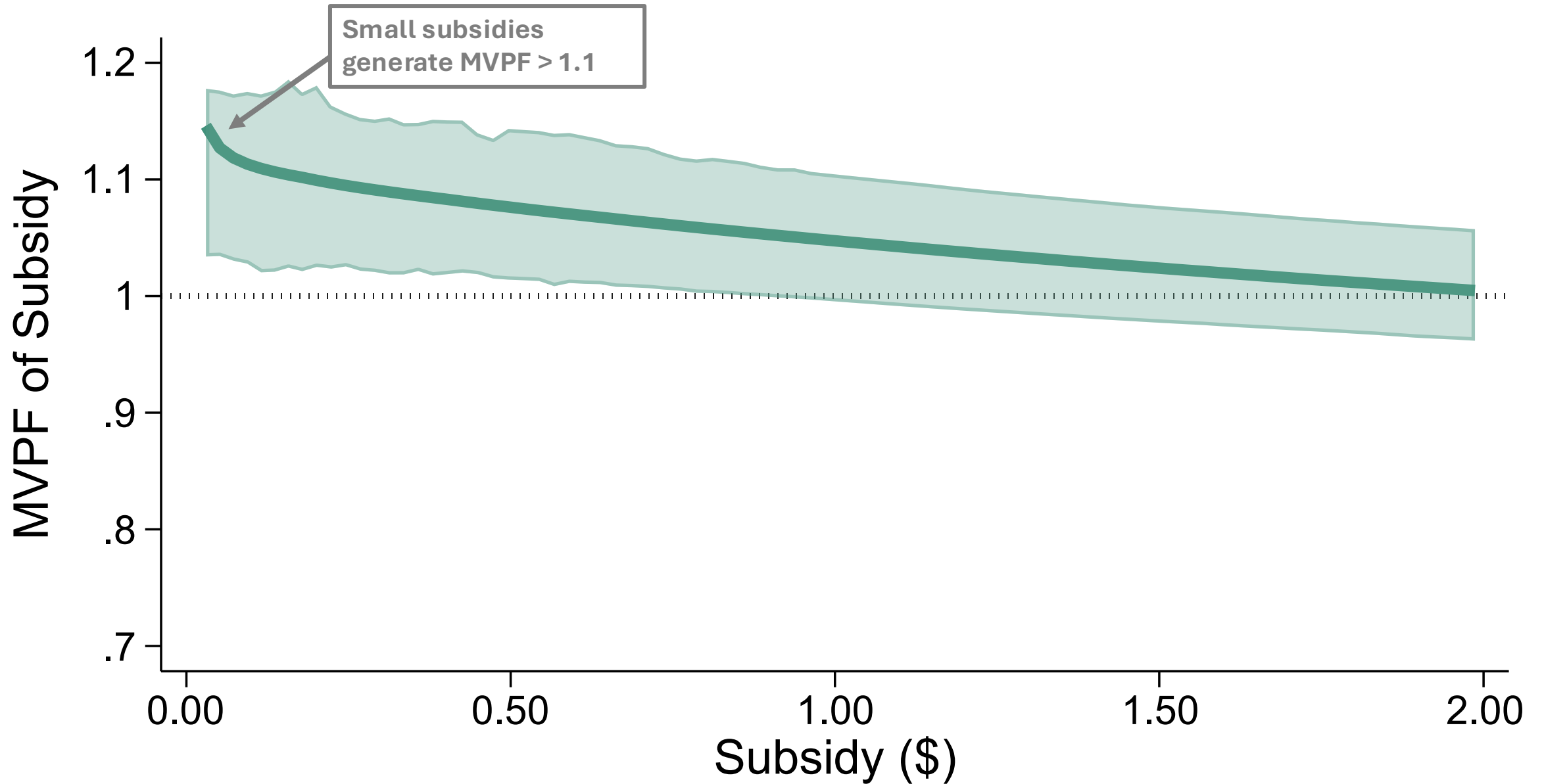
Net Govt Expenditure: The aggregate amount spent including fiscal externalities

- *Transfer* from government to hourly workers
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

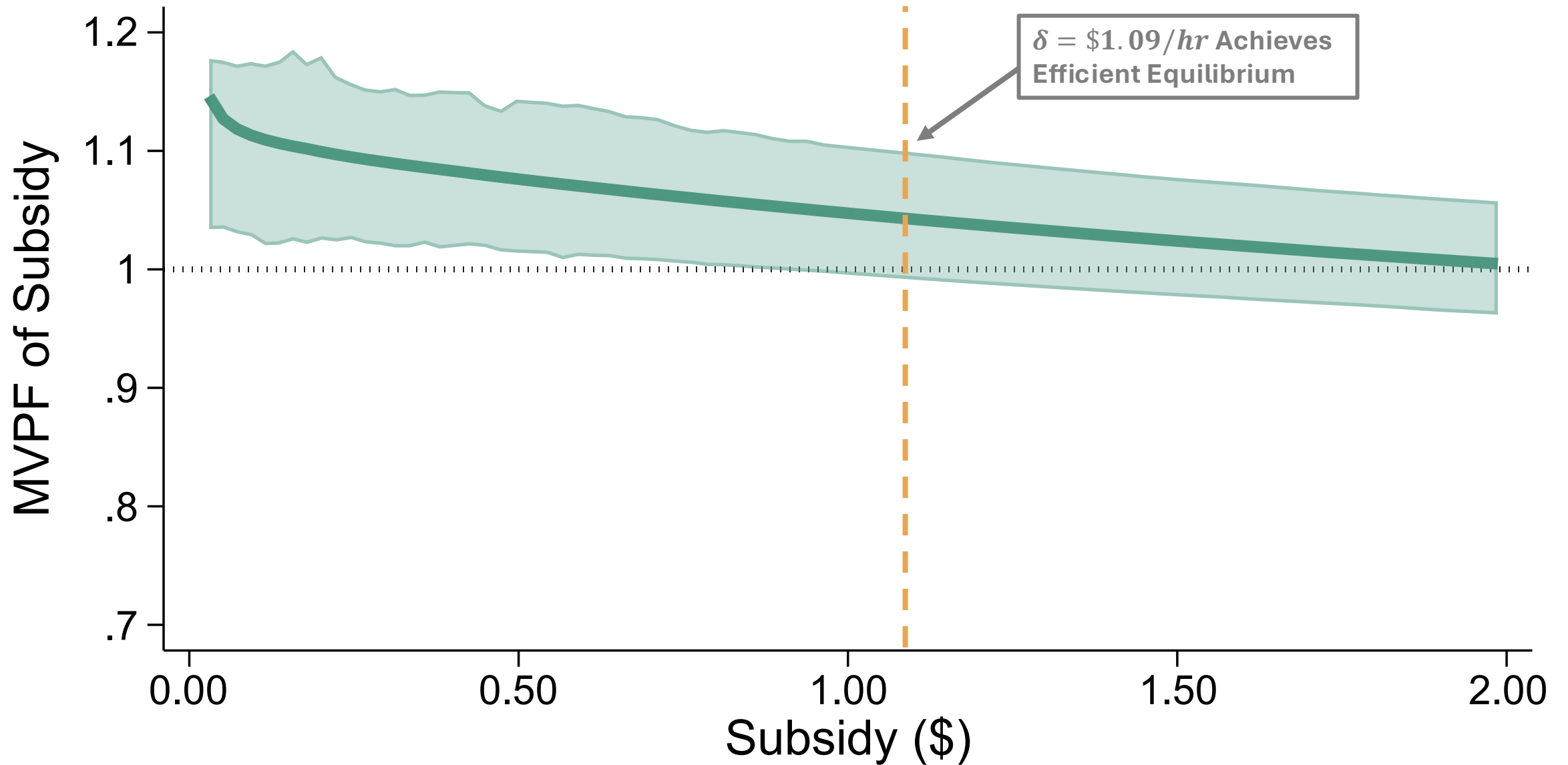
MVPF of Hourly Wage Subsidy



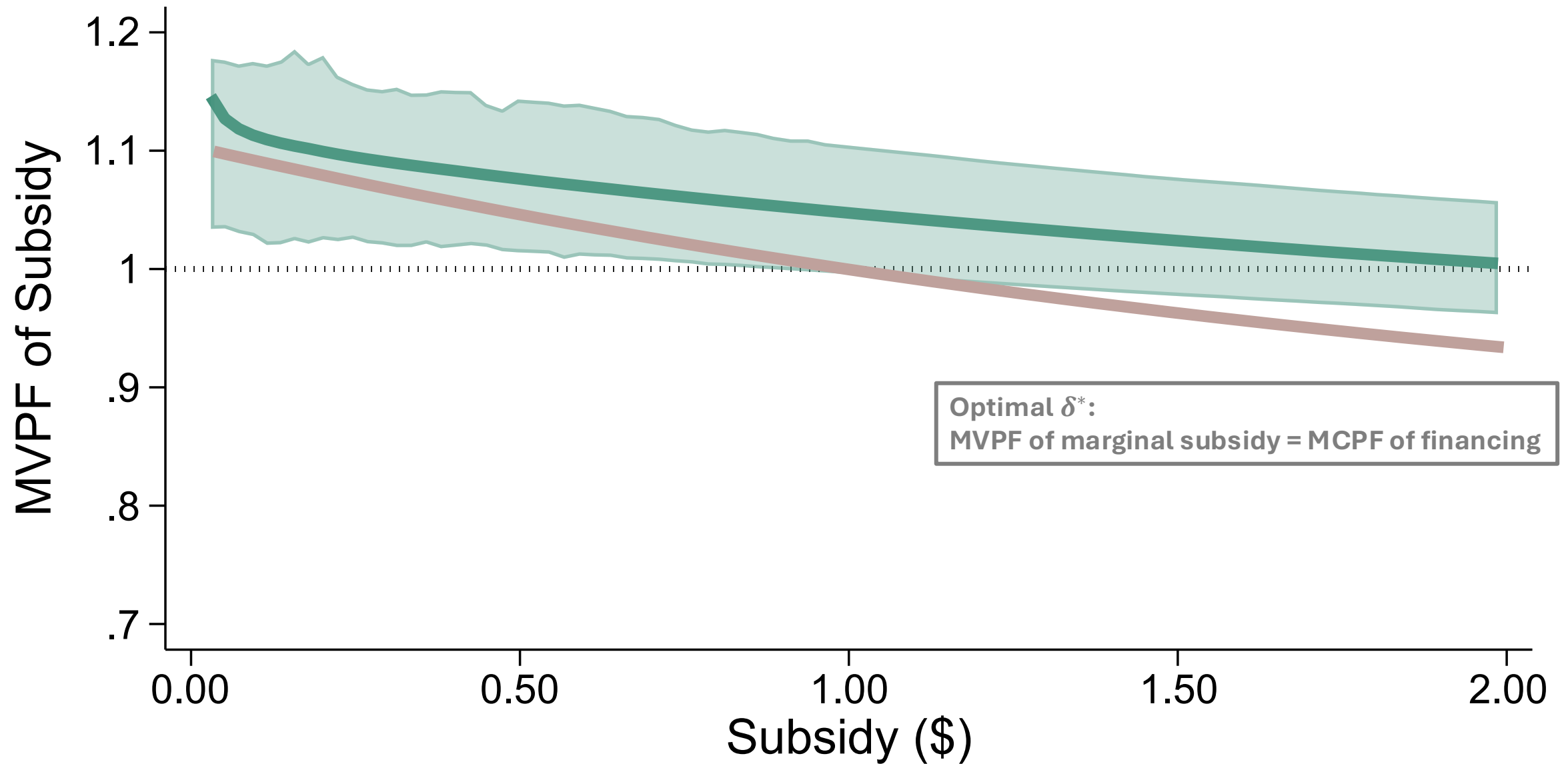
MVPF of Hourly Wage Subsidy



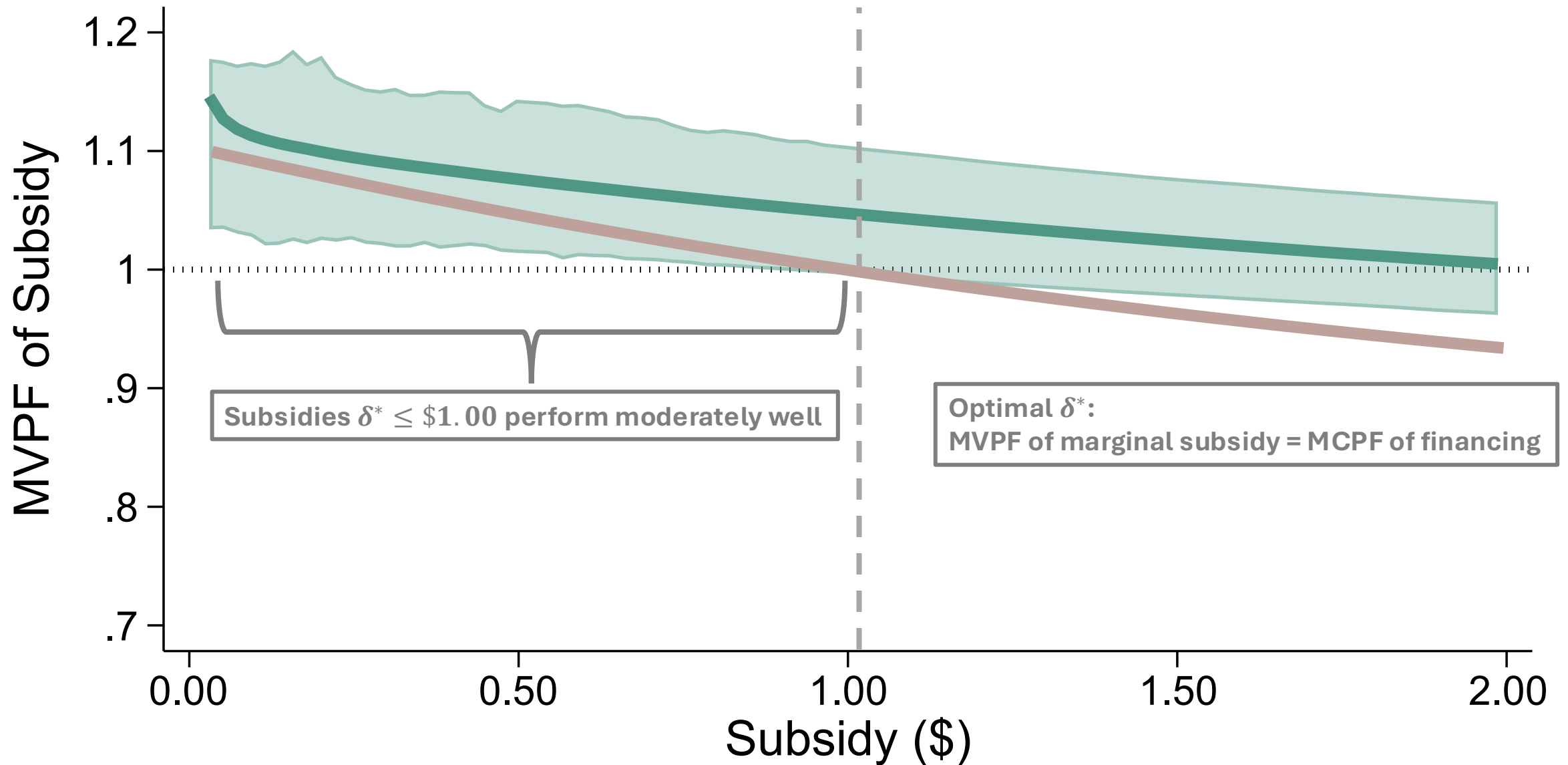
MVPF of Hourly Wage Subsidy



MVPF of Hourly Wage Subsidy



MVPF of Hourly Wage Subsidy



Policy #2: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{Welfare\ Loss}{Net\ Govt\ Revenue}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

Policy #2: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{Welfare\ Loss}{Net\ Govt\ Revenue}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

Socially efficient tax $\Rightarrow$ **lower** MCPF

Policy #2: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{\textit{Transfer} - \textit{Insurance Value}}{\textit{Net Govt Revenue}}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

- *Transfer* from piece-rate workers to government
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

Socially efficient tax $\Rightarrow$ **lower** MCPF

Policy #2: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{\text{Transfer} - \text{Insurance Value}}{\text{Transfer} - \text{Tax Loss from MH}}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

- *Transfer* from piece-rate workers to government
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

- *Transfer* from piece-rate workers to government
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

Socially efficient tax $\Rightarrow$ **lower** MCPF

Policy #2: Piece-rate Tax

Marginal Cost of Public Funds (MCPF) for a piece-rate tax:

$$MCPF_{tax} = \frac{\int_{\theta^P}^1 \rho MV_0(\theta) d\theta - \int_{\theta^{EQ}}^{\theta^P} (MV_1(\theta) - \bar{w}(\theta)) d\theta}{\int_{\theta^P}^1 \rho MV_0(\theta) d\theta + \int_{\theta^{EQ}}^{\theta^P} \tau MH(\theta) d\theta}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

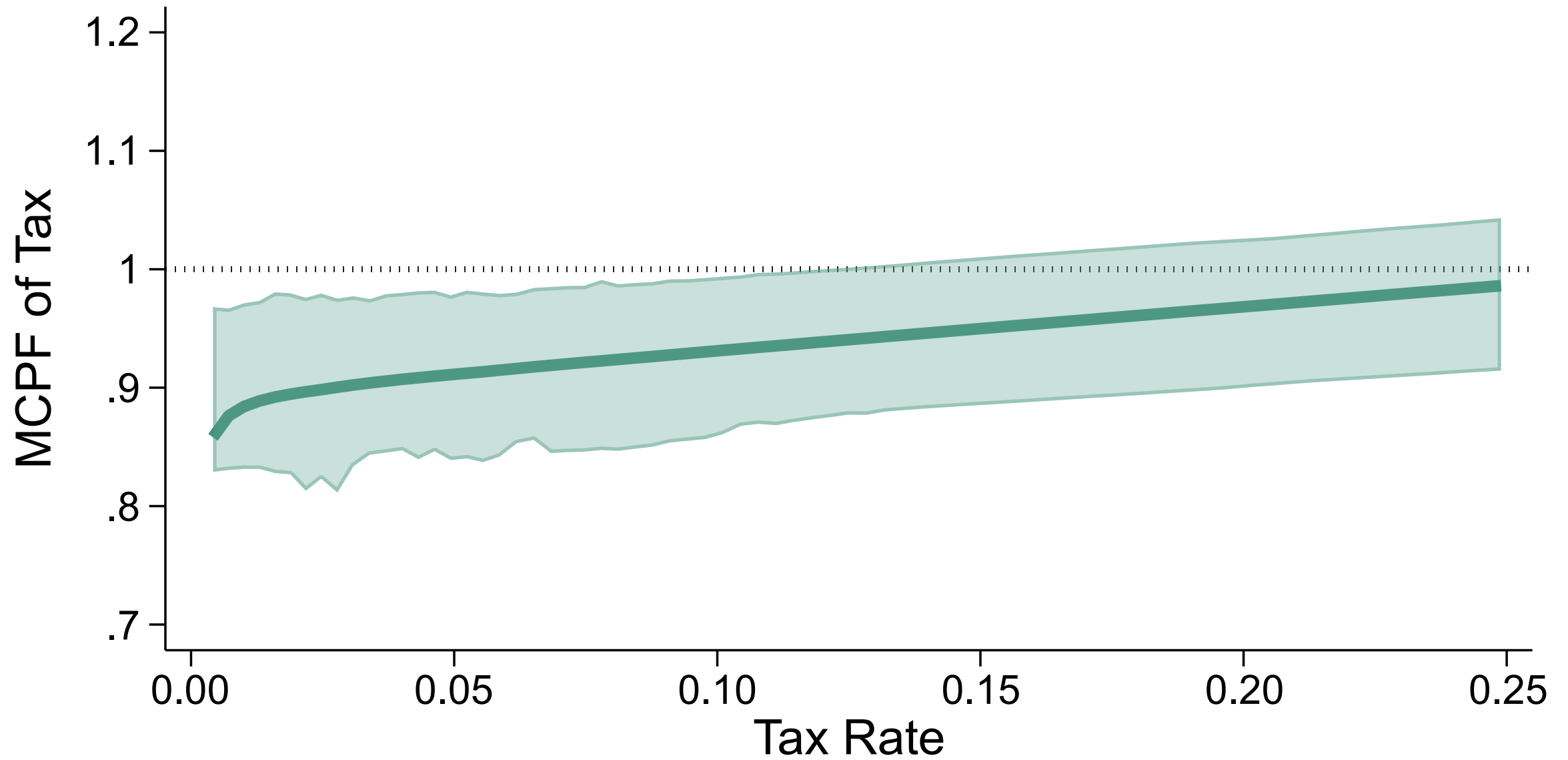
- *Transfer* from piece-rate workers to government
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

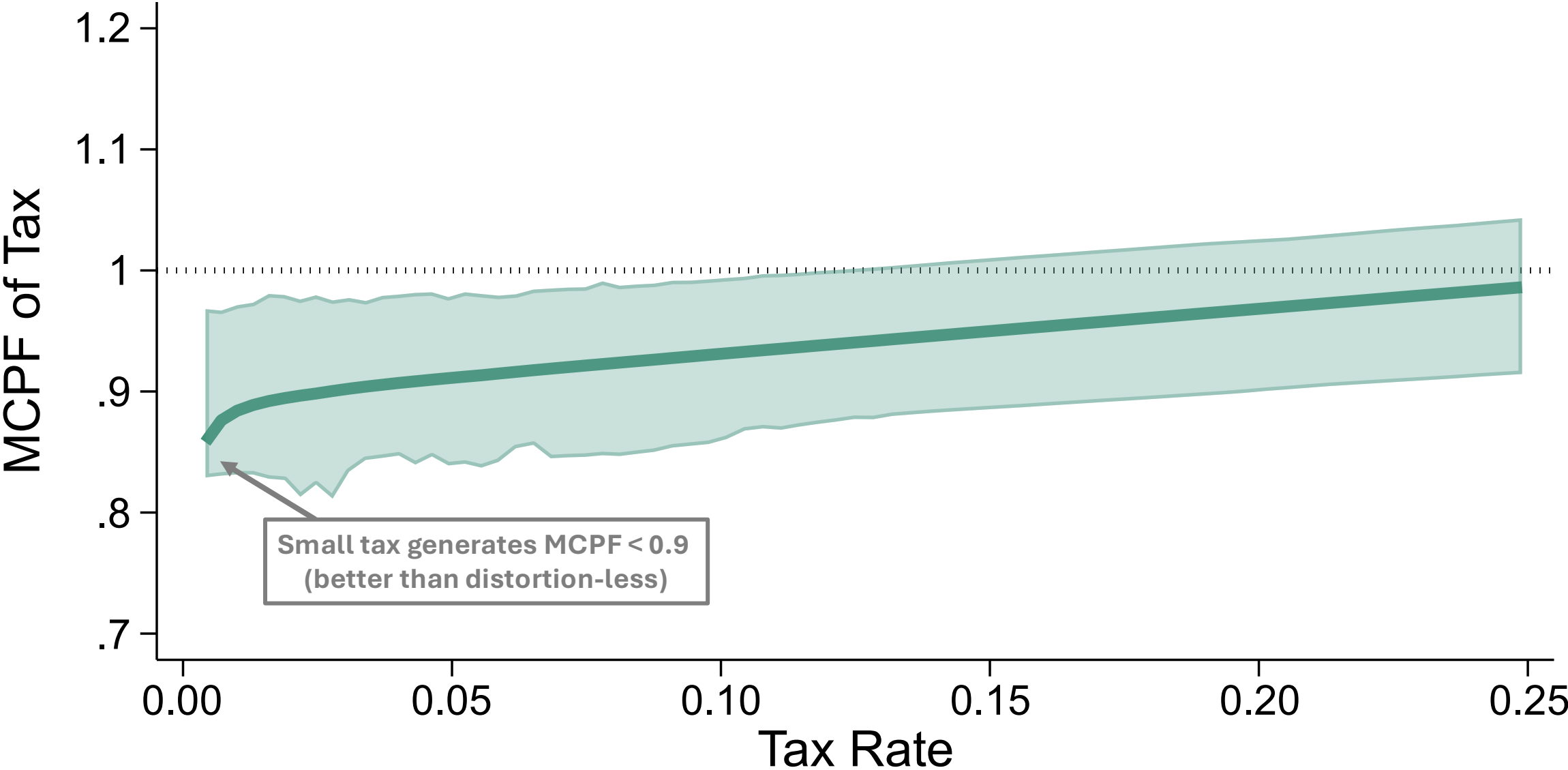
- *Transfer* from piece-rate workers to government
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

Socially efficient tax $\Rightarrow$ **lower** MCPF

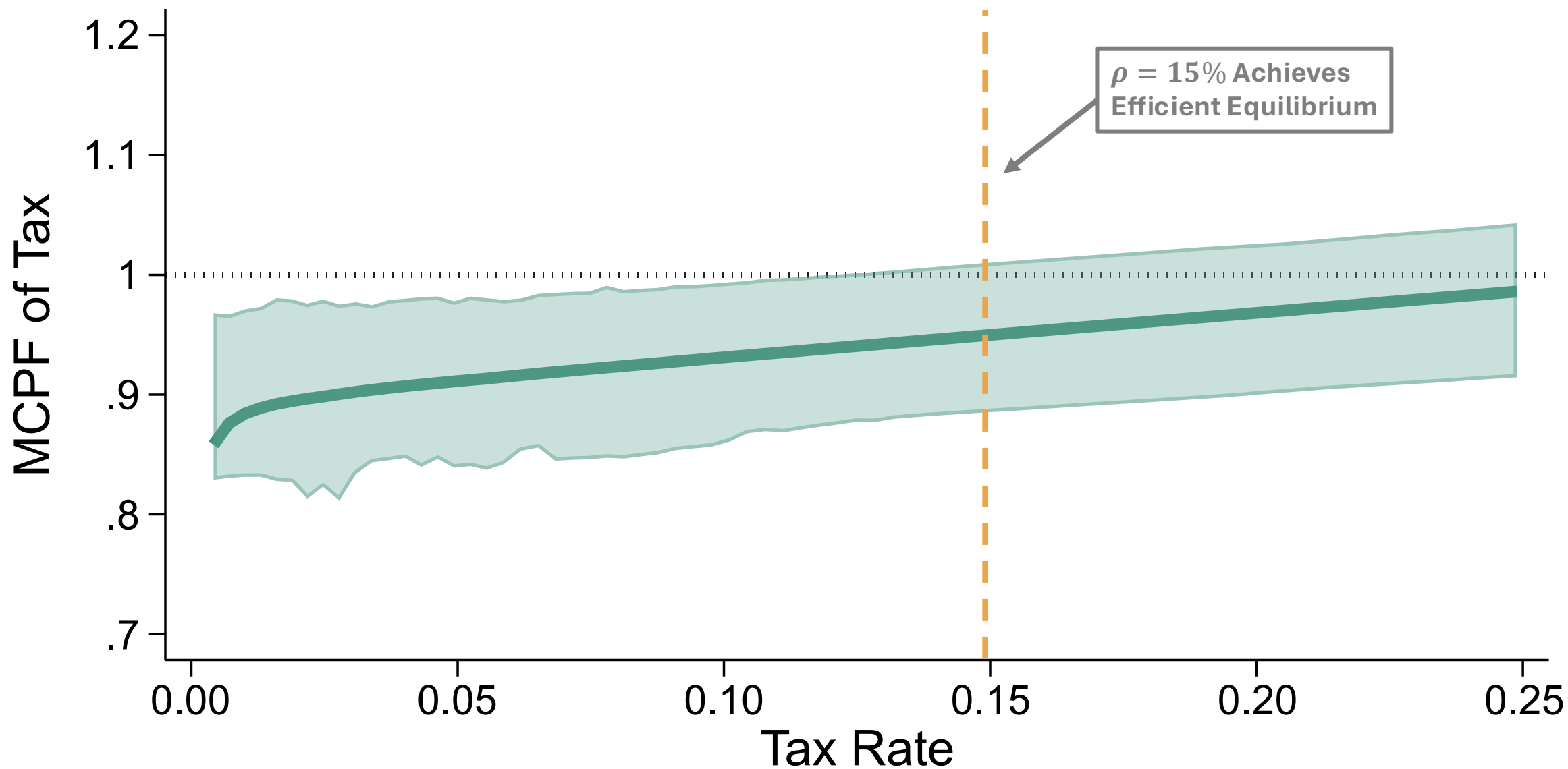
MCPF of Piece Rate Tax



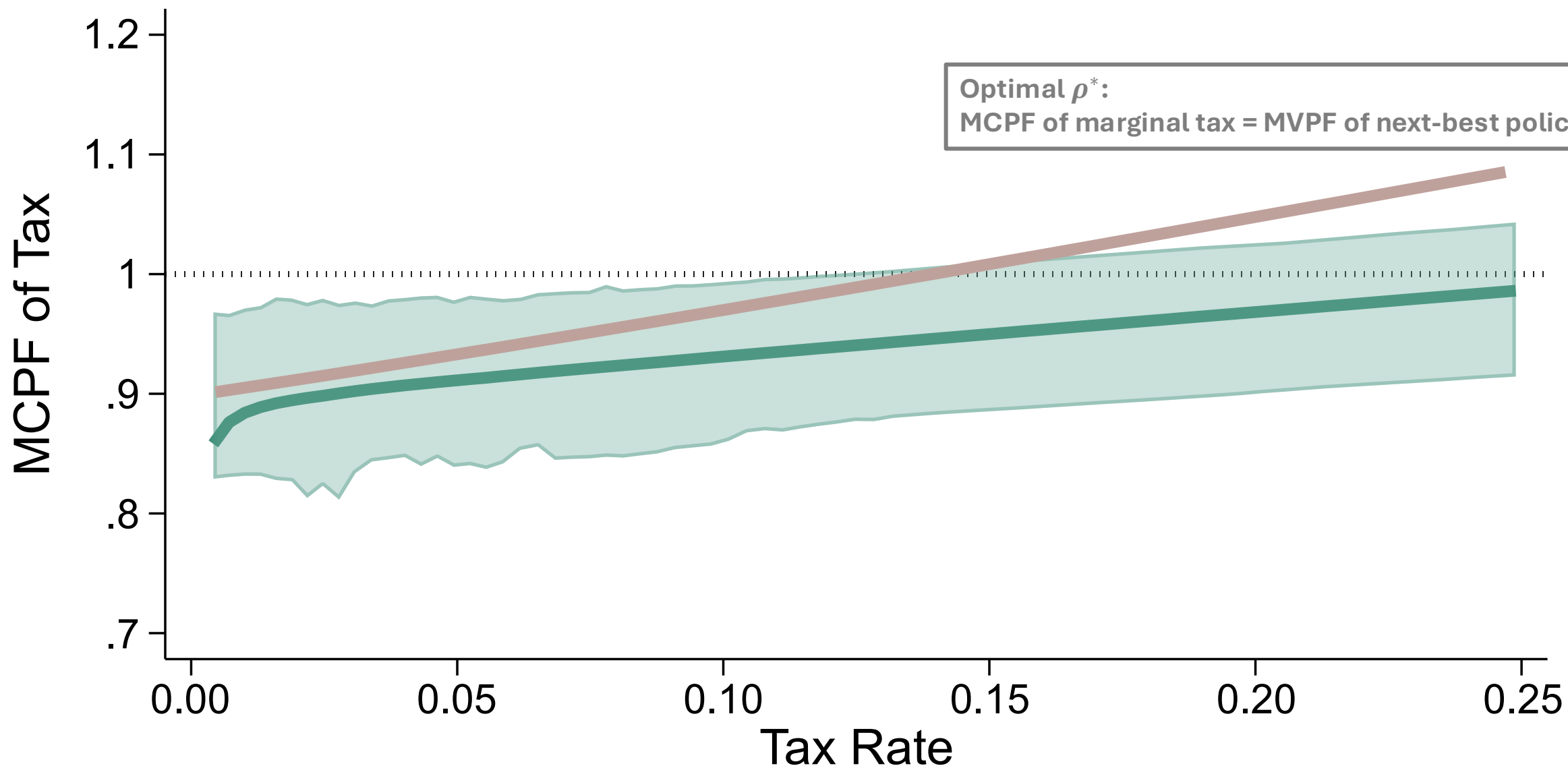
MCPF of Piece Rate Tax



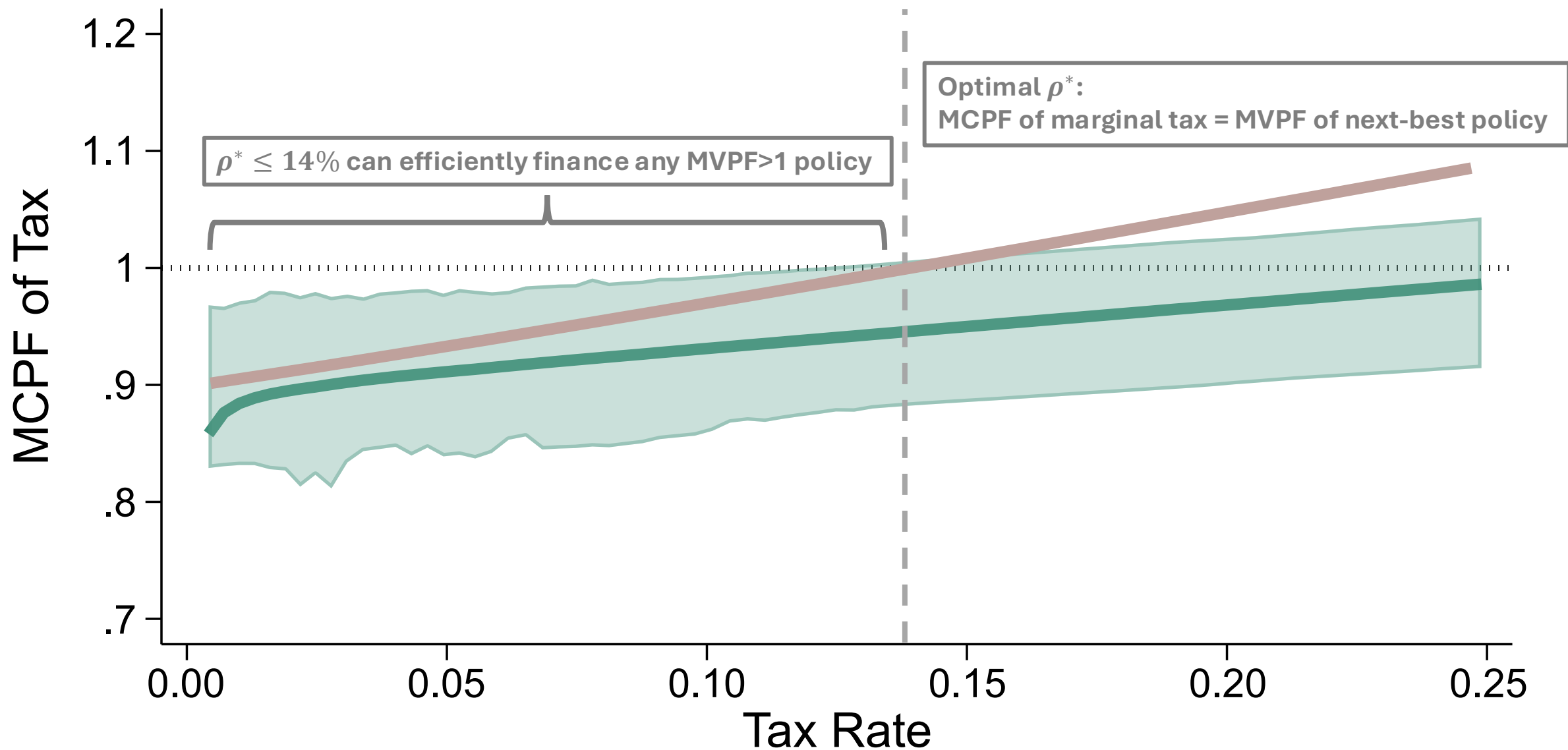
MCPF of Piece Rate Tax



MCPF of Piece Rate Tax



MCPF of Piece Rate Tax



Outline

- 1 Experimental Design and Results
- 2 Model of Asymmetric Information
- 3 Estimates of Marginal Value and Welfare Loss
- 4 Optimal Fixed-Wage Subsidy and Piece-Rate Tax
- 5 Extensions

Generalizing to Other Labor Markets

- Results are specific to online workers performing a data-entry task.
- Division of labor $\Rightarrow$ applied research on worker incentives usually comes from highly specialized settings
 - Rideshare drivers (Angrist et al., 2021; Cook et al., 2021)
 - Agricultural workers (Brune et al., 2022; Bandiera et al., 2010)
 - Cashiers (Mas and Moretti 2009)
 - Windshield repair (Lazear 2000)
 - Call centers (Mas and Pallais 2017; Nagin et al. 2002)
- Even so, I try to make the experimental task as generalizable as possible
 - 66% of US workers type on the job
 - Workers don't know it's an experiment (debriefed afterwards)
- Methods can be applied elsewhere, especially short-term labor markets

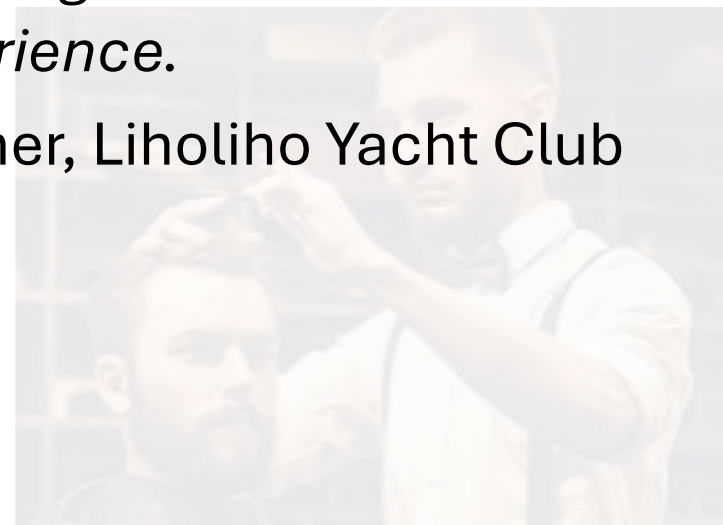
Extensions: Why do these jobs have unpredictable pay?

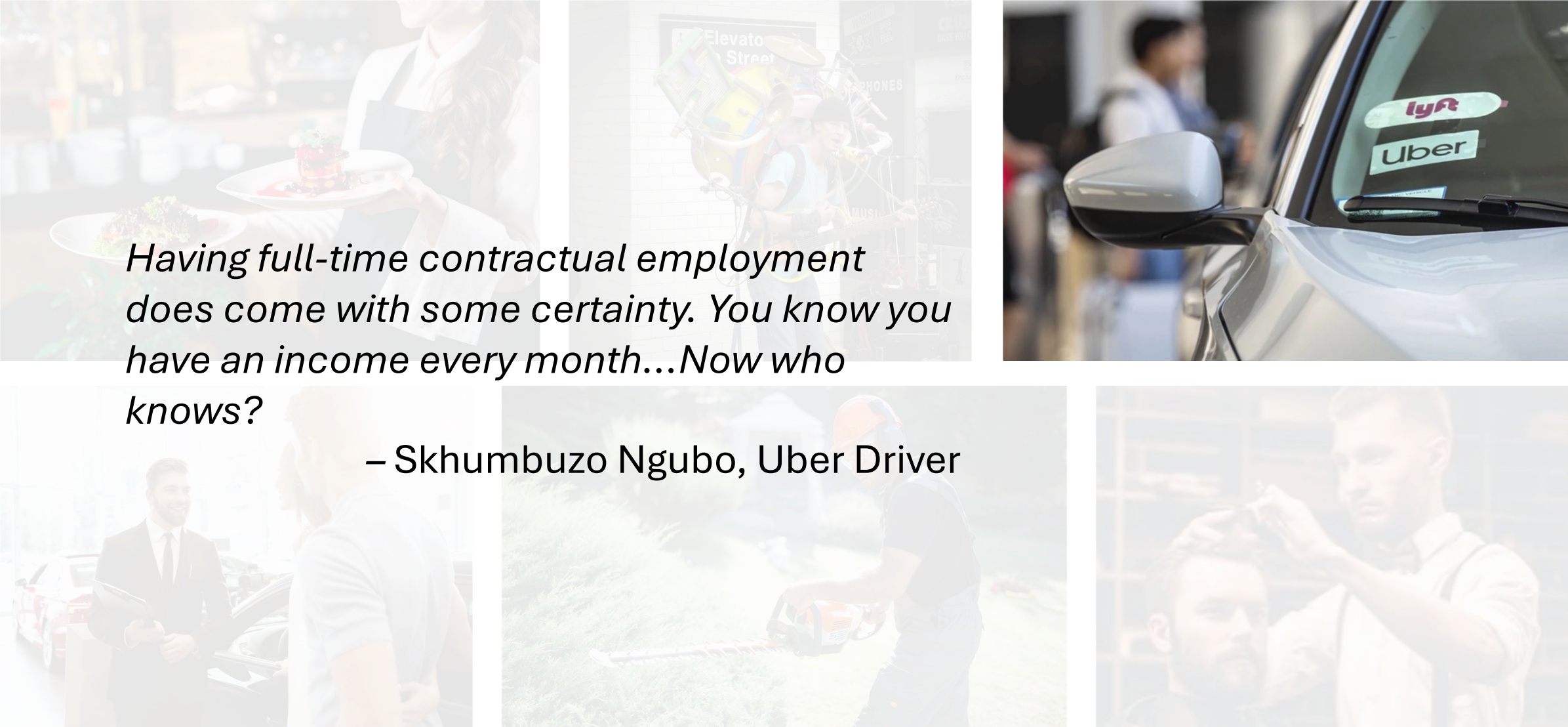




[After eliminating tips, server positions] have been harder to fill. Many veteran servers weren't interested, saying they could make double elsewhere with tips...As a result, many of the people who work in the dining room started with little to no restaurant experience.

—Owner, Liholiho Yacht Club





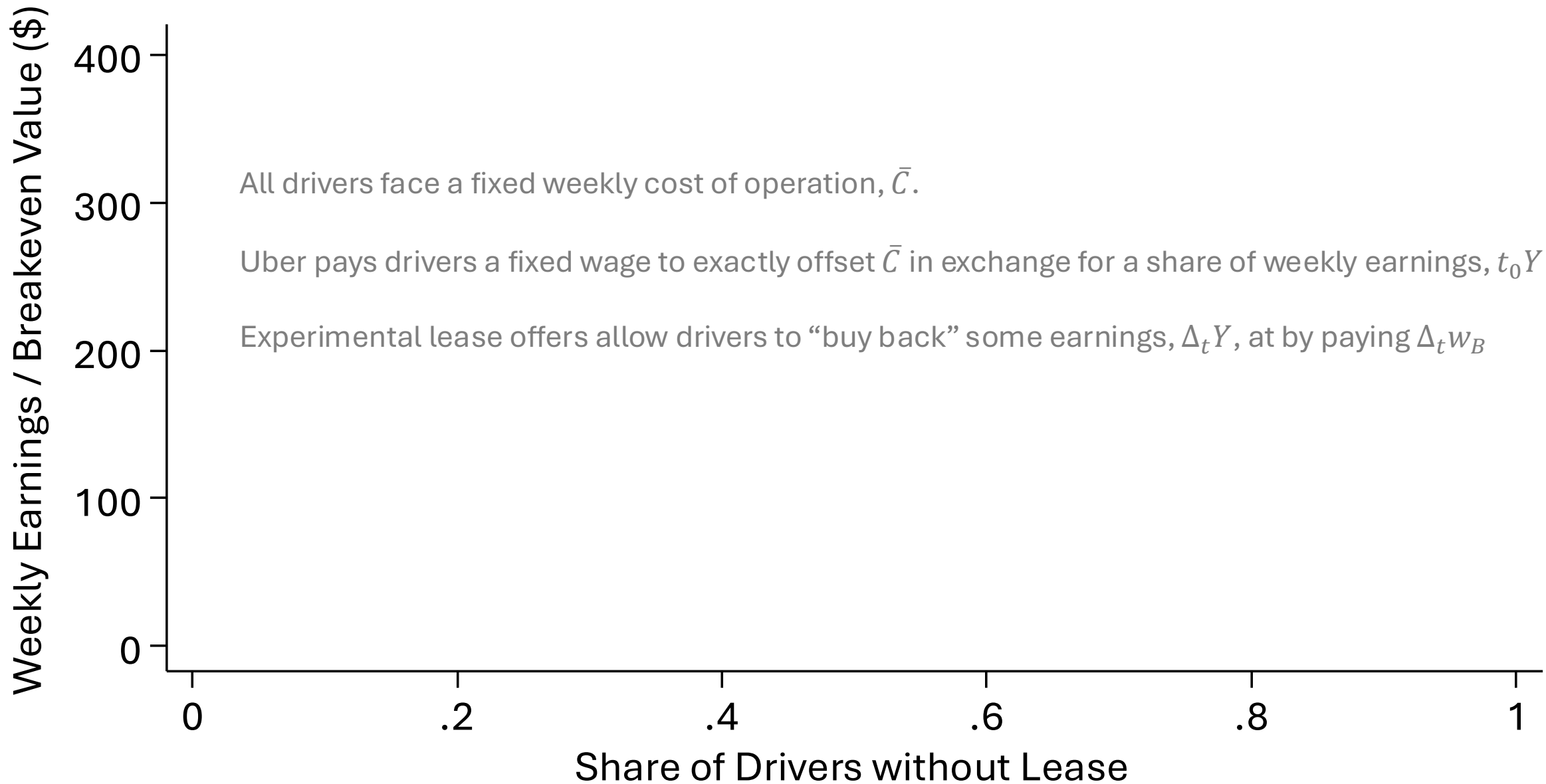
*Having full-time contractual employment
does come with some certainty. You know you
have an income every month...Now who
knows?*

– Skhumbuzo Ngubo, Uber Driver

Application to Rideshare Drivers (Angrist et al., 2021)

- RCT offers drivers larger commissions on ride fares in exchange for fixed “taxi-style” lease payments
 - ⇒ Driver can “buy back” some labor product
- Fixed-wage option $\approx$ “lease-free” driving (i.e., *decline* taxi offer)
- Evidence of moral hazard and adverse selection
 - MH: lease take-up increases labor supply (ISE = 1.2)
 - AS: more productive drivers have higher WTP for leases ($\Delta_{Y_0} = \$21$)
- AS+MH $\Rightarrow$ over-reliance on commissions? Welfare implications?
 - Calibrate my model using estimates from Angrist et al. (2021)

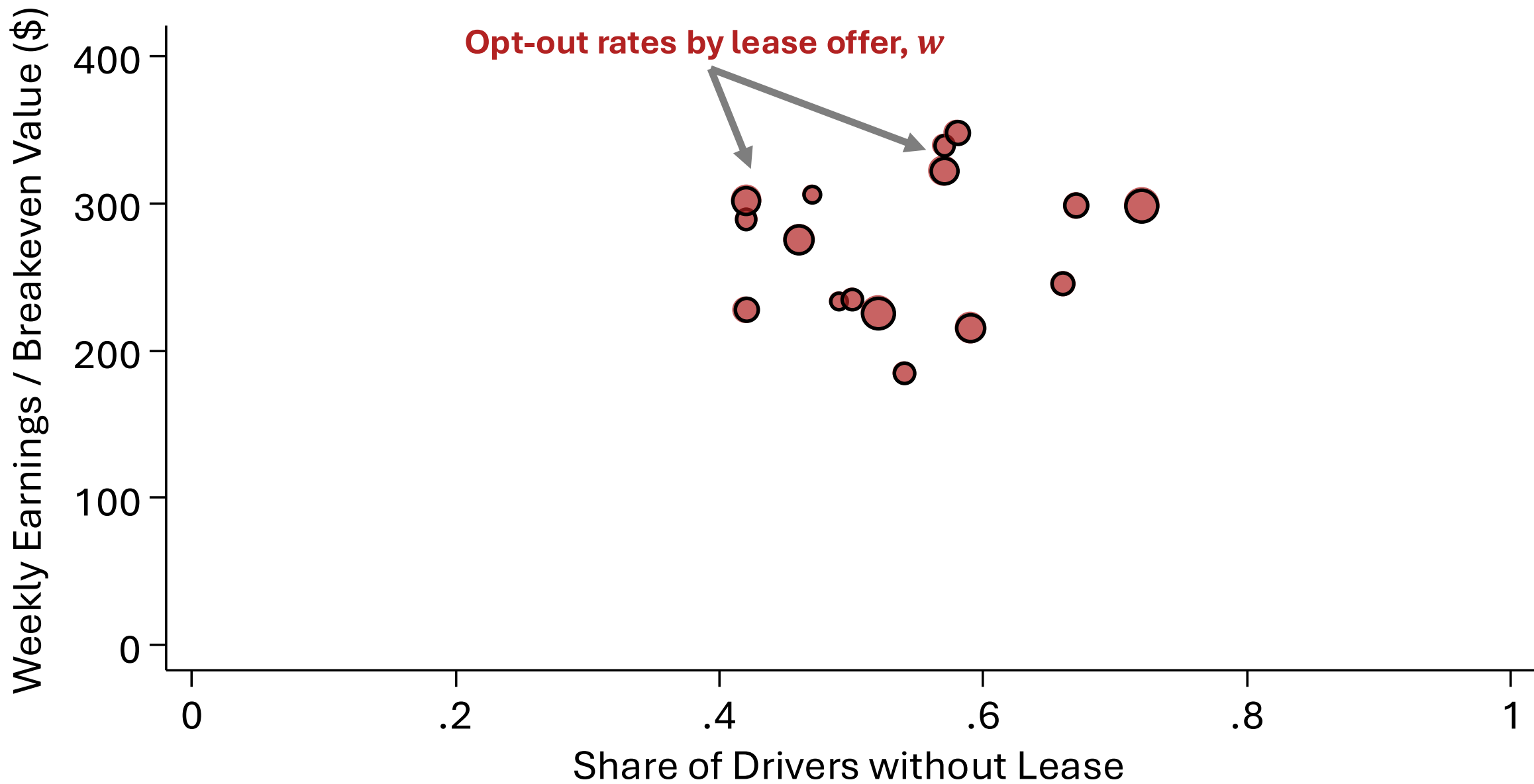




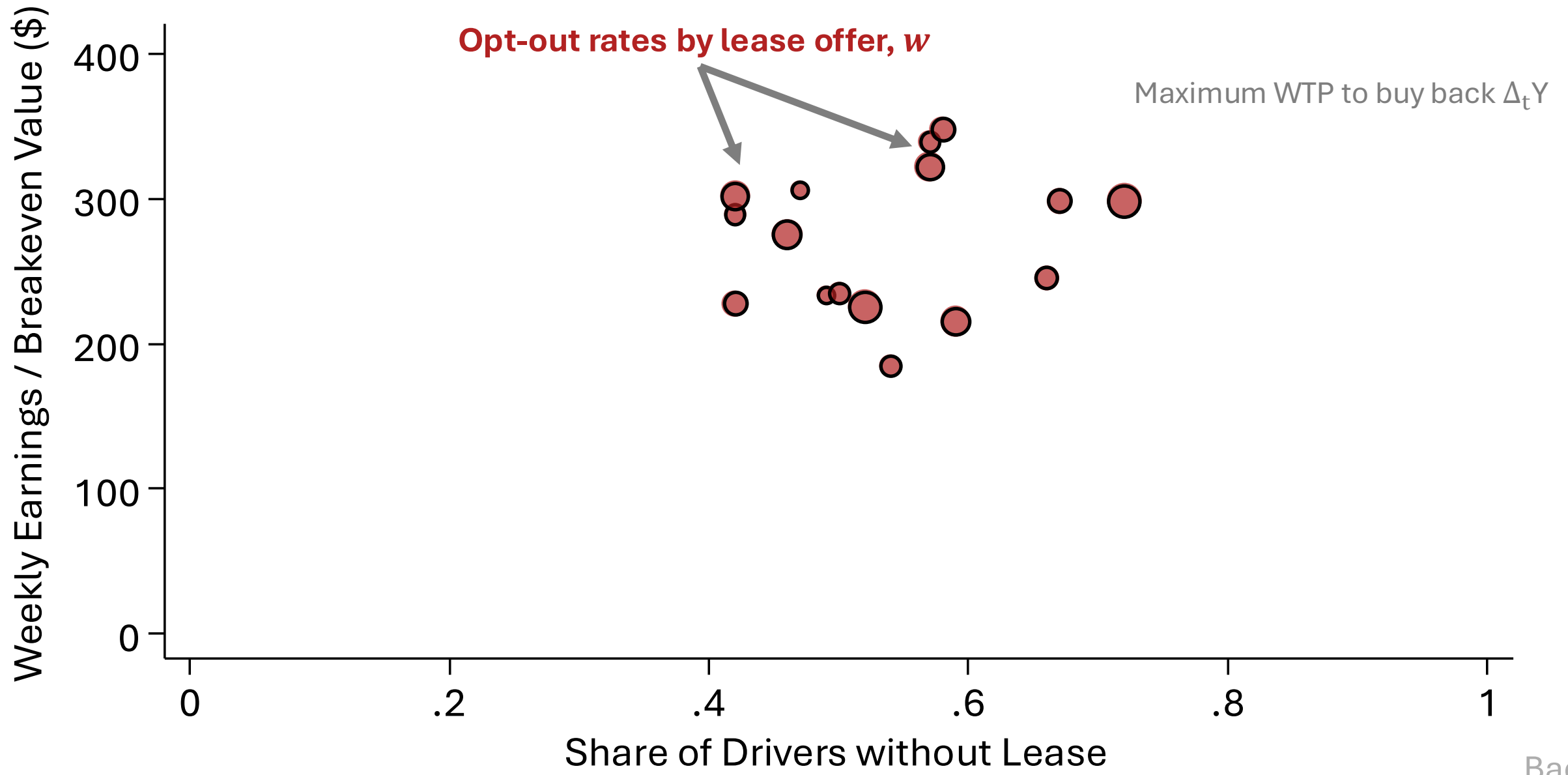




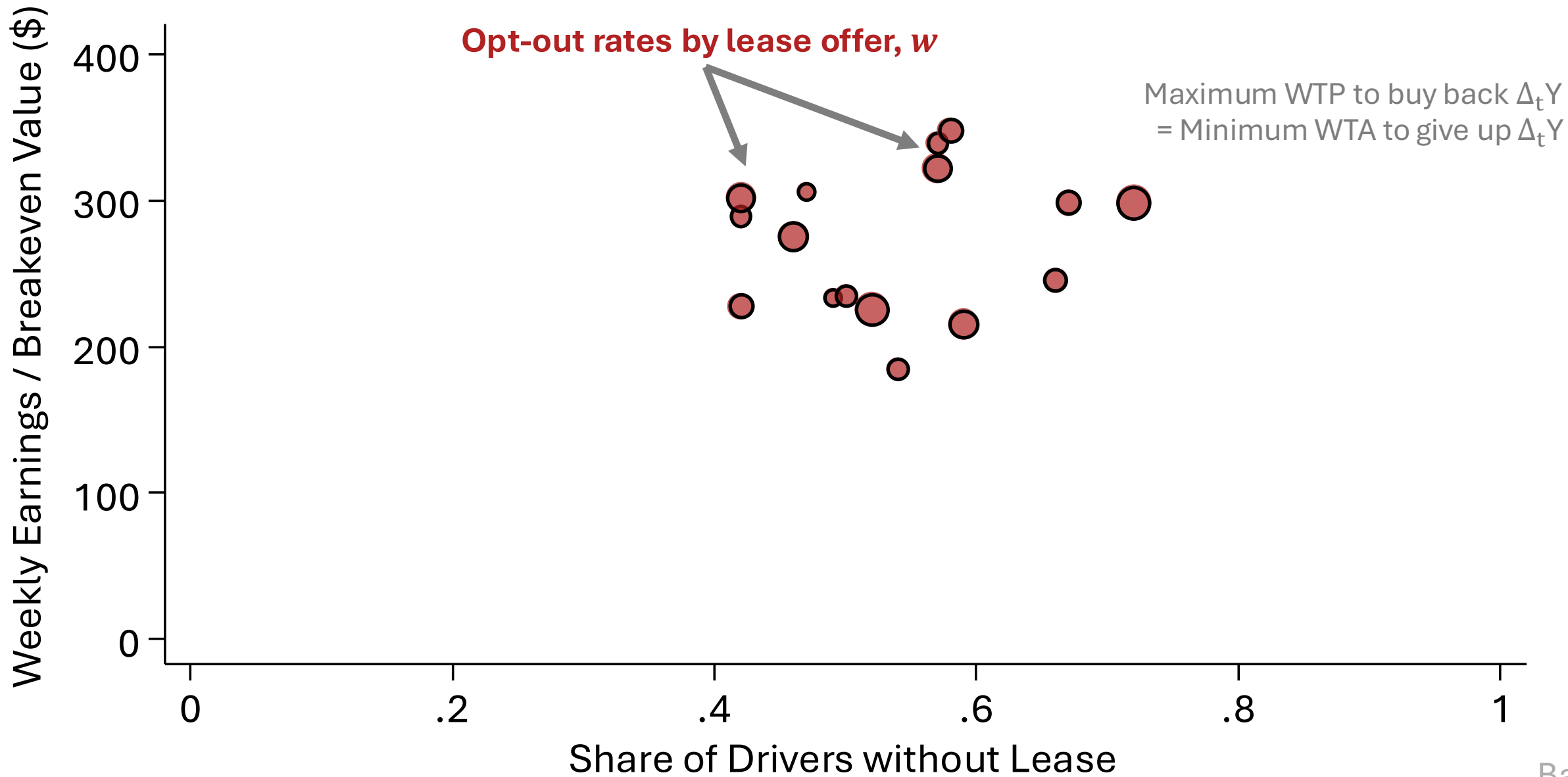
Opt-out rates by lease offer, w

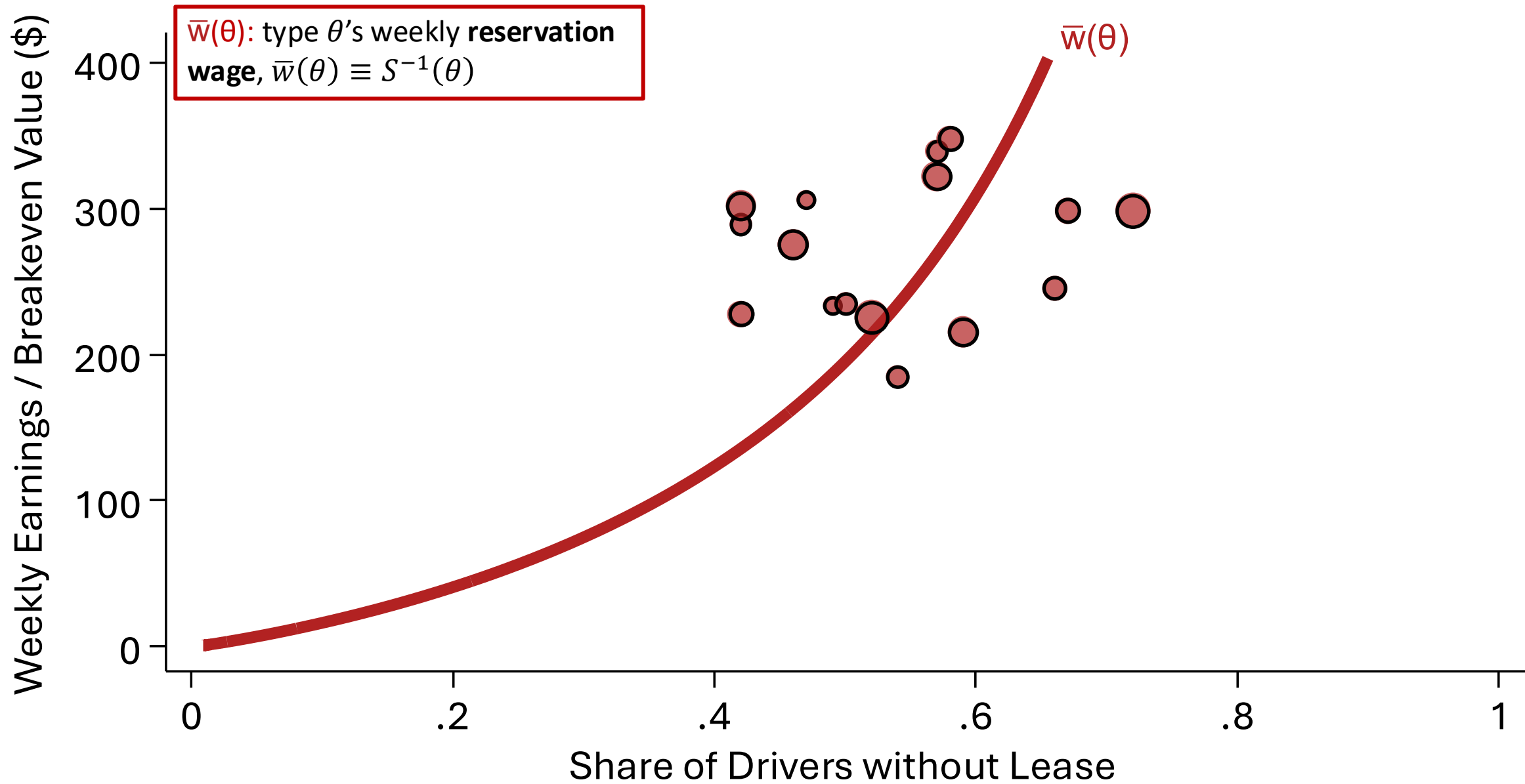


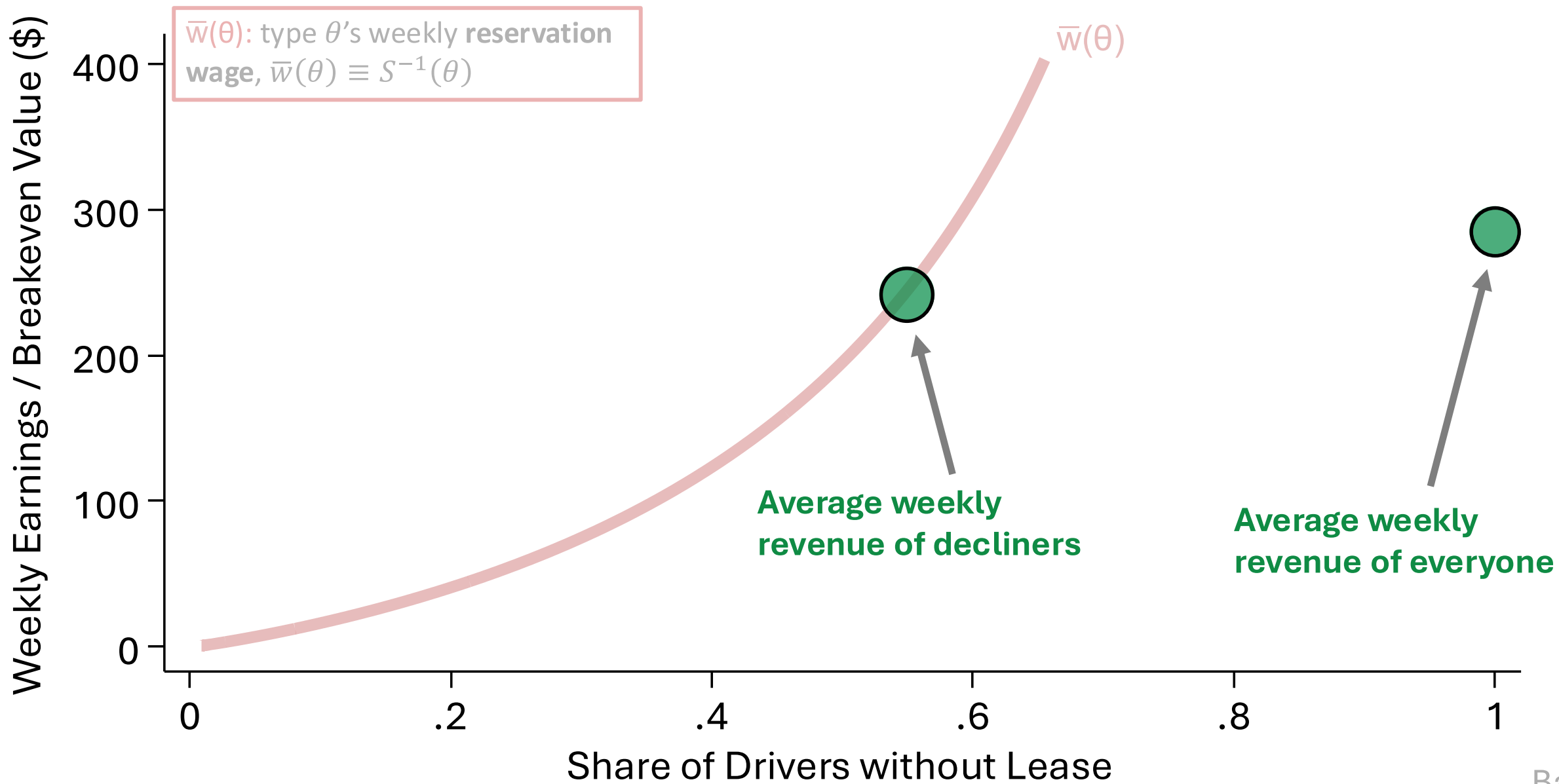
Opt-out rates by lease offer, w

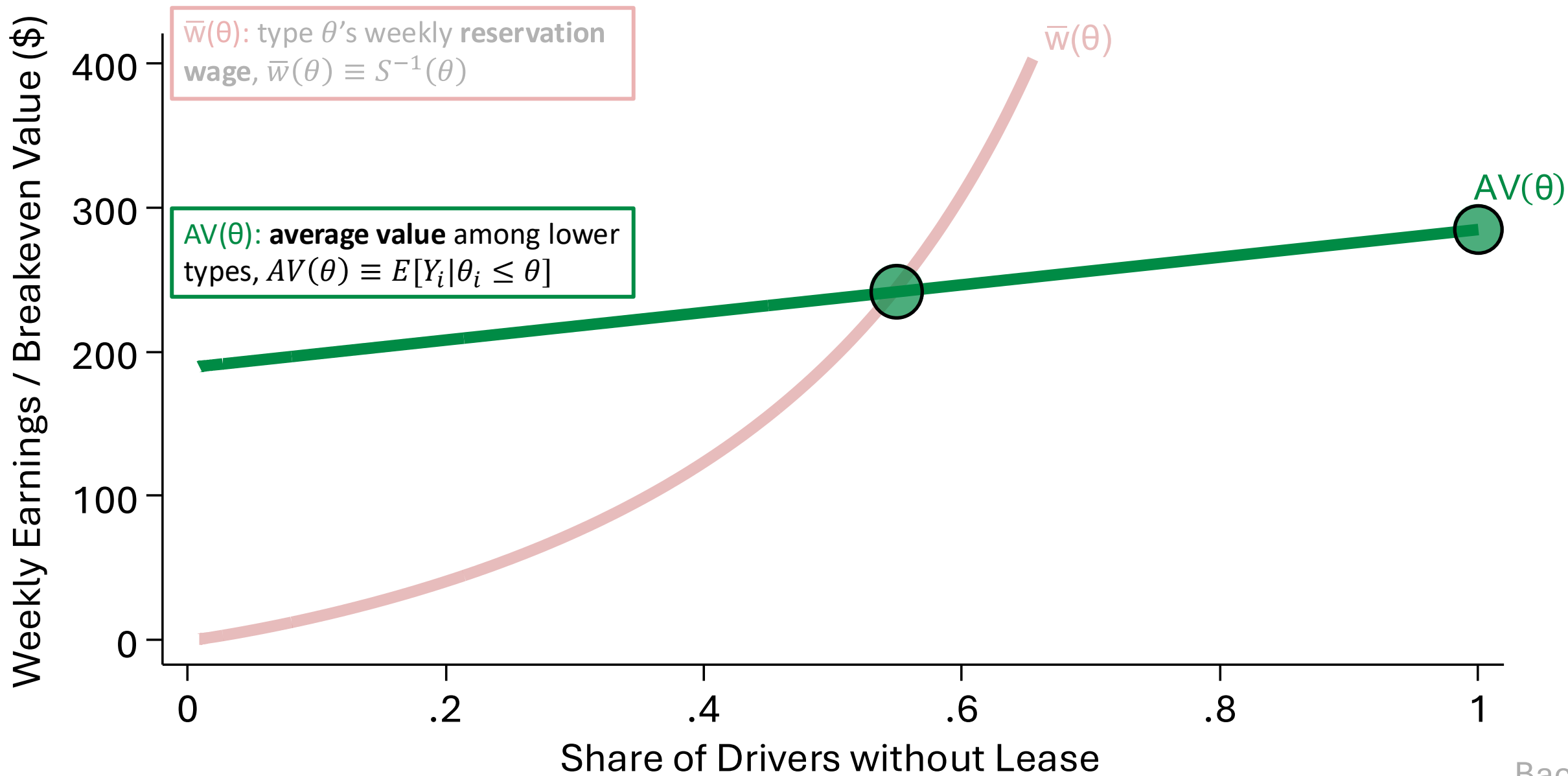


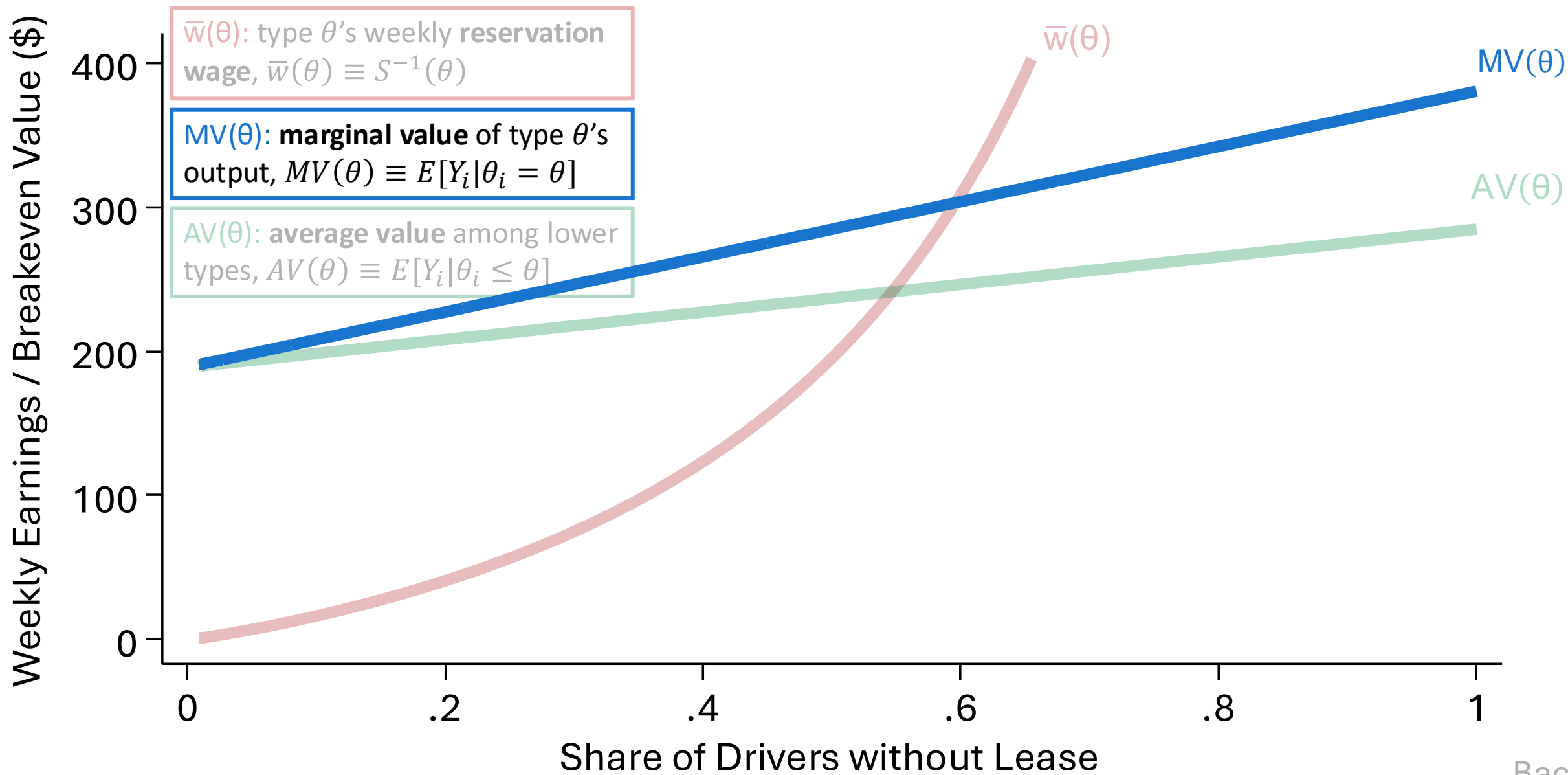
Opt-out rates by lease offer, w

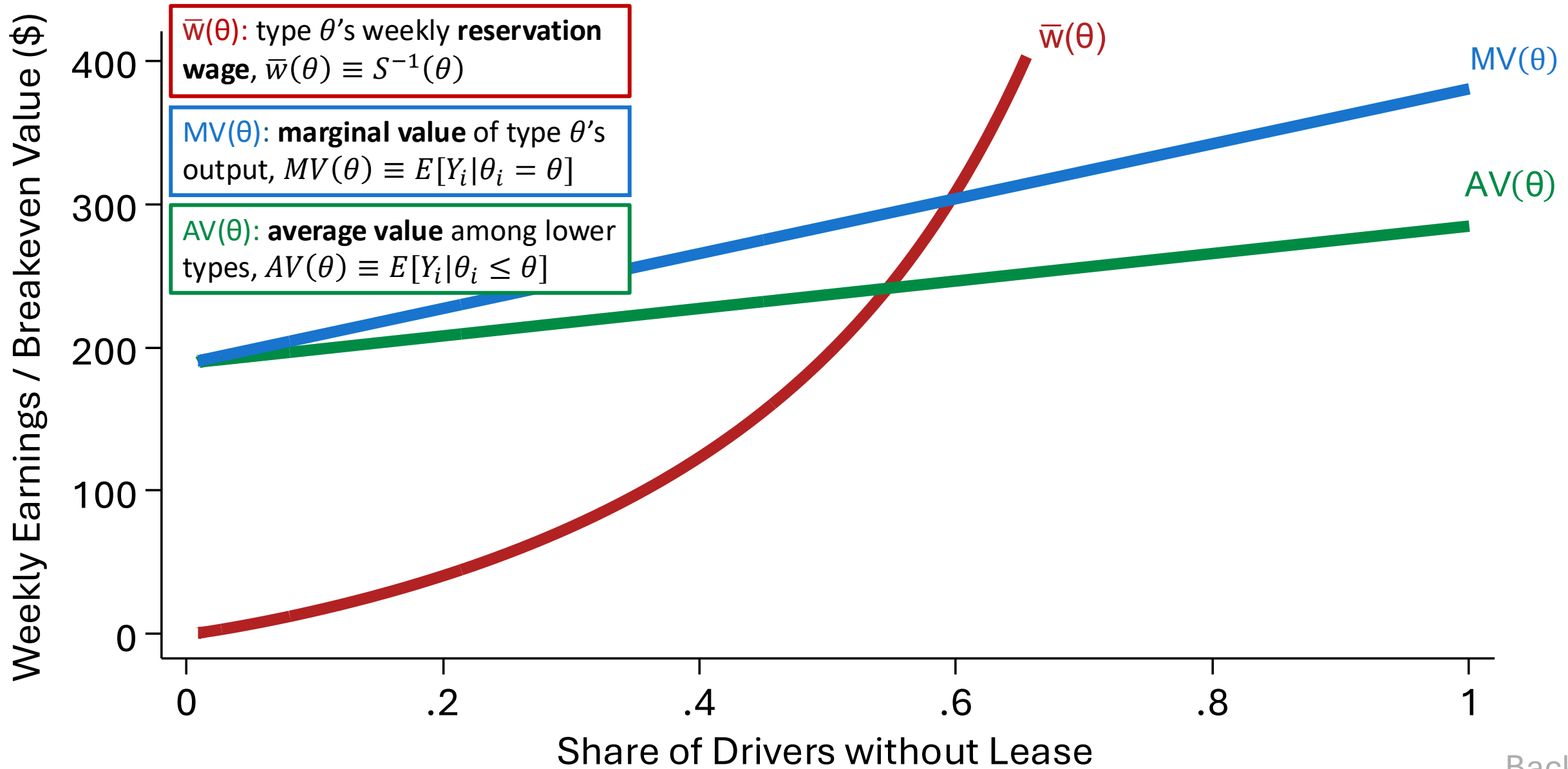


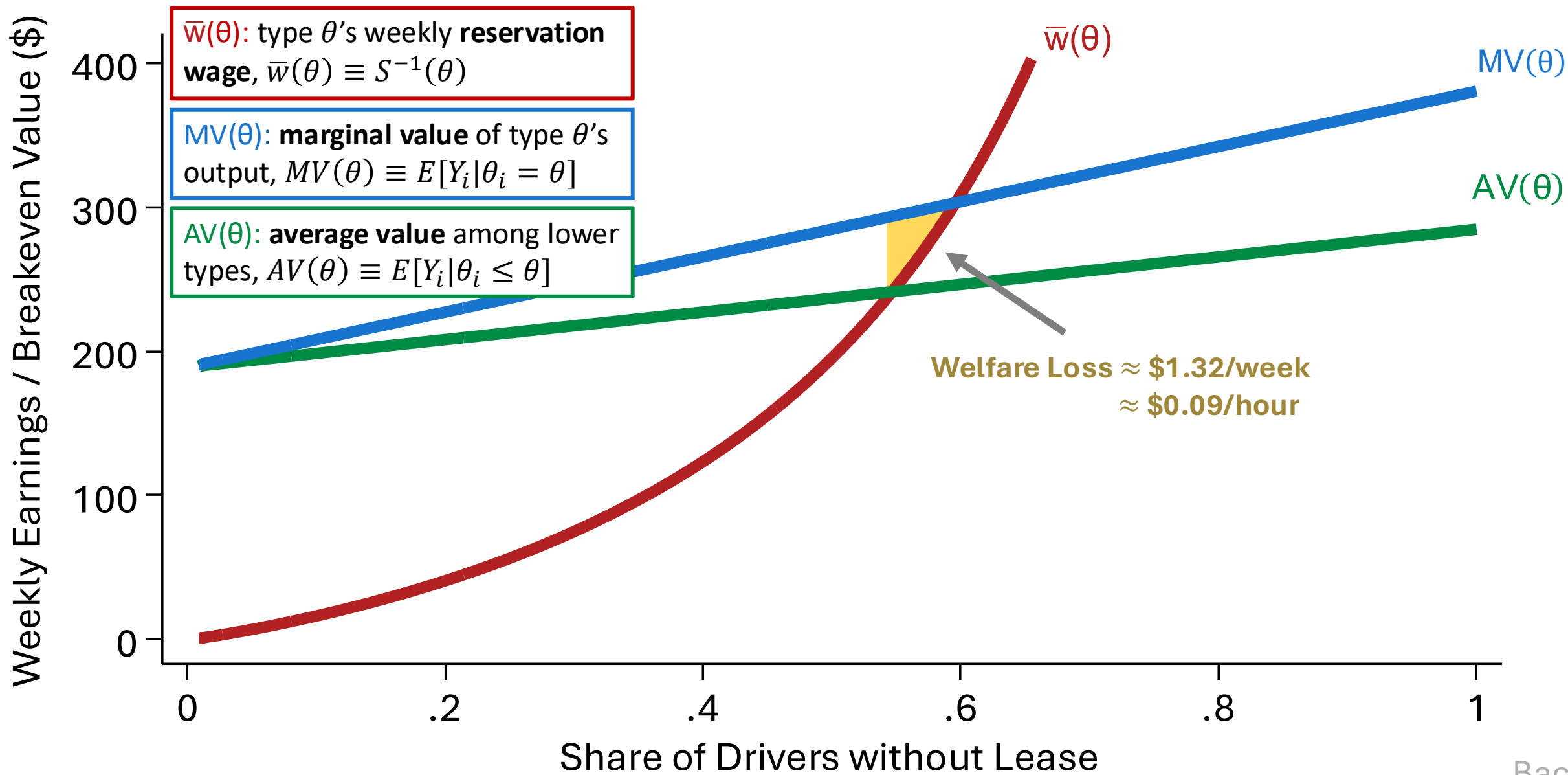




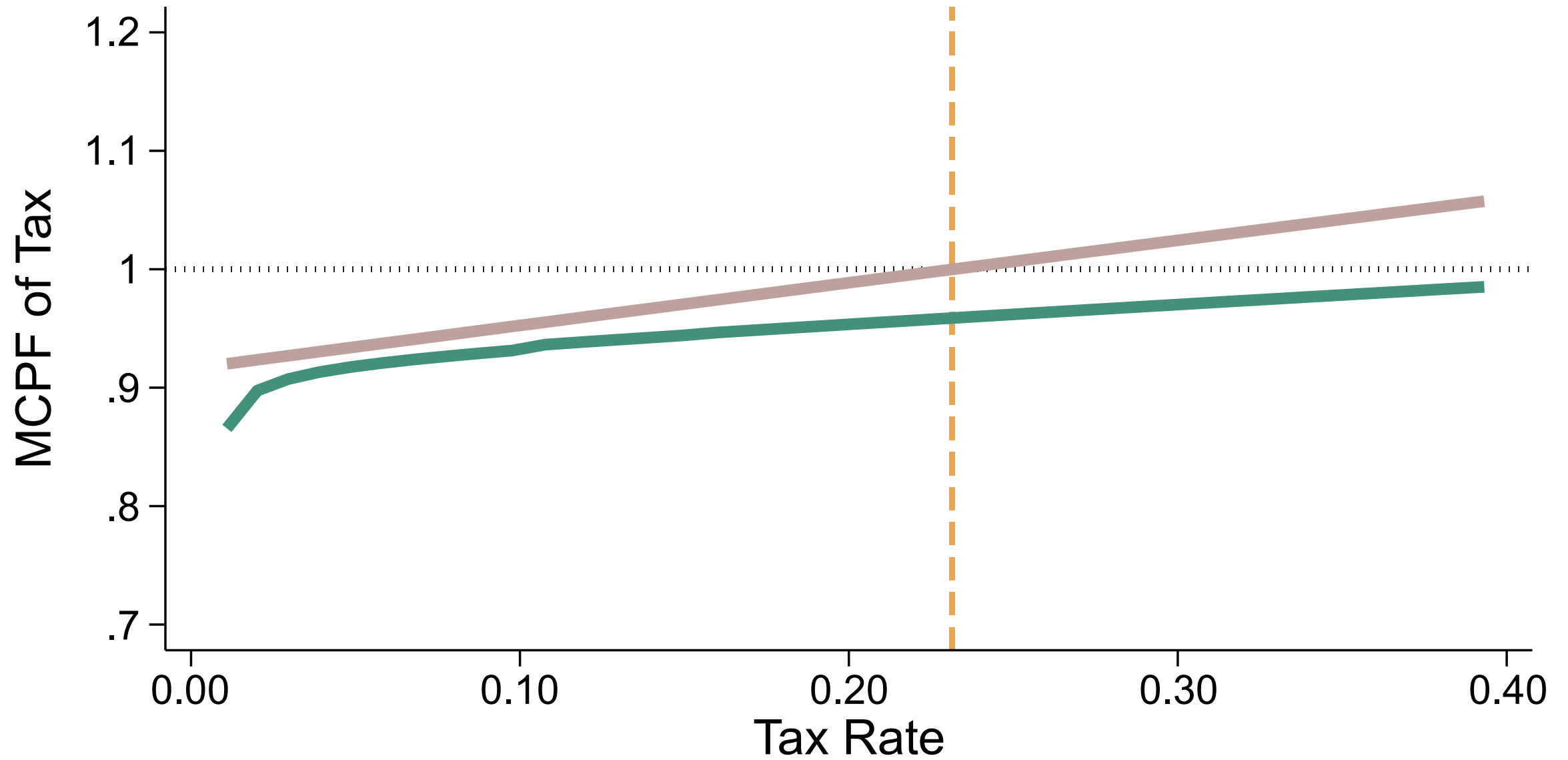








MCPF of Driver Fare Tax



Application to Rideshare Drivers

- Findings

- Sub-optimal equilibrium relies heavily on unpredictable fares.
- Welfare loss of \$1.32 per week driven.
- Taxing fare-based compensation mitigates adverse selection
- Can generate government revenue at with minimal social welfare costs.

- Limitations

- Linear extrapolation
- Experimental sample consists of *existing* Uber drivers.
- Excludes drivers with highest $\bar{w}$ —they're already working as taxi drivers!

Conclusion

- Experimental and theoretical framework for asymmetric information in wage contracts
 - Shows how moral hazard and adverse selection affect market equilibrium and welfare
 - Model correspondence to MTE estimands $\Rightarrow$ semi-parametric identification
- Apply framework to data-entry workers using an RCT that varies wage *offers*
 - Results imply an underprovision of fixed-wage contracts
- Policy prescription: increase taxes on piece-rate compensation (e.g., tips?)
 - $\tau \approx 14\%$ raises \$1 of government revenue at an estimated social cost of \$0.87 or less
- Extensions: use framework to estimate welfare loss in other markets
 - Experimentally vary compensation offers for other types of work (e.g., rideshare drivers)
 - Alternatively, use linked survey + administrative wage/productivity data

Any outcome-contingent contract can be vulnerable to moral hazard and adverse selection!

Thank you!

PAPER



SLIDES



Appendix

Experimental Design: Single Wage Offer

Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment***:

$$\frac{E[Y_i|W_i = 0] - E[Y_i|H_i = 0, W_i = 1]}{\pi} = \frac{(\pi E[Y_{0i}|H_i = 1] + (1 - \pi)E[Y_{0i}|H_i = 0]) - E[Y_{0i}|H_i = 0]}{\pi} = E[Y_{0i}|H_i = 1] - E[Y_{0i}|H_i = 0]$$

Identifying Selection Across Multiple Wage Offers

W_i : Randomized treatment-offer group

- $W_i = L$: Offered a choice between low hourly wage and piece rate
- $W_i = H$: Offered a choice between high hourly wage and piece rate
- π^W : Take-up of wage offer $W \in \{L, H\}$

Selection on Y_0 into offer H among those who would reject offer L :

$$\begin{aligned} E[Y_{0i} | H_i^H = 1, H_i^L = 0] - E[Y_{0i} | H_i^H = 0] \\ = \frac{1 - \pi^L}{\pi^H - \pi^L} E[Y_i | H_i = 0, W_i = L] - E[Y_i | H_i = 0, W_i = H] \end{aligned}$$

Selection on Y_1 into offer L among those who would accept offer H :

$$\begin{aligned} E[Y_{1i} | H_i^L = 1] - E[Y_{1i} | H_i^H = 1, H_i^L = 0] \\ = \frac{\pi^H}{\pi^H - \pi^L} E[Y_i | H_i = 1, W_i = L] - E[Y_i | H_i = 1, W_i = H] \end{aligned}$$

Balance Test

	(1) Experimental Wage Offer	(2) Output Value
Number of Previous Tasks/1000	−0.0478 (0.0343)	0.191*** (0.0305)
Age	0.00141 (0.00529)	−0.0683*** (0.00453)
Female	0.0909 (0.124)	0.366*** (0.108)
Minority	−0.0528 (0.125)	−0.896*** (0.109)
Employed	−0.202 (0.138)	0.142 (0.121)
Student	0.0685 (0.169)	−0.474*** (0.149)
F-statistic	1.019	36.446
p-value	0.426	0.000
N	3030	3030

Hourly Wage Take-up

	(1) Accepted Offer	(2) Accepted Offer	(3) Accepted Offer	(4) Accepted Offer
Log Hourly Wage Offer	1.198*** (0.0554)	1.202*** (0.0554)	1.212*** (0.0560)	1.245*** (0.0578)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
<i>N</i>	2728	2728	2728	2728

$$\Pr(H_i = 1) = \text{inv logit} (\alpha \ln W_i + \boldsymbol{\nu} \mathbf{X}_i)$$

Suppose supply and average value are linear in the wage offer:

$$Y_i = \xi_0 + \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \epsilon_i$$

$$H_i = \gamma + \delta w + v_i$$

Let $\bar{w}_i$ be worker i 's hourly reservation wage $\Rightarrow$ CEFs of hourly take-up and output value:

$$E[H_i | \bar{w}_i \leq w] = \gamma + \delta w$$

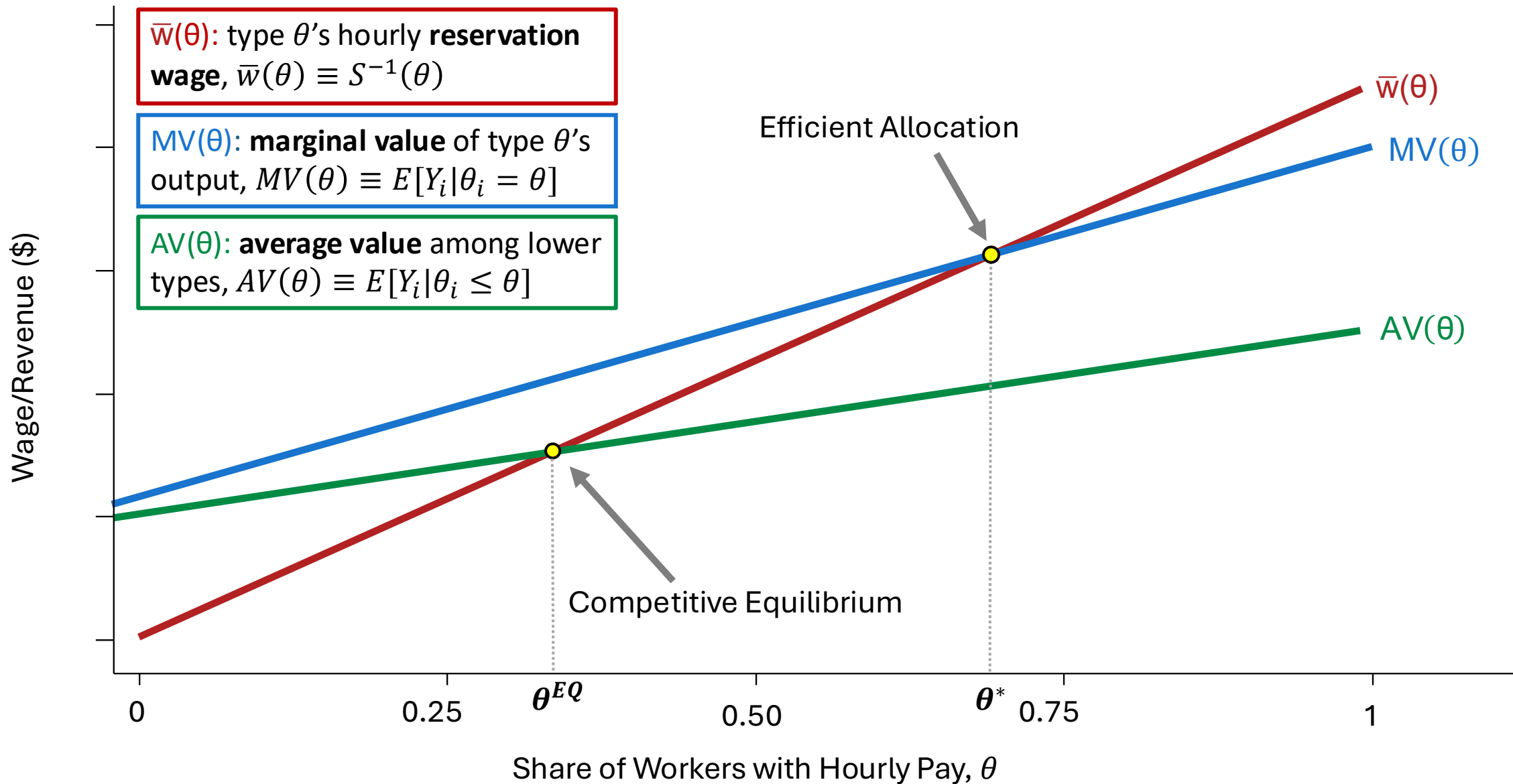
$$E[Y_i | \bar{w}_i \leq w] = \xi_0 + \alpha + \beta_1 H_i \times w$$

$$E[Y_i | \bar{w}_i > w] = \xi_0 + \beta_0(1 - H_i) \times w$$

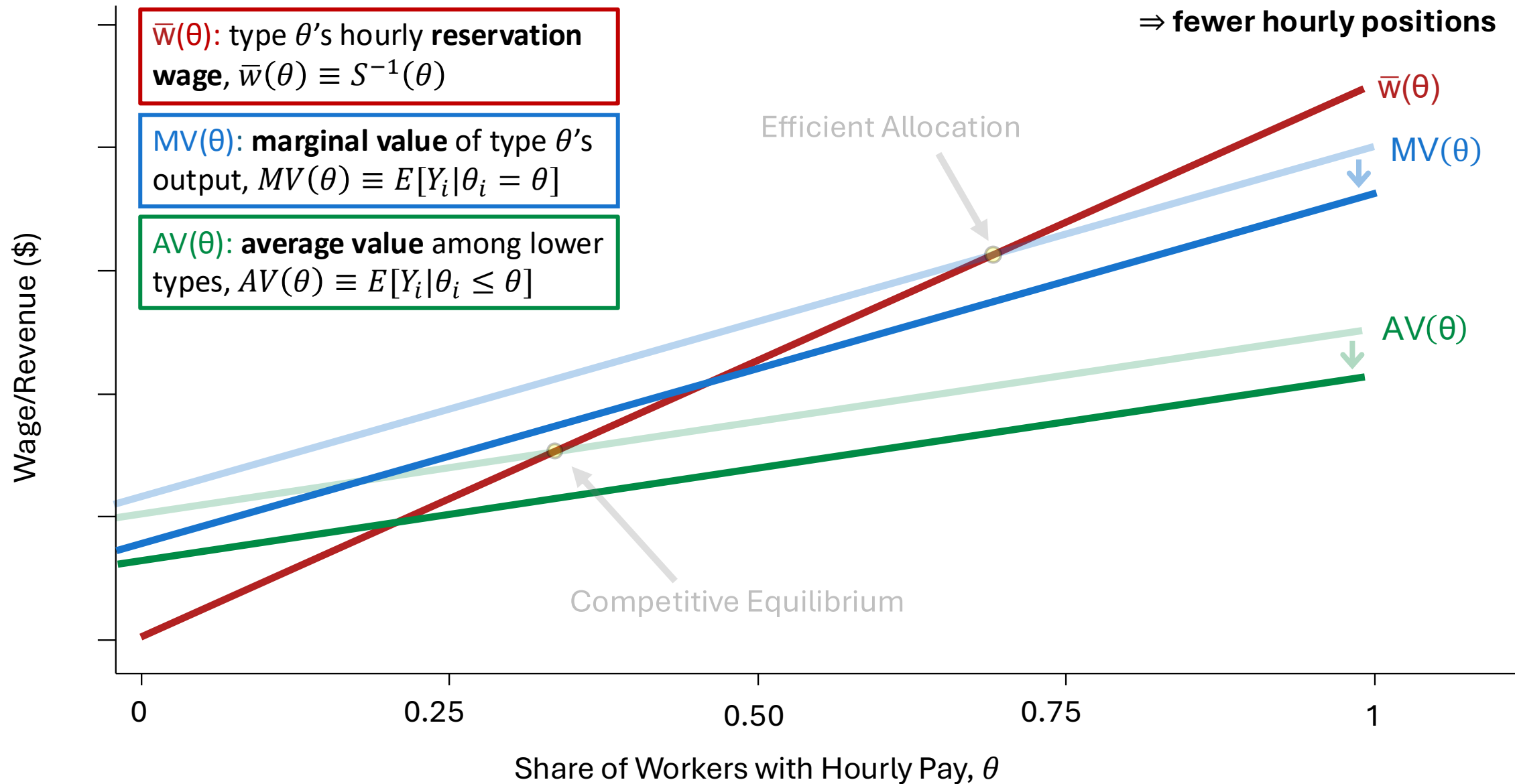
The *marginal* potential output values $E[Y_{1i} | \bar{w}_i = w]$ & $E[Y_{0i} | \bar{w}_i = w]$ are given by

$$E[Y_{1i} | \bar{w}_i = w] = \frac{\partial(E[Y_i | \bar{w}_i \leq w]S(w))}{\partial S(w)} = \xi_0 + \alpha + \frac{\gamma\beta_1}{\delta} + 2\beta_1 w$$

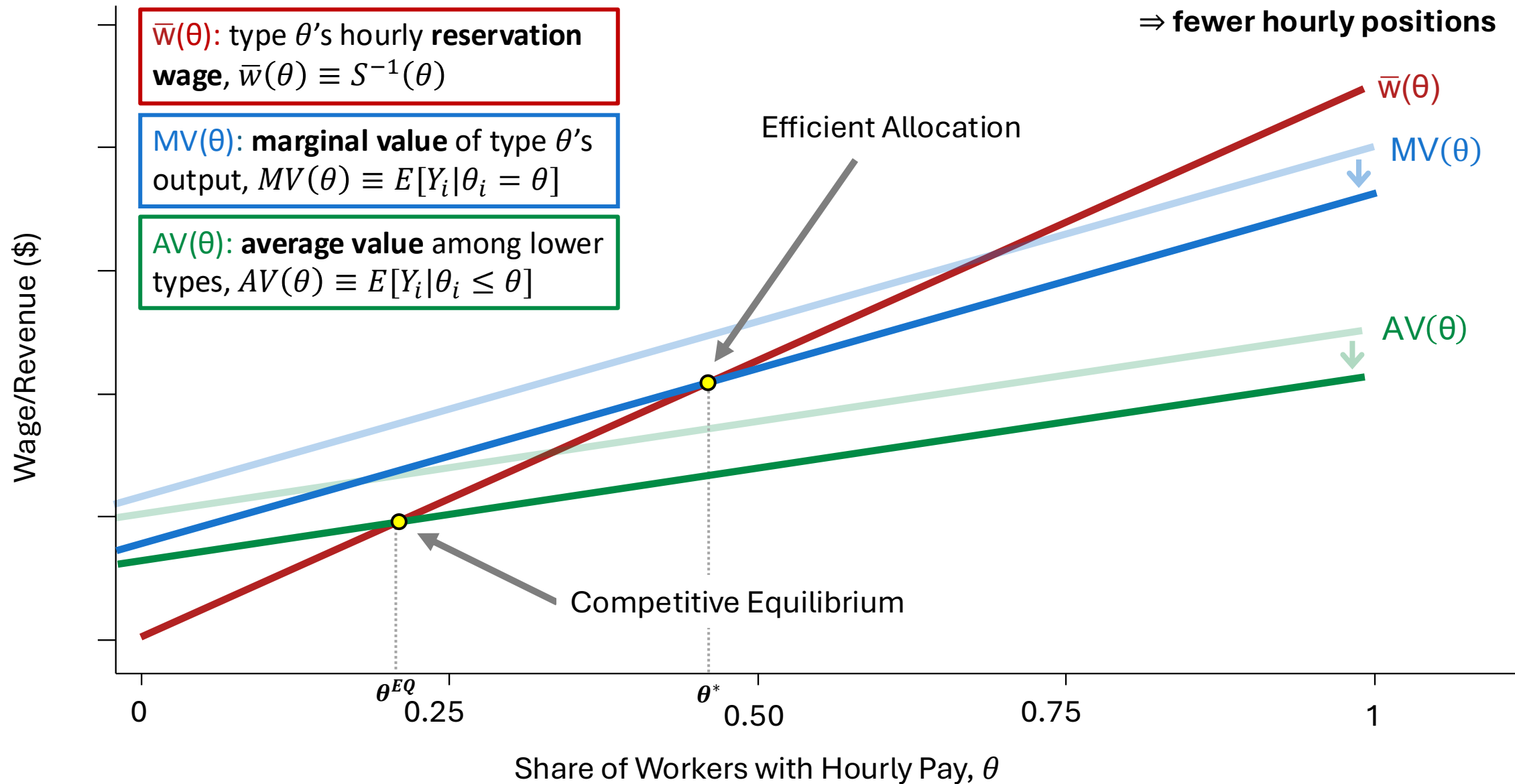
$$E[Y_{0i} | \bar{w}_i = w] = -\frac{\partial(E[Y_i | \bar{w}_i > w](1 - S(w)))}{\partial S(w)} = \xi_0 + \frac{(\gamma - 1)\beta_0}{\delta} + 2\beta_0 w$$

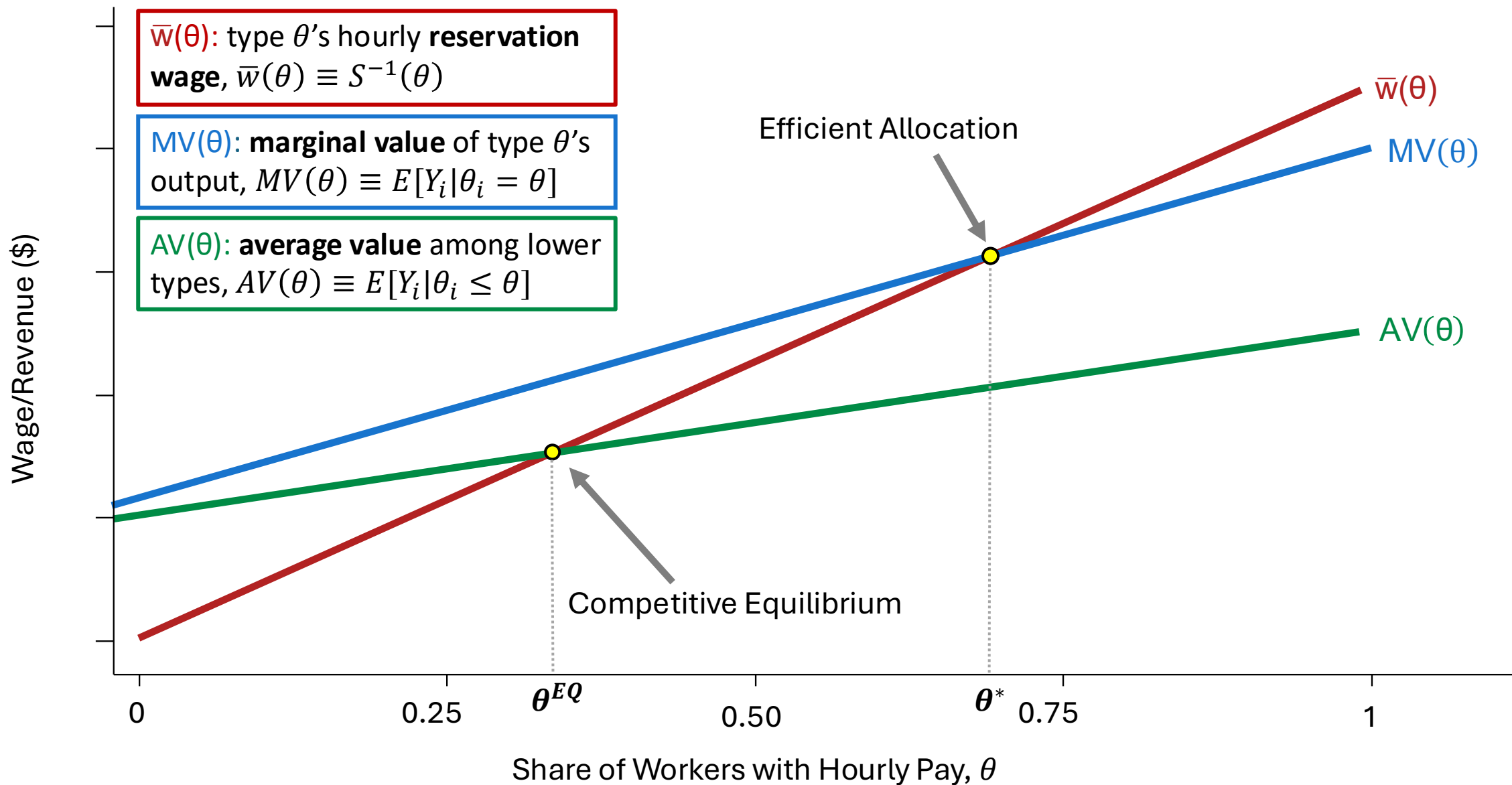


Input monitoring costs reduce firm's value of fixed-wage hire
⇒ **fewer hourly positions**

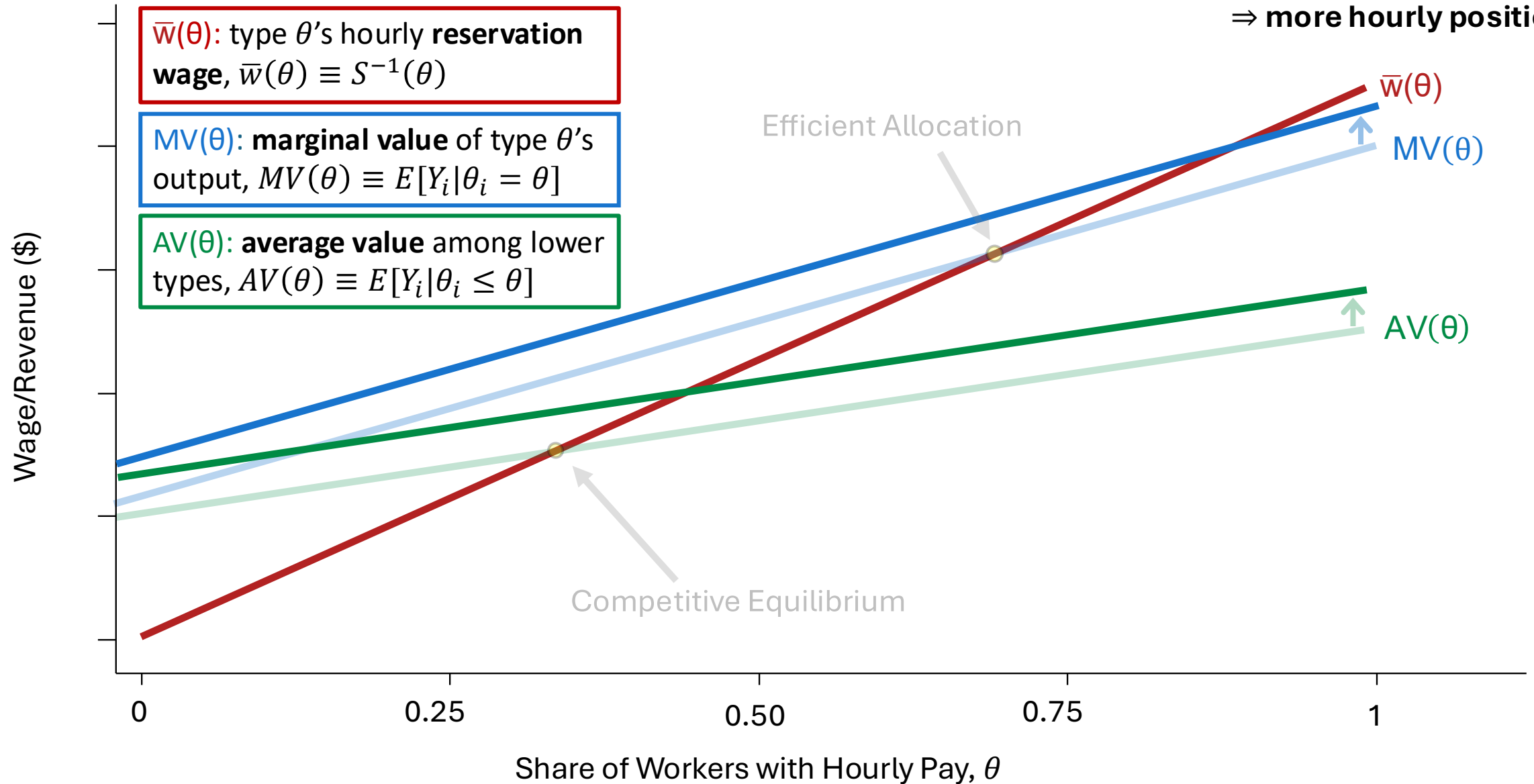


Input monitoring costs reduce firm's value of fixed-wage hire
⇒ **fewer hourly positions**

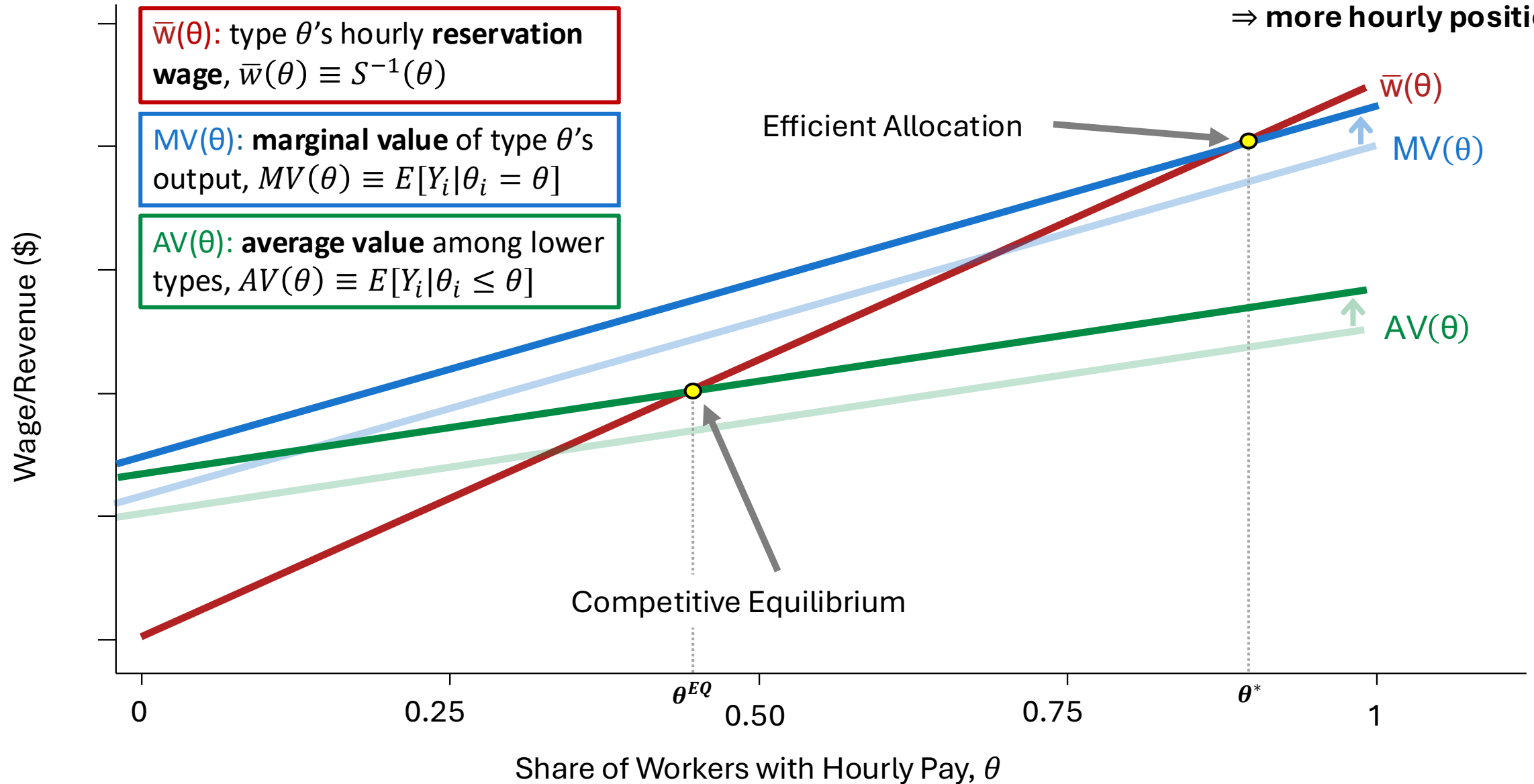


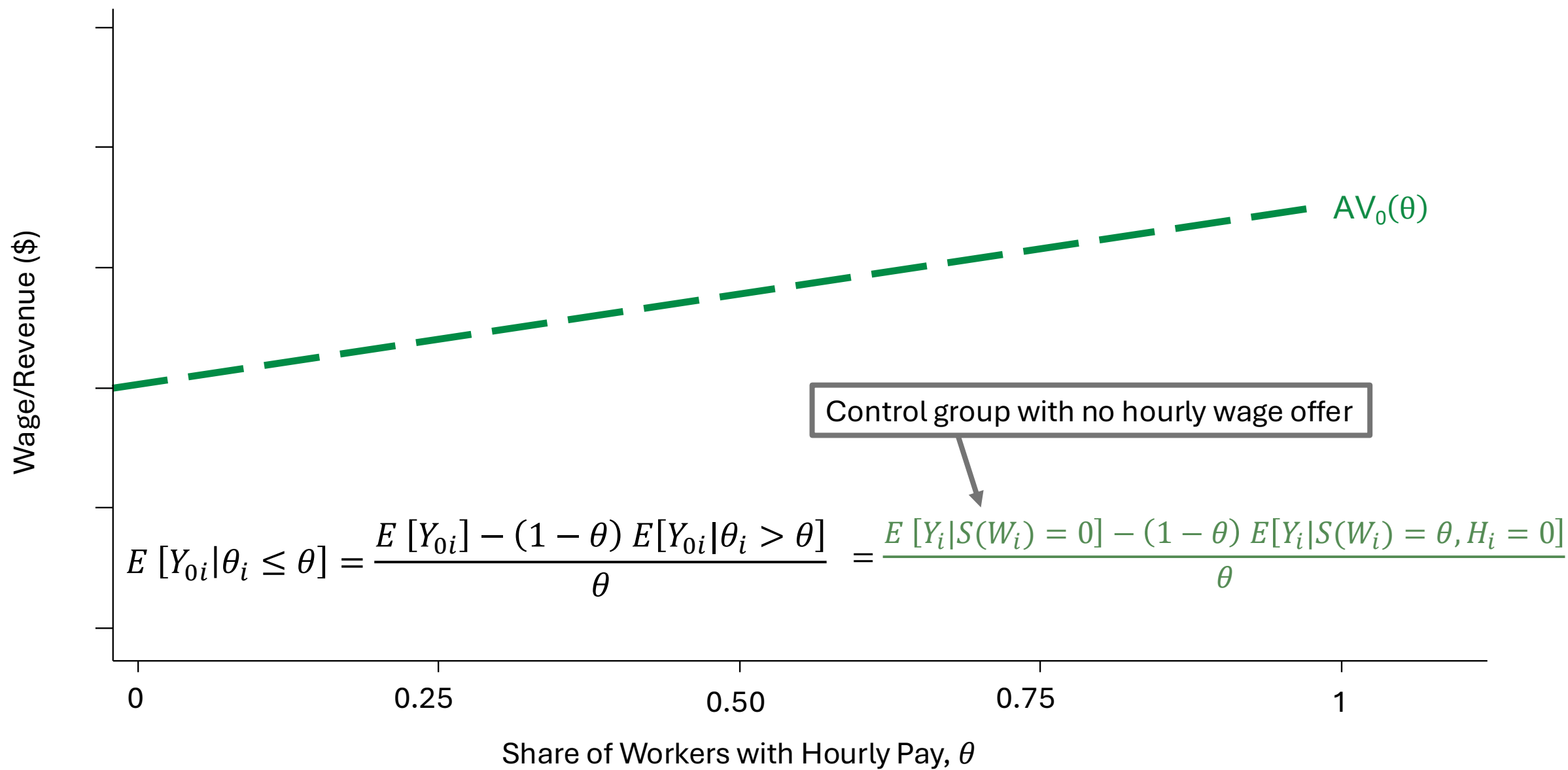


Output monitoring costs increase the firm's value of fixed-wage hire
⇒ **more hourly positions**



Output monitoring costs increase the firm's value of fixed-wage hire
⇒ **more hourly positions**





Estimation Details

Estimation Follows Carneiro et al. (2011):

1. Estimate $S(w) \equiv \Pr(H_i = 1 | W_i = w)$ using [logit regression](#)
2. Condition on X using double-residual regression (Robinson 1988)
 - X : Controls for number of previous tasks, task start time, and employment status
 - Assumes additive separability of X : $E[Y_{ji} | X_i = x, \theta_i = \theta] = \xi_j \tilde{X}_i + MV_j(\theta)$
3. Separately estimate $E[Y_i | S(W_i) = \theta, H_i = 1]$ and $E[Y_i | S(W_i) = \theta, H_i = 0]$
 - Local quadratic regressions of $\tilde{Y}$ on $S(W_i)$ for hourly and piece-rate workers ($bw=0.2$)
4. Differentiate with respect to θ

No Evidence of Earnings Bunching at Wage Offer

